Change-marked copy: additions relative to the 0f14fa3f compact-diagnosis draft are blue.

Conditional language generation in glyph image space

Imagized Language Model Project

Abstract

Written language Writing includes visual forms outside standard character encodingsthat character-identity interfaces need not preserve. We investigate continuous interfaces study a model that read and emit glyph reads prompt images, with generated generates answer images and rereads its emitted pixelssupplying subsequent history. Externally pretrained foundations provide Pretrained weights supply inherited language knowledge; tokenizers, lexical heads and decoding, font rendering lookup and a live teacher are absent from the deployed answer path. A restricted actor retaining 7.522B visual and contextual parameters gives sixteen clear complete literal answers among 32 reserved questions, versus eleven for a fixed-body control. Neither arm passes its qualification. A separate compact continuation experiment freezes a 1.409B contextual body and fits image interfaces on uses 13,785 complete 766 Chinese book and question–answer sequences and 41,covering 2.576 million target glyph positions298 updates. Two 8,172-update fits use 3.930 and 4.889 GiB peak allocated GPU memory. Both score zero Its final proportional model gives 7/32 strict complete exact QA answersout of , 14/32 at all five checkpointsreadable answers and 0/16 both-correct pairs, with no credited coherent complete book continuation. A wider subsequent joint image writer removes a measured velocity-rank restriction without solving language. Frozen-weight diagnostics show that writing increases original frozen physical-review credits from 599/1,040 to 970/1,040 but fails before generated feedbackthe per-group prerequisite. A pixel audit finds seventeen blank-flag errors, despite near-exact codec reconstructionincluding four credited empty images; additional sampling steps do not repair itoriginal grades remain frozen. Coupling image likelihood to the continuation state passes a real-source computational preflight within 17.795 GiB, but its language is unmeasured. The reader also has full continuation model retains 7.578 billion parameters; single-device execution establishes neither a measured affine target-alignment restrictioncompact foundation nor matched-quality efficiency. Cold replays verify Review uses one model reviewer with prior cohort familiarity and no independent pixel-feedback execution for fixed caseshuman adjudication. These results separate execution, physical The evidence separates visual writing and conditional from sustained languagesuccess. Reliable open-ended ; useful generation, historical-form transfer and a matched-quality efficiency advantage remain unresolved.

Written language is encountered as marks on paper, screens and historical surfaces. Its visual form carries information about script, handwriting and material history. Modern character encodings provide a powerful interface to many such marks , but a finite inventory cannot express every damaged, transitional or unencoded form. An image-based language modelcould retain that evidence throughout reading and generation, with transcription offered as an optional output viewWriting reaches a reader as visible marks. Script, typeface, handwriting and damage change those marks without necessarily changing their interpretation. Conversely, historical or unfamiliar forms may have no agreed character identity in a chosen encoding or tokenizer. A visual language interface would retain the marks while learning to

interpret and continue them. Its intended output is readable writing, including explanatory passages and, eventually, attested forms absent from the model's preparation fonts. This motivation does not establish that image representations carry more usable information per unit of computation or reproduce human reading mechanisms.

Pixel-based language modelling already has substantial Pixel-space language generation has established precedents. PIXAR demonstrates autoregressive language generation in pixel space and identifies the difficultyof generating readable text[1]. Continuous autoregression can use a diffusion loss in place of vector quantization[2], while Diffusion Forcing models continuous sequences under independently chosen noise levels[3]. These results motivate a visual language interface, but do not establish that a small glyph generator acquires linguistic knowledge or that image input automatically reduces training cost. Exact pixel factorization also has established precedents in PixelRNN and conditional PixelCNN[6,7]. Masked autoregressive networks provide normalized probabilities without a character dictionary[8]. Multilingual generation from pixels has also been studied[9]. studies autoregressive text generation from pixel representations and identifies readability as a central difficulty[1]. PixelRNN and conditional PixelCNN provide normalized image-density formulations[6,7]. Their existence motivates a trainable visual writing distribution, but neither an image likelihood nor a sharp glyph establishes that an answer follows its question. Replacing a character inventory with physical image variables also does not remove discrete computation: a binary raster and a categorical distribution over small pixel masks are discrete representations of appearance.

Recent continuous multimodal flow models share a generative objective across embeddings while retaining token-derived language encoders and text decoders[15]. OCG reports rendered-question and answer-image reasoning with teacher assistance on the restricted CLEVR task[16]. We focus on independent image-only inference and controlled tests linking conditional scores to autonomous writing.

We investigate the conditional generationproblem directly. The model receives raster history, predicts a distribution of future writing , emits pixels and rereads those pixels. Chinese provides a demanding initial case: stroke configurations must remain coherent while context selects among many similar written forms. The present study uses We study the coupling directly. A prompt image supplies context; a neural writer emits a glyph image; the model rereads the actual emitted pixels before deciding what to write next and when to stop. Typed questions may be rendered at the input boundary, and transcription may follow completed image generation. Neither operation supplies answer identities to the deployed writing loop. Offline text, teachers and pretrained foundations remain permitted sources of supervision and inherited knowledge. Independent inference therefore differs from learning a language entirely from images without textual pretraining.

The initial language setting uses Chinese writing in one canonical font to isolate this coupling before introducing handwriting or historical scripts. and ordered glyph cells. This isolates conditional shape generation before raw page reading, handwriting or historical-script transfer. Earlier compact complete-sequence fits produce zero strict complete answers at their fixed checkpoints. The subsequent full continuation trial recomputes all 36 answer layers while preserving frozen prompt memory, exposing complete QA and book targets rather than only isolated first glyphs. Its five prescribed checkpoints, paired questions, erased inputs, long continuations and retained successes separate physical fidelity from complete language. The final model improves some short answers relative to its initial checkpoint but fails the pilot.

The experiments progress from initial raster readers and denoisers to inherited contextual bodies with separately learned glyph interfaces.

The earlier architectures provide controls. Restricted short-answer experiments evaluate the complete-glyph loop shown in Fig. S37; a separate compact experiment then fits complete book and question–answer sequences and measures all five saved checkpoints. A controlled wider writer and frozen-weight interventions then test the sources of failure. The contributions are the The study contributes an explicit emitted-pixel inference interface, controlled language tests, reproducible single-GPU execution and measured writer/reader restrictions. None establishes a generally usable visual language model or a from-scratch language learner. complete-language comparisons and evidence that supplied-history writing and one-GPU execution are insufficient qualifications. The failed full trial motivates a separate conditional joint glyph writer. That head-only fit is complete and fails its physical prerequisite; its original review credits are preserved with detected blank-flag errors rather than treated as corrected physical accuracy. The probability diagnosis in turn motivates training the continuation state through complete-image likelihood. The exact state-gradient implementation passes a real-source computational preflight, but its completed fit and language remain unmeasured. The larger objective is useful visual language across written forms, with historical and unencoded writing assessed on its own provenance, shape and meaning. That objective remains open.

Results

An initial continuous visual interface preserves the written signal

In the initial control, each observed cell is a 32×32 ink image. An orthonormal transform maps its signed pixels into a 1,024-dimensional continuous field. The transform is invertible and contains no learned character dictionary. An ordered sequence of these fields enters a causal transformer. The deployed model receives no Unicode values, OCR transcripts, character IDs or vocabulary logits (Fig. 1). Host-side rendering uses encoded source text during data preparation. This distinction matters: the current benchmark studies learning from rendered Chinese, rather than discovering a script from raw archaeological photographs.

Ordered image history contains next-writing information

The strongest established reader result uses 2,048 deterministic development windows and an evaluator-only bank of 1,024 canonical glyph images. Full-history next-image top-1 is 19.97%, above the image unigram at 1.42%, the symbolic bigram at 12.26% and the prefix-shuffled condition at 18.36%. The ordered history improves target log probability by 0.174 nat over the shuffled control. This result supports bounded conditional prediction from image sequences.

These figures measure selection among completed images. The evaluator bank does not enter the model and cannot act as a deployed writer. Thresholding the predicted continuous field can mix strokes from several plausible futures. For squared-error prediction, the population optimum is $f^*(H) = \mathbb{E}[X \mid H]$. A conditional mean need not be a valid glyph when several futures are possible. Sharp copying, target ranking and sampled language generation therefore require separate measurements.

A glyph prior does not automatically bind to language

To separate visual synthesis from contextual prediction, we train an unconditional glyph prior and a noisy visual compatibility critic while freezing the causal reader. The combined runtime

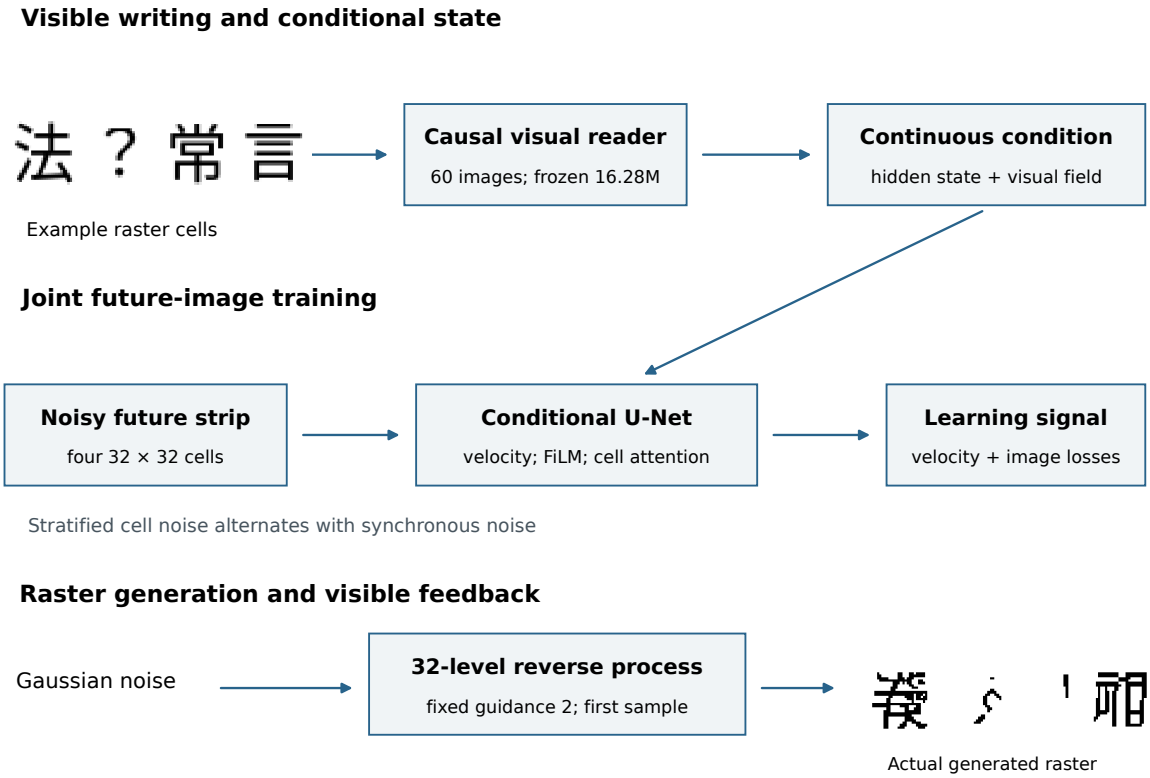


Fig. 1 | Initial visual language interface and its conditional generation test. a, Rendered or native writing supplies ordered image cells to a causal visual reader. Text rendering is an optional host interface. **b**, A conditional denoiser uses the frozen reader state and a noisy four-cell strip to predict diffusion velocity. Future cells exchange information through convolutions and pooled-cell attention. **c**, Sampling starts from noise, emits four visible cells and appends their pixels to the next context. Optional transcription follows generation and is absent from the model. The schematic shows the initial denoising control. Its generated content remains incorrect; the later complete-glyph runtime is shown in Fig. S37.

contains 23.27 million parameters. The critic scores an arbitrary noisy image against the visible history. In the ideal density-ratio limit,

$$\nabla_x \log p_t(x \mid H) = \nabla_x \log p_t(x) + \nabla_x \log \frac{p_t(x \mid H)}{p_t(x)}. \tag{1}$$

The construction is related to classifier-guided diffusion[4].

On 512 development windows, the clean critic reaches 21.875% top-1, versus 20.898% with shuffled history. Its first guided sample reaches zero target identities in 256 cases. Selecting the highest clean critic score among four samples reaches 1/256, with the same result after shuffling history. The generated glyph-like marks are diverse and nonblank, yet fail conditional identity (Fig. 2).

This experiment identifies a reader–writer gap without proving a unique cause. At the terminal corruption level, target signal power is 2.43×10^{-7}, and an exact conditional guidance term should vanish. That limit also occurs in successful conditional diffusion. It cannot alone explain the failure. Insufficient conditional-score accuracy, off-manifold gradients, limited prior quality and

finite sampling remain compatible explanations. We consequently test a directly conditional writer rather than interpret clean-image ranking as evidence of a generative solution.

Direct conditional block generation tests the missing coupling

The new writer models a joint four-cell future conditioned on 60 visible cells. It reuses the 16.28M-parameter reader and fine-tunes a 5.90M-parameter strip denoiser. A projected reader state modulates each residual scale, and a four-cell attention layer exchanges future information at the bottleneck. There is no separate guidance critic or candidate selector.

For signed clean writing x_0, noise ε and cumulative signal $\bar{\alpha}_t$, the training target is velocity:

$$x_t = \sqrt{\bar{\alpha}_t} x_0 + \sqrt{1-\bar{\alpha}_t}\varepsilon, \tag{2}$$

$$v^* = \sqrt{\bar{\alpha}_t}\varepsilon - \sqrt{1-\bar{\alpha}_t} x_0. \tag{3}$$

The clean estimate is $\hat{x}_0 = \sqrt{\bar{\alpha}_t} x_t - \sqrt{1-\bar{\alpha}_t}\hat{v}$. Training alternates independently stratified future noise levels with synchronous levels that cover the sampling distribution. Fixed visual-field and foreground losses retain the writing signal. Intermediate target-field alignment follows the motivation of representation alignment[5], using a fixed raster transform rather than a semantic teacher.

The run completes all 8,000 updates in 361.6 s, using 0.543 GiB peak allocated CUDA memory. Evaluation uses one primary sample from 256 prefixes under paired initial noise. Ordered first-position identity is 1.17%, compared with 1.17% after shuffling and 0.00% with a zero condition. Across all four future cells, identity is 0.586%, with an ordered-minus-shuffled difference of 0.098 percentage points and a record-bootstrap 95% interval of [0.000, 0.307]. Pixel F1 is 0.320. The registered complete-mechanism decision is `reject`. The raw image panel shows the first fixed examples without reranking or selection (Fig. 2).

A post-result motor diagnostic supplies the intended first-cell image to the same trained writer on 64 fixed prefixes. Exact plan and image-field supply raise pixel F1 from 0.378 to 0.459 and target identity from 1/64 to 5/64. The fixed inverse transform reproduces the supplied raster with F1 1.0. Because this test supplies the answer, it provides no autonomous-language evidence. It shows that the current learned writer still loses substantial shape information even with an exact visual condition.

Exact raster likelihood preserves copying but not conditional language

The diffusion writer's supplied-image failure motivates a distribution that retains the exact inverse visual field as aligned pixel logits. For binary image X, ordered history H and row-major pixel index p, we define

$$p_\theta(X \mid H) = \prod_{p=1}^{1024} \mathrm{Bern}[X_p; \sigma(6A_p + \delta_\theta(H, X_{<p}))]\,. \tag{4}$$

Here A is the inverse predicted field, scaled to signed pixels and clipped to $[-2, 2]$. A causal row transformer and within-row MADE supply the residual. The density contains 2,434,592 learned parameters, combined with the frozen reader for 18,711,841 runtime parameters. This is a conditional raw-pixel factorization[6–8], not a new language vocabulary.

With a supplied binary image, A_p is exactly ± 1. A zero-initialized residual therefore copies its greedy mode exactly before training. This is an architectural property, not acquired linguistic knowledge. Pure image NLL training retains that property: greedy copying is exact for all 256 development targets, stochastic-copy F1 is 0.99949, and a deranged supplied image reduces F1 to 0.358 (Fig. 3). These exercises supply the answer and cannot qualify autonomous generation.

On 512 natural targets, ordered NLL is 15.219 nats per image (0.02144 bits per pixel), versus 15.254 with earlier history shuffled. The paired advantage is 0.0355 nat, with source-record bootstrap 95% interval [-0.1014, 0.1693]. This interval includes zero. An independent-pixel distribution using the same predicted plan has NLL 846.09 nats. That comparison supports learned dependencies within the image; it does not isolate learned linguistic dependencies between images.

Autonomous sampling recovers 4/256 nearest target identities, versus 3/256 after shuffling, with pixel F1 0.350. The paired identity advantage is 0.391 percentage points, with interval [0, 1.210]. Greedy generation reaches 14/256 exact images but only 0.215 F1 and 0.330 target-relative ink, consistent with frequent sparse forms in the visible gallery. Four-step feedback identity is 4.297% ordered and 3.125% shuffled; this small secondary difference does not reverse the failed primary language tests. The registered decision is `reject`.

Joint learning improves the conditional image distribution

A matched experiment tests whether the same raster law benefits from learning its reader under image NLL. Both arms start from the exact-density EMA and receive identical new training draws. One continues to freeze the reader; the other differentiates likelihood through its hidden state and predicted field. Reader dropout remains disabled in both, so gradient flow is the controlled change. The unused contrastive temperature stays fixed.

After 8,000 updates per arm, joint target NLL is 7.026 versus 7.579 for the frozen control. The paired gain is 0.553 nat, with interval [0.450, 0.665]. Joint ordered history improves NLL over earlier-shuffled history by 0.185 nat, with interval [0.128, 0.244]. The frozen continuation also gains an earlier-history effect (0.119 nat, interval [0.052, 0.186]); extra training alone therefore explains part of the progress. The matched comparison isolates the additional likelihood benefit of updating the reader.

Joint first-sample identity is 9/256, versus 6/256 for the matched control and 8/256 after shuffling. The joint-minus-control identity difference is 1.172 percentage points, with interval [-0.397, 2.893]. Both arms retain exact greedy copying of all 256 supplied images. Joint sampling has F1 0.362 and isolated-ink fraction 0.00744. The joint arm passes six of eight gates, but fails sampled quality and ordered-generation criteria; its decision remains `reject` (Supplementary Note 13 and Fig. S2). This is evidence of improved conditional learning, not a qualified language generator.

Raw-pixel search improves choice but exposes repetitive rollouts

Known-form validity and target correctness separate two remaining problems. A post-result audit of the original saved images finds exact known-form membership in 43.750% of joint first samples, versus 33.594% for matched frozen training. The paired gain is 10.156 percentage points [4.219, 16.240]. Joint greedy images reach 98.438% known-form membership, but only 7.422% exact target match (Supplementary Note 14). Clear writing alone is insufficient.

We next search the same normalized pixel law with eight partial-raster beams, without a glyph inventory or weight change. On 2,048 new deterministic development windows, ordered exact next-image match is 11.621%, versus 8.252% for original pixel greediness and 9.863% after earlier-history shuffling. The beam-minus-greedy gain is 3.369 points [2.351, 4.412]; the earlier-history gain is 1.758 points [0.966, 2.552]. Every primary beam output matches a known canonical raster after generation.

Eight-step feedback on 128 prefixes reaches 6.641% exact target images, versus 4.883% greedy, but adjacent repeated images rise to 25.446%. The actual target rate is 0.112%. Thus the registered decoder decision is `reject_decoder`: five of six gates pass, but repetitive rollout fails (Supplementary Note 15 and Fig. S4). This outcome is compatible with known limits of probability-maximizing language decoding [10]. No repetition filter or output replacement is applied.

Natural questions expose the gap between selection and complete answers

To test use of language beyond continuation, we add a copy distribution over images in the current prompt, following the established pointer–generator idea [11]. Repeated identical images contribute summed attention mass. A learned gate combines copying with the exact raster law, and a separate stop hazard controls answer length. Only observed pixels and causal states enter this 19.01M-parameter model. No reference answer, symbolic identity or persistent writing inventory enters generation.

Training uses 3,211 natural Chinese questions with full paragraphs from CMRC 2018 [12]; article overlap is removed across training and development. Both copy and no-copy arms make 4,000 updates from the same V55 weights, retaining natural continuation and motor exercises. On 256 fixed development questions, the copy arm gives 10 exact stopped answers (3.906%), versus one (0.391%) with question pixels blanked and zero for the no-copy control. Raster F1 is 15.718%, 5.553% and 6.582%, respectively (Fig. 4). The paired question-dependent F1 gain is 10.165 points [7.043, 13.371].

Complete reading remains inaccurate. Only one of 393 question pairs sharing a paragraph has both answers correct. Output reaches the 17-cell safety cap in 22.656% of primary cases. All 2,677 emitted primary cells use the copy branch; the free writer is retained but not exercised in these answers. All 256 supplied motor images remain exact. Four of seven gates pass, failing answer accuracy, F1 and paired-question correctness; the decision is `reject_reading_model`. A post-result image-only diagnostic finds a correct first glyph in 43 cases, but only 10 become correct complete answers. Fourteen outputs contain a complete reference prefix followed by extra output. Both evidence selection and answer structure need improvement (Supplementary Note

16).

Image-block answers improve structure but retain selection errors

We next normalize a joint start–end law over contiguous observed image blocks of one to sixteen cells. Equal image sequences share probability mass, and the selected end position terminates the answer. Training marginalizes all matching source locations using target images alone. This 19.60M-parameter model retains the free raster writer and natural/motor exercises.

After 4,000 additional updates from the V57 copy checkpoint, exact answers reach 25/256 (9.766%), versus 3/256 with question pixels blanked and 21/256 for matched additional glyph-answer training (Fig. 5). Block F1 is 27.033%, versus 9.169% blanked and 22.140% for the matched control. The block-minus-control F1 gain is 4.893 points [0.469, 9.047], but the exact-answer difference is unresolved: 1.563 [-1.509, 4.652]. Extra training explains much of the improvement over V57; it cannot all be attributed to the block law.

Both answers are correct in only 4/393 paragraph pairs. Six of seven registered gates pass; paired-question correctness fails. Motor copying remains exact on 256/256 images. The block arm trains in 211.0 s at 0.967 GiB peak allocated memory on the same GPU, excluding previous training. Native prompt-PNG replay through the inference-only export exactly reproduces output pixels and boundary events. The result establishes limited extractive reading, not useful general free writing (Supplementary Note 17).

Frozen pretraining is not sufficient under a shifted visual interface

A separately registered probe compares frozen V58 and released MIXAR-477M[9] features using identical fresh 789,520-parameter block readouts. Both see the same prompt images, pair-aligned states, fixed continuous projections and fine image features. The student readout reaches 24/256 exact answers and 25.340% F1; MIXAR reaches 6/256 and 11.588%. Its F1 difference is -13.752 points $[-18.247, -9.094]$. No MIXAR pair has both answers correct, versus 12/393 student pairs. The frozen-foundation probe is rejected. This comparison includes different prior task training and a font/layout shift from MIXAR's native renderer; it does not disprove the released model's language competence. External pretraining is not local single-GPU work (Supplementary Note 18 and Fig. S5).

Moving the question before the passage is insufficient

For a causal reader, placing the question first allows each later passage state to depend on it. A registered input-order comparison starts both arms from identical V58 weights and retains the same images, parameters and 4,000-update budget. Question-first reading yields 21/256 exact answers and 23.765% F1, versus 27/256 and 26.908% for continued context-first training. The F1 difference is -3.143 points $[-6.544, 0.446]$, unresolved. Both questions are correct in 5/393 versus 16/393 paragraph pairs. Four of eight intervention gates pass, and the change is rejected. Motor copying remains exact in both arms. This result limits a simple ordering explanation of the selection failure; it does not establish that question-conditioned source representations are ineffective in general (Supplementary Note 19 and Fig. S6).

Native free writing fits pixels but fails reading

We next adapt every parameter of the released MIXAR-477M native pixel-language model[9], rather than selecting source-image spans. The actor receives prompt images and predicts unrestricted binary image blocks, with no output glyph inventory, OCR or external inference call. All 477,145,984 parameters receive answer-image likelihood gradients. The 1,000-update run takes 1,501.1 s on one RTX 4090 D at 9.912 GiB peak allocated memory, excluding loading and final serialization. External foundation pretraining is attributed separately.

Held-out answer-pixel NLL improves from 0.6842 to 0.1548 nats, yet generated answers remain inaccurate (Fig. 6). Exact answers are 0/256 and both-correct paragraph pairs 0/393. Exact-raster F1 is 0.149%, while aligned foreground F1 is 12.264%. Question blanking reduces foreground F1 to 8.473%, but the strict-raster F1 difference does not resolve zero. Five of ten registered gates pass, all concerning implementation, resources or termination rather than the required answer quality. The decision remains `reject_native_free_reading`.

Native PNG inference reproduces answer tensors, PNG bytes and stopping events from a self-contained learned bundle in a fresh offline process with empty model/code caches. This establishes an independent runtime, without certifying its language. The Noto CJK cell renderer differs from the released model's native font interface. A blinded answer-only OCR diagnostic is poorly calibrated even on reference images, recognizing only 40/256 exactly; it cannot replace a semantic evaluation (Supplementary Note 20).

Correlated pixels improve fit without establishing answers

We next replace the single product-Bernoulli head with eight conditional experts, each defined over a complete 32×64 bitmap. A shared latent expert permits pixel covariance without a character inventory. Both arms fully adapt the same native foundation for 1,000 updates on identical sampled examples; the eight-head actor has 490.00M parameters versus 477.15M for the single-head control. The comparison matches training budget, not capacity.

Held-out answer-pixel NLL is 0.1248 versus 0.1542 nats for eight and one experts, respectively (Fig. 7). Strict-raster F1 increases from 0.234% to 0.815%, a 0.580-point difference [0.089, 1.094]. Foreground F1 increases from 12.205% to 15.827%. Nevertheless, both arms produce zero exact answers among 256 questions and zero both-correct pairs among 393 shared-paragraph pairs. The eight-head normal-minus-blank raster gain is only 0.295 points [−0.006, 0.623], leaving question dependence unresolved. Foreground F1 also has no resolved question-dependent gain. The registered density-intervention gate and both bounded reading gates fail.

Training takes 1,887.5 and 1,972.8 s, with 9.904 and 10.174 GiB peak allocated memory for one and eight experts. Both independent bundles reproduce native PNG tensors, image bytes and decoder events offline. The eight-head gate entropy averages 1.812 nats, with five experts selected across the primary continuations, including termination. These usage diagnostics describe the mixture; they do not demonstrate semantic specialization (Supplementary Note 21).

Direct assignment changes reference scores but weakens free writing

To test conditional binding, we continue the eight-expert generator from one shared learned checkpoint. Two questions about a paragraph provide distinct accepted answer images. Both matched 489,998,208-parameter actors score all four correct and swapped question–answer streams. The control uses positive image likelihood alone; the aligned arm adds a smooth penalty on the assignment margin. Both consume identical pair draws for 2,000 updates. The extra loss cancels additive question-only and answer-only offsets, but this property does not guarantee useful conditional generation.

The aligned actor reaches 2.417% strict-raster F1 versus 3.229% for control (Fig. 8). The difference is -0.812 percentage points, article-bootstrap 95% interval $[-1.504, -0.202]$. Exact answers are one and two of 256 respectively; both-correct pairs remain zero of 393 for both. Normal-minus-blank-question raster gains resolve positively in each arm: 1.677 points [0.492, 3.033] for aligned and 1.765 [0.504, 3.251] for control. Thus both use question pixels, while neither meets the absolute reading requirements. The assignment intervention and both reading gates fail.

When accepted answer images are supplied, control chooses the correct pair assignment in 324/393 cases and aligned in 330/393. These descriptive scores cannot establish a statistically resolved improvement, and supplied answers are unavailable to autonomous writing. All-development pixel NLL is 0.113901 for control and 0.114186 for aligned. Training takes 7,355.2 and 6,737.3 s, with 10.301 GiB peak allocated memory in both. Complete sample journals match; standalone offline tensors, PNG bytes and stopping events replay exactly (Supplementary Note 22). Timing variation in one sequential run does not establish comparative efficiency.

Separate output coordinates preserve forms

To isolate a writing failure from conditional language selection, we cross the inherited reader features and a lossless pixel control with flow and direct image reconstruction. All four writers receive the same 64 training blocks, initial weights and 2,000-update exposure. Direct reconstruction recovers 63/64 blocks exactly from inherited features and 64/64 from pixels; neither flow arm recovers an exact block (Fig. 9a,c). This fitting experiment identifies a failure of the tested flow route, but supplies the answer image and cannot demonstrate language.

A broader direct writer then samples 160,000 blocks from a 12,976-example Chinese/English pool, consuming 12,972 distinct examples. After 10,000 updates, the fixed 128-case reconstruction audit reaches 56.563% foreground F1 versus 18.520% with shuffled conditions, with three exact blocks. The conditioning gain passes its original requirement, but fidelity misses 90%. The run uses a fitting-stage warm start and is not a matched broad-corpus comparison with the earlier flow writer. Raw outputs and failed criteria are retained.

A separate continuous writing codec removes the requirement to invert the semantic reader. With its original frozen weights and no additional learning, two 768-dimensional image codes reconstruct the same 128 development blocks at 99.998% F1 and 126 exact blocks, versus 21.413%

F1 and zero exact blocks with shuffled codes (Fig. 9b,d). The encoder receives target images; the decoder receives only continuous coordinates. Thus the result qualifies supplied-code fidelity, not prediction of an unknown answer.

The complete successor retains all 27 visual and 36 contextual layers while predicting this separate writing space. It has 7,539,239,282 total parameters and 9,351,937 trainable parameters, with 16.953 GiB peak allocated memory. All 24 actual execution checks pass, including early/late adapter gradients, future-image isolation and exact optimizer/RNG replay. Its two-update output remains fragmented. These preflight measurements establish execution, separately from the subsequent instruction-learning test (Supplementary Notes 24–26).

Conditional prediction remains inaccurate

The complete actor then finishes its registered 10,000-update instruction stage on all 12,976 admitted Chinese/English training examples. Fresh corpus permutations yield 20,000 recorded draws, including every admitted example once in the first epoch. The final adaptation and production checkpoint agree exactly; all recorded draws match the seeded order. This adapts 9,351,937 parameters within the inherited 7.539B-parameter actor, using 16.965 GiB peak allocated memory on one RTX 4090 D. External foundation and codec training remain part of its provenance (Supplementary Note 27).

The final actor generates answers to 128 Chinese questions from 64 shared-passage pairs and 64 English instructions, each with a fixed blank control. All 384 raw streams and input layouts are independently verified. Chinese foreground F1 is 32.671%, versus 19.508% with question pixels blanked (Fig. 10a). The paired difference is 13.163 percentage points, with 95% article-bootstrap interval [10.933, 15.730]. Nevertheless, no Chinese answer is exact and no pair has both answers correct. All 128 normal Chinese outputs stop through the learned decision. The three absolute answer-quality gates fail while question dependence and stopping pass (Fig. 10b,c). English recorded-reference agreement is also poor, at 9.763% F1 and zero exact matches; that metric does not credit every alternative valid answer.

A fixed 32-case diagnostic then separates supplied-code reconstruction, prediction with exact reference history, and autonomous writing. Each split/source group contains eight examples. Supplied-code F1 is 99.947–100%, but prediction with supplied history and answer length reaches only 21.805–35.554%, including training examples. No first predicted block is exact (Supplementary Note 28 and Supplementary Fig. S9). The initial pixels and stop probabilities agree between paths starting from the same prompt. The failure therefore precedes generated-history drift. This assisted diagnostic does not identify a unique learning repair or qualify language; the original autonomous decision remains rejection.

Numeric controls retain bounded visual reading

We test the original imported body through numeric chat/task vectors with its vocabulary absent. A separate helper temporarily applies the source text projection. Among eight Chinese development questions, native-budget images give four exact answers versus zero with questions erased;

original-scale images give one, and matched text gives three. All 742 retained helper decisions replay as exact argmax choices (Supplementary Note 29 and Fig. S10). This reused sample supports bounded reading, without qualifying image writing or isolating the scale effect.

The successor conditions a 38.652M-parameter causal writer on frozen prompt-only visual states. A joint flow predicts two-tile codes; the codec emits pixels, which are re-encoded for history. Real-example future-target exclusion, bit-exact optimizer/RNG resume and physical feedback checks pass. Full-corpus training completes 40,000 updates and 640,000 draws over 12,976 examples. The autonomous audit gives zero exact Chinese answers out of 128 and 29.386% foreground F1, versus 21.873% with questions erased. Four of six Chinese requirements fail (Supplementary Note 30 and Fig. S11).

A post-hoc diagnosis identifies an obstruction in the velocity head: its final affine span has rank 769 in 1,536 writing coordinates. In real arithmetic the remaining 767 components of initial noise are invariant before normalization. Actual-weight CPU trajectories support this prediction. A full-dimensional velocity control improves fidelity but fails its worst-image requirement on fresh noise. Predicting clean writing codes instead, with an FP32 final projection and analytic noise contraction, reproduces one learned Chinese image exactly on all 256 fresh draws (Supplementary Note 31). Clean-data prediction follows prior denoising work[20]; the known-image control establishes neither reading nor conditional language competence. The complete clean writer subsequently fits 32 seen training questions: 29 original answer images are exact, versus 31 frozen-codec reference images, with 99.997% original-image F1. This is a paired learning-capacity control, not unseen-question generalization (Supplementary Note 32).

Full-corpus clean writing still fails

A fresh 64.520M-parameter clean writer trains on all 12,976 examples for 40,000 updates, completing 49 corpus passes. Its autonomous audit produces 2/128 exact Chinese answer images and 37.631% foreground F1, versus 23.609% with question ink erased. Both questions are exact in 0/64 passage pairs; three original requirements fail. The question effect is resolved at 14.023 F1 points [10.861, 17.487], and all normal Chinese answers stop by their learned rule. Dependence on question pixels does not establish correct answers. A separate 12-request Chinese/English/numeric copy probe also gives no exact outputs (Supplementary Note 33).

The full writer fails even on the earlier 32 training questions: free generation reaches 45.771% F1 with no exact answer. Correct prior image codes and supplied answer length raise F1 to 58.045%, still with no exact stream. Supplied-code rendering reaches 99.9953% F1. Thus error is already present under correct histories. The clean predictor also slightly corrupts constructed near-clean target codes. This identifies an additional writing defect, without showing that it causes the much larger free-answer failure. The earlier small fit received 2,000 passes over its selected examples; neither that memorization result nor theoretical time-coverage calculations demonstrate a useful general language model.

A matched head-only refinement broadens denoising-time supervision while holding the causal body fixed. It improves assisted near-clean block F1 from 99.0658% to 99.8602%, but free-stream

F1 falls from 48.726% to 45.028%. Both arms remain at 0/32 exact training answers. This intervention fails all four prespecified repair criteria (Supplementary Note 34); better near-clean reconstruction is insufficient for prompt-conditioned generation.

An offline source-decoder diagnostic recovers 8/12 exact copy answers and 19/32 exact training answers from the same physical prompt images, compared with zero exact image outputs in both groups (Supplementary Note 35 and Fig. S12). All 44 normal prefill memories match the original 52-state BF16 suffixes exactly. The first source projection matches the correct answer prefix on 22/32 training questions, excluding the paired answer in each case. Thus some answer information is available before the new writer. Later source decoding uses the full layerwise visual/context KV history, which the image writer does not receive. These measurements motivate examining the compact-memory and conditional-readout boundary; they do not demonstrate sufficiency of the compact state or a usable image-language model.

A key-visibility intervention now tests that boundary directly. After identical prefill, permitting attention only to the final 52 task positions reduces exact source-helper copy answers from 8/12 to 0/12 and training answers from 19/32 to 2/32 (Supplementary Note 36 and Fig. S13). Original cache positions, rotary coordinates and first decisions are preserved. This rejects the tested frozen suffix interface without proving a lower bound on learned compression. The next continuous image actor retains all 36 native layers and full prompt KV. Initial forward, gradient, causal-mask and emitted-image feedback checks use 16.638 GiB on one GPU, with zero optimization updates. Its first writing block still depends on the initial reader state alone. The complete trained actor has 7.583B parameters including externally pretrained weights; 46.900M image interface parameters are trainable.

Subsequent training completes 40,000 updates over all 3,193 Chinese examples, using 18.298 GiB peak allocated memory, but gives 0/128 exact development images, 36.504% foreground F1 and 0/64 both-correct pairs (Supplementary Note 37 and Fig. S14). Erasing the question reduces F1 to 25.404%; the paired gain is 11.101 percentage points, with a 95% article-bootstrap interval of [8.578, 14.018]. All streams stop, but stopping and input sensitivity do not establish accurate answers. On the same 32 training questions, free generation gives 46.235% F1 and correct-history prediction gives 50.241%, with no exact images in either path. Supplied-code decoding gives 99.9953% F1 and 28/32 exact images. Thus conditional prediction remains inaccurate under correct history; generated-history errors cannot be the sole explanation. These physical-image scores are not a separate measurement of semantic accuracy.

A matched readout repair adds an initial continuous query and trains rank-8 suffix adapters while retaining the full prompt cache. After 2,000 updates per arm on those same 32 training questions, free F1 is 71.177% for continued control and 72.748% for adaptation; neither produces an exact complete image (Supplementary Note 38 and Fig. S15). Correct preceding history gives 77.372% and 79.629%, again with no exact images. Supplied-code fidelity is unchanged at 99.9953%. The adapted model fails three of four declared training-fit criteria. The small increase in image agreement does not resolve conditional writing or qualify general language capability.

Decoder-sensitive reweighting also fails (Supplementary Note 39 and Fig. S16). From identical adapted weights, 2,000 matched updates lower autonomous F1 from 85.984% to 70.009%, with 0/32 exact in both arms. Correct-history F1 is 92.430/75.116%; supplied codes give 99.9953%. All four repair criteria fail; scaling is rejected.

A separate native image-generator adaptation learns partial copying (Supplementary Note 40 and Fig. S17). Two fresh rank-16 adapters train on the same 32 Chinese/English copies for 512 updates. Removing exactly white bottom padding preserves every source pixel but changes frozen-VAE features. Training invocation time falls from 795.624 to 287.182 s, with peaks of 9.501 and 8.437 GiB. On the matched eight-case screen, OCR exact match changes from 2/8 to 3/8. Complete strip observation yields 6/32 training and 10/32 validation strings, against clean-reference OCR scores of 27/32 and 28/32. Raw omissions, repetitions and unwanted marks independently reject the 30/32 reliability requirement. All 64 outputs are retained; eight are original observations and 56 are additional calls. This model has 3.900B deployed transformer parameters plus an 84.046M VAE, with 24.359M trained adapter parameters. No natural QA is optimized in these copying runs. Faster copying execution therefore does not establish language understanding or efficiency at useful language quality.

Frozen recognition supervision also fails reliable copying: a matched 512-update treatment gives 6/32 seen and 9/32 held-out OCR-exact strings (Supplementary Note 47 and Fig. S24). A subsequent 32-update continuation supervises actual final images generated from independent noise, combining reader CE with balanced pixel error and differentiating through all 50 native steps[24]. Seen copying rises from 6/32 to 15/32 and held-out copying from 9/32 to 13/32, meeting an exploratory progress rule but failing the 30/32 writing floor (Note 48 and Fig. S25). Training takes 1,135.452 s at 12.057 GiB peak allocation. At initial assessment, the continuation lacks an additional-training control and changes input-encoding precision as well as the loss; the later comparison below holds precision common. Raw omissions and repetitions remain. It establishes neither semantic answering nor a quality-matched efficiency advantage.

The subsequent old-objective continuation gives 5/32 seen and 10/32 held-out OCR-exact copies on matched example draws; both it and the final-image pilot retain only 3/16 held-out Chinese copies (Note 49 and Fig. S26). This retrospective comparison differs in objective and compute. Continuing the pilot for 128 balanced updates, four per training input, raises held-out copying to 17/32, including 4/16 Chinese, but reduces seen copying to 13/32. The prospective Chinese and seen-retention conditions fail; only English retention passes. Training takes 4,500.117 s at 12.032 GiB. Omissions, repeated characters and copied instructions still reject reliable writing, and semantic QA remains untested. This result does not support further unchanged copying-budget scale-up.

A matched physical-loss replacement then increases held-out OCR-exact copying from 16/32 in control to 19/32, including 4/16 to 6/16 Chinese; seen copying is 16/32 versus 18/32 (Note 50 and Fig. S27). Both arms use identical starting weights, optimizer states, inputs and noise over 32 updates. The prospective exploratory rule passes, but raw omissions, repetitions and copied

instructions still reject reliable writing. No semantic QA is optimized in these arms. A separate full-page preflight fits two real discarded updates at 12.203 GiB; four native source reconstructions retain 97.834–98.180% foreground F1, while the fixed earlier checkpoint answers none of four normal questions correctly (Note 51 and Fig. S28). Physical representation and actual gradients have not established language learning.

A native unified backbone provides a separate visual-access diagnostic (Show-o2[21]; Supplementary Note 41 and Fig. S18). Its frozen source gives no exact answers on twelve image prompts, while the same body recovers all eight requested copy contents from their original text. Gated visual-only adaptation then trains 8.490M parameters for 4,096 updates on 128 copies and 256 natural questions. The native language body and Unicode output head stay frozen. Complete held-out text readout gives 27/32 exact stopped copies (11/16 Chinese and 16/16 English), but 0/32 correct natural answers. All 32 question responses collapse to two unrelated years. With matched passage seeds, 15/16 question pairs receive the same answer and erasing the question leaves 29/32 normal outputs unchanged; no pair is both correct. Training takes 2,758.904 s at 6.894 GiB allocated peak on one RTX 4090 D, excluding source pretraining, cache preparation and separate evaluations. The adapted source contains 2.839B parameters plus a 126.893M codec. These are auxiliary text-head measurements, with no trained image-answer result. Improved short-string reading therefore coexists with failed question-dependent answer selection.

The subsequent full-page curriculum and geometry intervention refine this interpretation (Supplementary Note 42 and Fig. S19). Another 2,560 updates raise exact copying to 29/32, but Chinese and English natural QA remain 0/32 at the original geometry. Full transcription improves on four short copying pages and fails on all four complete Chinese QA pages. The source interface had extended the official 432-square reading configuration to 1024-square pages with interpolated spatial positions. Resizing the same whole English pages to 432 recovers 11/32 exact stopped answers from the frozen source, versus 0/32 at 1024, without optimization. Nine of these eleven correct answers disappear when the passage is erased; two remain correct without it. Both learned adapter checkpoints reach only 5/32 at 432. All eighty calls per condition retain the same questions, references, seeds and decoding. The pixel/position change is joint, so the particular geometric mechanism remains unresolved. This recovers inherited source reading capability, while rejecting the existing adaptation as a demonstrated repair. Direct image answers and useful independent image-language learning remain unachieved.

A subsequent source-preserving visual-state correction improves English QA from 11/32 to 16/32 and English copying from 7/16 to 16/16, but Chinese QA remains 0/32 and copying 0/16 (Supplementary Note 43 and Fig. S20). Four of six reader requirements fail. A separate seen-prompt check corrupts all eight Chinese strings despite previous training exposure. The earlier dense checkpoint retains 13/16 Chinese copies on its original 1024-by-256 strips; adding only white bottom padding reduces copying from 29/32 to 7/32 overall, with every ink pixel and model weight fixed. Tight whole-request cropping also fails. All 480 layout-control calls are retained, including exact replay of the 160 original responses. A separate contextual-attention learner

retains 29/32 copies but leaves QA at 0/32 in both languages. A further 128-call feature/context intervention does not restore Chinese copying by shortening the square context alone; strip features in the long context recover 16/16 English but only 2/16 Chinese copies (Note 43). These observations identify geometric brittleness and failed Chinese fit, without isolating a unique internal cause or qualifying image-answer generation.

A separate contextual source-pixel component copies all 32 new Chinese/English strings and fits all 32 tested training questions exactly, but gives only 9/64 exact held-out Chinese answers and 1/32 both-correct paragraph pairs (Supplementary Note 44 and Fig. S21). Question erasure gives 2/64 and passage erasure 0/64. Three of four registered component requirements fail. Four actual native-image calls reproduce its cached descriptors and answer pixels, including the wrong first question. This is learned source-region selection with exact pixel copying, rather than free glyph synthesis. Its 2.378M trained head retains a 7.531B externally pretrained runtime. The clean copying and seen-data fit therefore expose a remaining selection/boundary generalization gap without qualifying a lightweight general image-language model.

A matched continuation uses the same four-pass schedule over 3,193 admitted Chinese QA images and 64 copies, with 3,257 updates per arm. Head-only control gives 23/64 exact unseen answers; additionally adapting 6.193M contextual attention parameters gives 26/64. Both retain 32/32 new copies, but answer both questions correctly in only 3/32 paragraph pairs (Supplementary Note 45 and Fig. S22). Both fail two of four reader floors, and the three-answer gain with no additional complete pairs misses the prospective improvement rule. Independent scoring reconstructs all 512 native outputs, and eight standalone calls replay pixels and events exactly. These checks verify execution and the failed comparison rather than free image-language synthesis. A separate retained-history audit finds only 2/75 intermediate image boundaries with a matching saved text-prefix state (Note 46 and Fig. S23); the distances do not establish an alignment repair.

Full-page answer learning remains unsuccessful

A complete natural-QA adaptation of the image writer fails all six prospective exploratory conditions (Supplementary Note 52 and Fig. S29). Four permutations of 32 training questions give 128 updates. Initial and final models each produce 0/32 source-accepted OCR-exact seen answers, 0/32 held-out answers and 0/16 both-correct held-out passage pairs. Both erased-input controls also give 0/32. Held-out copying falls from 19/32 to 16/32, exceeding the allowed loss of two exact strings. Training takes 12,782.358 s at 12.204 GiB peak allocation. All 384 original assessment pages are inspected, with malformed writing, extra fragments and omissions retained (Fig. S30). Imperfect reference OCR remains material. These observations reject scaling the unchanged recipe and do not establish an independent language model.

A separate native image-answer flow adaptation completes 1,536 updates on 384 image pairs using 37.381M low-rank adapter parameters within a retained 6.240B-parameter pretrained system (Supplementary Note 53; Fig. S31). The complete matched comparison yields 0/32 source-accepted OCR-exact seen answers, 0/32 held-out answers and 0/16 both-correct held-out passage pairs in both arms. Copying rises from zero to 2/32 seen and 3/32 held-out exact strings.

All 384 output pages are authenticated and inspected. Training takes 3.18 h at 15.697 GiB peak allocation, including loading, saving and fixed monitoring. Reference OCR remains imperfect. A small conditional-field signal and failed guidance-one diagnostic do not qualify answer generation; eight saved target-codec reconstructions preserve four seen answers. These results do not establish useful language learning.

Precise image fitting does not establish language generalization

Additional controls separate optimization from useful answering (Supplementary Notes 54–56; Figs. S32 and S33). A 128-update reader-query fit still gives no exact answers on two seen and two excluded questions. Supplying correct text to the native writer yields 1/4 source pages at guidance four, falling to 0/4 when the VAE reference is removed; adapted writing remains 0/4. A separate 3.453M-parameter causal pixel head learns from frozen image-only contextual states. After 1,200 updates, even correct preceding pixels leave Chinese training-piece accuracy at 76.831%. Restoring the same untrained head and extending optimization to 16,000 updates gives 512/512 exact autonomous training first blocks, but 0/128 Chinese and 0/64 English development first blocks. Active head training takes 279.131 s at 0.825 GiB peak allocation, excluding the pretrained reader and cache. Thus precise emission for learned conditions is achievable in this component; conditional generalization and complete language answering remain unresolved. Physical fitting is not a semantic benchmark.

Complete image-answer fitting and failed transfer

The native actor learns complete answer images (Supplementary Note 57; Figs. S34 and S35). Full prompt context conditions a causal pixel density and learned stop; emitted images become history. The externally pretrained 7.544B actor adapts 13.856M parameters on 32 natural Chinese questions from 16 passages. Exact complete training answers increase from 0/32 to 17/32, 29/32 and 32/32 at updates 0, 1,000, 2,000 and 4,000. The registered exact-fit stop triggers at 4,000; fresh-PNG evaluation confirms all 32 answers and all 16 passage pairs. However, reused development gives 0/16 exact Chinese answers and 0/8 English transfer answers, with foreground F1 0.240 and 0.034. Development outputs contain broken glyphs and unrelated training fragments. All outputs stop below the safety cap. Active optimization takes 63.2 min and total execution 81.6 min, at 18.191 GiB peak allocated memory, excluding inherited pretraining. Five cold-loaded PNG replays preserve complete pixels and stopping exactly. Complete image fitting succeeds; new-prompt answering fails. Changes in initialization, decoder, precision and schedules preclude attributing this gain to the likelihood alone.

The learned writing interface loses usable reading capability

Increasing the cohort from 32 to 512 questions gives 495/512 complete pixel-exact training answers after 16,000 updates, but no exact normal development answers (0/16 Chinese and 0/8 English; Supplementary Note 58, Fig. S36). All 512 training first images are exact; none of the 24 normal development first images matches a reference. The mismatch is therefore already present before feedback from an earlier complete generated image, although within-image autoregression

remains a possible contributor.

Scoring fixed owning checkpoints separates an erroneous learned distribution from a decoding-only explanation. Every supplied correct development first image is less likely than the corresponding saved emission. At the final state, the mean best-reference minus emitted-image log probability is -751.24 nat for Chinese and -941.82 nat for English. A maximum-likelihood decoder cannot select any of these supplied references over the existing emission. This does not rule out other correct wordings or fonts. Relative question conditioning nevertheless transfers: 14/16 Chinese reference events favor their owning question and all eight paired assignment contrasts are positive. Relative preference alone does not establish useful output.

The original source reader, using the same normal and question-erased PNGs, reproduces all 32 ILM image-memory fingerprints. A temporary helper-only source text head then gives 12/16 literal normal answers, zero erased answers and 11/16 source-text answers. The two date questions receive the correct years; the learned pixel writer's corresponding forms remain damaged. Some literal mismatches are aliases or longer relevant wording. These are reused development cases without independent semantic grading, and the symbolic helper generates no ILM images. The comparison localizes substantial lost capability to the writing interface without uniquely separating its initial query, conditional density and image-history projection.

We consequently separate glyph formation from language continuation (Fig. S37). A continuous CNN encoder and decoder learn training-only semantic anchors for 4,087 distinct canonical forms. Direct decoding is exact for 4,048 forms (foreground F1 0.999940); actual-image roundtrip is exact for 3,697 (F1 0.999634). The 80.22-second primitive run peaks at 0.728 GiB, excluding the large contextual body and its pretraining. Semantic alignment is weaker than reconstruction, with mean cosine 0.77735 on the first 128 forms. Clear reconstruction neither validates unseen glyphs nor demonstrates language understanding. The proposed continuation loop preserves the native assistant state, predicts one complete glyph and re-encodes its actual pixels. Fitting only its readout/stopping on 508 supported questions yields readable new complete strings corresponding to 288 items and the year 1874 (Supplementary Note 59). Nevertheless, only 1/16 normal Chinese development images is canonical-exact, and eleven cap; the one exact full year string occurs in training. These are partial examples, not a semantic-accuracy estimate.

A frozen intervention replaces passage dates on two templates selected from those outputs (Fig. S38). Eleven of fourteen prospectively fixed new dates have clear matching output-first transcriptions, two are canonical-exact and all fourteen stop naturally without a target length. Three damaged/wrong digit streams remain failures. Erased-question controls give no correct date. Six clear outputs compose full year strings absent from supported training answers. This supports restricted input-dependent composition through actual image feedback, while broader unseen-answer reliability remains unresolved.

Sharp known-form generation does not recover uncertain glyph identity

Additional controls distinguish a useful motor from correct conditional writing (Supplementary Note 60; Fig. S39). On sixteen normal Chinese initial states, an allowed reference first form is

nearest in semantic cosine for nine, but only one first image is pixel-exact. A spectral calculation gives a 0.060109 lower bound on uniform anchor-code MSE for a rank-2048 affine output; distance to the trained output span and bias is 0.305180. These are regression limits, not pixel-error bounds. Synthetic-state readout transfer and an added contrastive loss both fail their physical glyph requirements. Noise-trained decoding and two joint motor flows also fail fixed mixed-code tests. A 3.89M-parameter spatial flow generates the 4,087 known clean-condition forms with mean foreground F1 0.998260, but mixed-condition F1 is 0.810537. None of these rejected motors undergoes autonomous language assessment. Confident known-form generation remains distinct from recovering the intended form and producing a correct complete answer.

Improved glyph copying does not restore complete-answer generation

Training on genuine copy-prompt states separates physical writing from language decisions (Supplementary Note 61; Fig. S40). Removing source-anchor regression from the writer objective improves held-wording glyph F1 from 0.951778 to 0.972856 and the number reaching F1 0.95 from 194/256 to 229/256. Complete development-answer transfer nevertheless fails: the resulting image-feedback model gives no canonical exact answer among sixteen reused questions. On a fresh C3 cohort, neither the original nor the improved writer produces a clearly correct complete answer phrase among 32 conditions. A separate source-policy diagnostic isolates a sequence-interface problem: with identical image prefills, native source-unit feedback gives twelve literal matches among sixteen questions, whereas both exact-character and image-encoded character feedback give three. These helper trajectories use a source lexical head and are not ILM answers. The comparison motivates adapting sequence dynamics, but does not establish that frozen contextual states cannot support an independently trained writer.

Contextual adaptation improves autonomous image answers

Adapting the final four contextual layers, together with the image writer and stopping head, improves complete answers relative to a broader-data readout-only control (Supplementary Note 62; Fig. S41). Both conditions use 3,176 training questions and 18,000 updates. On 32 prospectively reserved questions from 16 contexts, conservative clear complete literal matches are 16/32 versus 11/32; both different questions are correct in 5/16 versus 2/16 contexts. Canonical pixel-exact complete answers are 2/32 and 1/32. Neither condition meets the predefined pilot requirement of 24/32 matches and 10/16 paired contexts, and both retain a failed training-pixel qualification. The result demonstrates partial image-conditioned answer transfer, not a reliable general language model. Training first-glyph fidelity is lower than later reference-conditioned fidelity, so generated-image feedback cannot explain every failure. A separate cold replay reproduces the full image-feedback execution on one fixed input without runtime lexical, font or answer-bank access.

An additional balanced writer refinement isolates a training-fit problem. With the learned reader fixed, batched first/later supervision raises training F1 from 0.963524 to 0.999811 and passes physical qualification. On a different reserved cohort, however, clear complete answers are 18/32 versus 19/32 for the unchanged parent, with two canonical exact answers each. The one-case

reading difference is a damaged initial glyph, not an established semantic regression. Neither passes the answer pilot (Supplementary Note 62; Fig. S42). Stronger reference-conditioned fitting alone has not established a generalization gain.

Historical glyph fidelity and resolution

We encoded and decoded 27 catalogued oracle, bronze and seal glyph rasters from sealed source families. The rasters had no exact match in the original historical training partition for the fixed codec. Twenty round trips were pixel-exact, and mean foreground F_1 was 0.9981 (Fig. 11).

The bronze source B12643 retains only ten ink pixels at 32×32 even though its codec reconstruction is exact. Its 128×128 raster reveals thin strokes lost at the smaller size. Five tall bronze forms represented as four vertically stacked tiles averaged foreground $F_1 = 0.9389$ after decoding and reassembly. All five changed pixels. The historical reading and stage-selection tests remain unevaluated with a trained language checkpoint.

Complete-sequence adaptation exposes a compact writing failure

We next test a smaller contextual foundation on complete continuation and question–answer sequences. The deployed contextual layers contain 1,409,410,048 externally pretrained parameters from a Qwen3 1.7B archive[18], with vocabulary embeddings and the lexical output head absent. A frozen 7.42M-parameter codec maps each visible glyph to 768 continuous coordinates. A learned image interface groups up to sixteen glyphs, supplies one to three contextual updates per span, predicts the next span's length and writes its image. Subsequent context receives the emitted pixels. This experiment transfers inherited language knowledge to a visual interface; it does not establish language acquisition from images alone.

The corpus contains 13,785 training sequences: 4,762 public-domain book passages from fourteen works and 9,023 complete CMRC question–answer pairs[12]. The targets contain 2,576,239 glyph positions. Whole works and passage groups define the development partition; exact source, text and pixel overlap checks pass, while exposure during inherited pretraining is unknown. Offline teacher extraction supplies 2,380,620 causal state targets. One complete fit makes 8,172 updates over all 108 shards, adapting 12,872,213 image parameters and an 8,388,608-parameter training-only projection. The contextual body and codec remain frozen. Extraction and fitting take 1,783.41 and 1,759.83 seconds, including resource waits of 770.72 and 110.11 seconds. Peak allocated GPU memory is 13.396 and 3.930 GiB, respectively. These are preparation and adaptation costs; they exclude the inherited foundation's original training.

The fixed evaluation uses 32 questions in sixteen held-out passage groups and twelve prefixes from two held-out books. All normal inputs run at steps 0, 1,956, 4,049, 6,152 and 8,172. Endpoint erased-input controls bring the total to 372 generated pages. Every checkpoint scores 0/32 complete exact QA answers and 0/16 both-correct pairs (Fig. 12). The source-matched Qwen3 1.7B native-text reference gives 8/32 exact answers and 2/16 both-correct pairs on the same normal prompts. Its run is on CPU; the comparison does not establish a matched-hardware efficiency advantage.

One model-based reviewer inspected opaque answer images before opening the mapping to cases and checkpoints or the scoring references. The reviewer had prior dataset familiarity; no independent human adjudication was performed. Thirty pages were blank and 342 had uncertain whole-response transcriptions. Individual characters or punctuation were sometimes recognizable, but no reliable complete reading was credited. Consequently, no readable complete book continuation was established. This limits the interpretation of its coherence score. All raw streams remain in the denominators. The same first QA pair and book prefix, fixed before fitting, are shown at every checkpoint in Figs. 13–15.

The final normal cohort has seventeen predicted stops and twenty-seven budget-limited streams, compared with forty-four predicted stops initially. Blank normal responses decrease from nine to two. Neither change establishes language learning. Endpoint question-erased and passage-erased QA counts also remain zero. The registered passage-group bootstrap returns [0, 0] intervals for the constant complete-answer observations and their paired differences. These degenerate sample intervals do not establish zero population success.

A separate traced final-checkpoint process reproduces the fixed 128-glyph book stream, its PNG, stopping/cap status and all 128 emitted-pixel feedback events. Required prompt and weight opens are verified, with no forbidden training/reference paths or IP connections observed. This checks the standalone boundary for that invocation, while its answer remains a language failure.

The writer has an identifiable capacity restriction. Its final linear velocity projection has rank 384 in a 768-dimensional codec space. For a complementary projector P, $PW = 0$ gives $Pv = Pb$, so the eight-step Euler endpoint satisfies $Pz_1 = Pz_0 + Pb$. At least 384 Gaussian directions cannot be contracted or conditioned by this continuous velocity model. Terminal normalization can restrict the varying output further. The result is an exact-arithmetic capacity statement, not a hardware-exact sampler identity or proof of the sole cause of failed answers. A separately registered 1,024-channel writer comparison removes this tested restriction, as evaluated below.

Removing the writer rank restriction is insufficient

The controlled comparison changes the writer width from 384 to 1,024 while preserving the corpus, exposure, six objectives, optimizer schedule and seed. The initial encoder, length head and teacher projection match the compact control. The wider writer's output projection, centered map and three tested local velocity Jacobians have rank 768 before and after fitting. Its image parameters increase to 48,547,733, with the same 8,388,608-parameter training-only projection. One pass again completes 8,172 updates. Fitting takes 1,945.56 seconds, including 150.14 seconds of resource waiting, with 4.889 GiB peak allocated GPU memory.

All five wider-writer checkpoints still give 0/32 complete exact QA answers and 0/16 both-correct pairs (Fig. 16). Another opaque review retains all 372 outputs: twenty-four are blank and 348 have uncertain whole transcriptions. No readable complete book continuation is established. The final normal cohort contains thirteen predicted stops, thirty-one budget-limited streams and four blanks. All erased-input QA scores remain zero. The separate trained cold child reproduces its fixed pixel stream and feedback without observed forbidden accesses. Its wrapper terminated

before writing verification, for an unknown reason; the completed original child traces permit CPU-only recovery without rerunning inference. Removing the rank restriction therefore does not solve language under this recipe.

Conditional writing fails before emitted feedback

We localize the failure with frozen weights in both final checkpoints. The fixed diagnostic contains eight QA and eight book records from each of the training and development partitions. At most eight following spans per record give 201 spans, containing 122 training and 131 development glyphs. Every writer arm receives the reference next-span length. Controlled history and count supplies separate physical writing from autonomous inference.

Correct image history with eight sampling steps gives foreground F1 of 0.3801/0.3477 on training/development targets for V1 and 0.3720/0.3518 for the wider writer (Fig. 17). Direct codec reconstruction exceeds 0.9998 F1 and exactly reconstructs 117/122 and 128/131 glyphs. Increasing the writer to 64 sampling steps does not repair writing. A native-embedding bypass, which changes the conditioning distribution, also remains poor. These supplied-target F1 values measure pixel reconstruction, not semantic answer quality.

The first target spans contain twenty training and twenty-six development glyphs. Both models have zero exact matches under correct history. Their first outputs are identical to the emitted-history arm, before feedback can diverge. Generated-history drift alone cannot explain these failures. Later emitted-history arms deteriorate further. All reference lengths remain supplied, so none of these arms is a new autonomous language result.

The reader has a separate restriction. Its 384-channel terminal-normalized feedback head has a 383-dimensional affine family for 2,048-dimensional native targets. Even after relaxing preceding image-network constraints, the best possible relative squared error on selected source positions is 0.5705/0.5768 for V1 and 0.5685/0.5754 for the wider fit, compared with achieved errors around 0.75 (Fig. 18). These are target-alignment bounds under the fixed trained heads, not language-quality bounds. No complete prompt grouping exactly matches the reference in either model, despite high local count accuracy and contextual cosines. Those representation diagnostics do not establish comprehension.

An initial prefix-consistency interruption led to a separate one-case precision probe. Full versus truncated BF16 evaluation gives relative RMS error 0.05029; FP32 arithmetic on the same rounded weights reduces it to 2.59×10^{-6}. Same-shape future erasure is bit-equal. The completed diagnostic uses matching cached execution and retains the original consistency tolerance. It takes 260.16 seconds with 3.401 GiB peak allocation and no optimizer updates. The stopped attempt, probe and CPU preparations bring these registered stages to 467.24 seconds. The evidence rejects an unchanged compact fit or a sampler-only repair, without identifying a single cause of all failed answers.

Input presentation changes retained answers but fails retention

To test whether the compact failures discard previously demonstrated image answers, we

evaluate the unchanged earlier contextual-tail actor on the same cohort. This actor retains 7,522,463,984 inherited visual/contextual parameters and uses a deterministic complete-glyph writer with actual emitted-pixel feedback. It gives 2/32 strict complete exact QA answers, both from one passage pair, and 3/32 readable QA outputs. Neither compact final checkpoint retains the two exact answers. The third readable output omits a canonical parenthetical alias and fails unchanged strict matching; this does not establish semantic falsity. All twelve book continuations reach the cap, with no credited readable coherent complete continuation. Different bodies, writers and adaptation histories prevent a causal model-size comparison.

The frozen full-prompt comparison changes input font, spacing and wrapping together while preserving every passage and question. There are no new optimizer updates or answer-centered crops, and the output cap remains 128 glyphs. Median page height changes from 704 to 512 pixels at 1,024-pixel width. Complete exact QA rises to 6/32, with a pointwise 95% passage-group bootstrap interval of 6.25–31.25%. Whole-response readability rises from 3/32 to 11/32, and QA caps decrease from 24/32 to 13/32 (Fig. 19). However, both-correct pairs fall from 1/16 to 0/16. The exact-answer gain is 12.5 percentage points, with interval 0–25; the paired difference is minus 6.25 points, with interval minus 18.75 to zero. Both difference intervals include zero. The previously exact area answer becomes unreadable, while the location answer and shorter instrument list remain readable. The registered retention rule therefore fails.

All 120 fixed-cell and 96 proportional-input outputs are retained. Separate opaque reviews freeze their complete readings before reference scoring, with prior cohort familiarity and no independent human adjudication. Both erased QA conditions give zero exact and zero readable answers in both layouts. Readable concise answers, omitted aliases and one digit/letter discrepancy are preserved without post-hoc transcription repair or a substituted semantic metric. This repeated development cohort supplies no untouched confirmation or independent training-seed replication. The comparison tests presentation transfer of inherited capability, not language acquisition from images alone.

The fixed-cell run takes 1,589.88 seconds for 120 inputs; the proportional run takes 1,179.29 seconds for 96. Both peak at 17.786 GiB allocated memory. Their different input counts prevent a matched speed comparison. Separate fixed-book and fixed-question cold invocations reproduce signed pixels, contextual states, stopping and actual feedback, with no observed forbidden input access or IP connection. Those replay checks establish execution for the tested cases, not reliable language. Neither input presentation qualifies a usable visual language interface.

Native answer access exceeds glyph strict matching

The completed native-reader diagnosis authenticates 448 calls and reproduces all 192 adapted image-prefill states bit-for-bit from actual glyph generation. An external native lexical decoder receives those states; its tokenizer, output projection and previous-symbol embeddings never enter the ILM. Adapted native continuation gives 4/32 exact fixed-cell answers and 10/32

proportional answers, versus the glyph actor's 2/32 and 6/32 (Fig. 20a,b). Both-correct pairs are 1/16 in each adapted native image arm, versus 1/16 and 0/16 through glyph output. The proportional native-minus-glyph exact difference is 12.5 percentage points, with a pointwise 95% passage-group interval of 0–25 points. These are reused-development contrasts, not untouched confirmation.

Seven proportional cases are exact only through native continuation, three only through glyph output, three through both and nineteen through neither. Six native-only cases have uncertain whole glyph readings; the seventh omits a canonical alias. The three glyph-only cases instead have native wording or punctuation differences, without establishing native semantic error. The lost area answer remains exact through native continuation while its proportional glyph stream is unreadable. Retained native access therefore exposes a writing/continuation gap without identifying a unique mechanism.

Disabling contextual adapter outputs changes native exact scores to 2/32 fixed-cell and 13/32 proportional. Complete text controls give 11/32 adapted and 10/32 with adapter outputs disabled. Proportional-minus-fixed native gains are 18.75 points [6.25, 31.25] adapted and 34.375 points [21.875, 50] with adapter outputs disabled. Layout brittleness thus extends to native reading; mixed adapter effects do not justify globally replacing the deployed body. All eight question- or passage-erased image arms give zero exact answers. The helper produces no new ILM images or optimizer updates.

The subsequent CPU audit authenticates all 192 original QA glyph streams and their first-state bindings. Before scoring, the rasterizer reproduces every one of 4,096 primitive characters and all 745 canonical target tiles exactly. All 32 first forms have primitive coverage. Supplied-image reconstruction is 31/32 pixel-exact at those first positions, with mean foreground F1 0.999934. Original normal first images instead have mean canonical F1 0.492173 fixed-cell and 0.632918 proportional, with 0/32 and 3/32 exact bitmaps. These are physical comparisons, not character or semantic accuracy: both exact complete fixed-cell answers have nonidentical first pixels, and valid alternative wording can begin with another form.

Across full canonical answers, 743/745 positions have training-primitive coverage; two absent forms occur later, leaving thirty complete answers fully covered. Supplied reconstruction is exact at 726/745 positions, with mean F1 0.999220. The covered subset has mean F1 0.999933; the two absent forms give 0.734417. Coverage describes exposure rather than a hard decoder vocabulary. No stream stops before its first image. These controls reject missing first forms or poor direct reconstruction as a general explanation for initial damage, while retaining wording, later writing, stopping and feedback limitations. Complete glyph-language scores remain unchanged.

Direct-context motor and decoded loss fail their gates

The next writer conditions directly on the retained 4,096-dimensional state and generates the frozen decoder's 1,024-dimensional motor feature. This bypasses the old predicted semantic point while preserving reading, adapters, image encoding, decoding and stopping. All 216

initial states match their parent states bit-for-bit. Under unchanged output rules, fixed-cell exact QA remains 2/32 and both-correct pairs remain 1/16. Proportional exact answers change from 6/32 to 7/32 and pairs from 0/16 to 1/16 (Fig. 21a,b). Readable QA changes from 3 to 4 fixed and 11 to 14 proportional; natural stops change from 8 to 4 and 19 to 18, respectively. Both original fixed exact answers remain, but only five of six proportional answers remain. Two newly exact answers do not compensate for this loss under the registered retention rule. The pilot requires at least eight exact answers, two correct pairs, eleven readable answers and retention of all earlier exact cases; it fails. No normal book continuation is credited as readable, coherent and complete, and all QA erasures score zero exact.

The proportional exact difference is 3.125 percentage points, with pointwise 95% passage-group bootstrap interval $[-6.25, 12.5]$; the paired difference is 6.25 points $[0, 18.75]$. These small reused-development comparisons have no untouched confirmation or independent training-seed replication. All 54 opaque boards and twelve book prefixes are viewed before reference scoring. Of the 216 whole readings, twenty are clear and 196 uncertain. One model reviewer has prior cohort familiarity, without independent human adjudication. Strict canonical mismatch does not establish semantic falsity, and partial uncertain readings are not repaired after reference access.

A separate cached-state diagnosis selects 32 first and 32 later training positions and reuses all 64 retained development first images. First/later training median foreground F1 is 0.986408/0.974518 for the flow writer versus 0.990277/0.997523 for its parent; 32/32 and 23/32 flow tiles reach 0.95, versus 20/32 and 27/32 parent tiles. Supplied reconstruction is exact for 31/32 and 32/32. Each training pool must have median F1 at least 0.99 and at least 30/32 tiles reaching 0.95. The flow writer fails both gates (Fig. 21c,d). One later nonterminal training state also predicts stopping with probability 0.99542. The diagnosis explicitly writes one glyph regardless of that decision. It tests supplied-state writing rather than autonomous language.

Conditional target exposure differs from codec coverage: the first/later pools contain 870/2,141 unique target forms, with 2,245 combined versus 4,087 codec bitmaps. Six canonical development first positions lack first-pool targets, and one lacks a target in either pool. Exposure counts alone do not explain failures on common forms. Canonical development first-image mean F1 is 0.447771 fixed and 0.661558 proportional. Native/canonical first characters agree in 5/32 and 18/32 cases; their corresponding means are 0.770103 and 0.817164. Different wording and learned stroke variation remain possible, but visible damage also precedes generated feedback.

A paired objective adds late-time balanced decoded-image BCE while preserving initialization, data exposure, random draws, architecture and sampling. It raises first/later training medians to 0.987034/0.976561 but leaves counts at 0.95 unchanged at 32/32 and 23/32. Exact bitmaps change from 4 to 6 first and 0 to 1 later, while later mean F1 falls from 0.954535 to 0.953115. Both gates still fail. Its canonical development first-image means are 0.451322

fixed and 0.658197 proportional. These local changes do not qualify consistent writing or language. The registered prerequisite withholds its full language trial; whole-answer scores are unmeasured. Neither extending the same fit nor selecting another sampling seed follows from this result.

Realized writing gains do not establish complete-answer transfer

A paired continuation supervises the endpoint actually reached by the glyph sampler, rather than a local endpoint estimate on a training interpolant. Both arms start from the same original flow EMA and receive 2,000 additional updates with matched samples and random states. The candidate adds balanced decoded-image loss through all 32 sampling intervals; its control retains velocity loss alone. First/later training median F1 is 0.990581/0.978614 for the candidate versus 0.986350/0.977090 for the matched control (Fig. 22a). Candidate counts at F1 at least 0.95 are 31/32 and 25/32, versus 32/32 and 24/32 for the control. Each pool requires median at least 0.99 and thirty qualifying tiles. The candidate passes first only; its control fails both. Exact bitmaps are 6/32 and 2/32 versus 5/32 and 0/32. Neither complete-language trial is run under the fixed prerequisite; its whole-answer quality remains unmeasured. The full-sampler objective is insufficient for stable writing under this recipe.

An earlier deterministic balanced writer already passes its full physical qualification. Its previously reported clear-answer result uses another cohort and metric. We therefore freeze that checkpoint before evaluating it on the current unchanged inputs, with no new fit or sample selection. On the same supplied states, all 32 first and 32 later training glyphs are pixel-exact (Fig. 22b). Development first-image means are 0.479076 fixed and 0.658382 proportional, with 0/32 and 4/32 exact canonical tiles. Canonical disagreement can include different wording or stroke variation, so these first-image measurements do not score answer meaning. Sampled nonterminal states no longer cross its stopping threshold, including the previous false stop, but this does not test ending on generated histories.

The complete 216-stream balanced comparison gives 2/32 exact fixed-layout answers, 1/16 both-correct pairs and 3/32 readable answers. Proportional inputs give 6/32 exact, 0/16 pairs and 16/32 readable (Fig. 22c,d). These compare with 2/32, 1/16 and 3/32 fixed, and 6/32, 0/16 and 11/32 proportional for the unchanged parent. Against the original flow's seven proportional exact answers and one correct pair, greater readability does not improve complete-answer accuracy. All 216 initial reader states match the parent bit-for-bit. Readout and stopping change together; the result does not isolate a writer-only effect.

All parent exact cases remain, but only 5/7 flow proportional cases and 6/8 union cases are retained. Two flow successes are lost while a parent success missing from the flow is recovered. The prospectively fixed pilot requires at least ten proportional exact answers, two both-correct pairs, sixteen readable answers and complete union/fixed retention. It fails. The proportional exact difference from the original flow is minus 3.125 percentage points, with pointwise 95% passage-group bootstrap interval $[-12.5, 6.25]$; the paired difference is minus 6.25 points, interval $[-18.75, 0]$. Exact outcomes are identical to the parent on this sample; a

degenerate zero bootstrap difference does not establish population equivalence.

Fixed-layout normal QA has 6 stops and 26 caps among 32 inputs. Proportional QA has 18 stops and 14 caps. None of twelve normal book continuations is credited as readable, coherent and complete; four stop and eight cap. All twelve erased book prefixes stop after unreadable output. Every QA-erasure arm scores zero exact; one proportional response with the question erased is readable, while the other arms have no readable response. All 54 opaque boards and twelve original book prefixes are inspected before new mapping/reference scoring. The 216 frozen readings contain twenty clear and 196 uncertain transcriptions, including uncertain readable fragments followed by damaged continuation. Pixel repetition for enlarged inspection does not repair outputs. One model reviewer has prior cohort familiarity, without independent human adjudication or untouched confirmation. Strict canonical mismatch alone does not establish semantic falsity.

Correct image history improves conditional writing but leaves failures

The frozen balanced writer permits a direct history intervention on the same 64 normal QA inputs, with 32 questions per layout. At each reference boundary, the model predicts the next glyph after receiving the correct preceding glyph images. The reference length is supplied and predicted stopping is recorded but bypassed. This diagnosis therefore supplies answer information; it is not autonomous language generation. All 64 initial states match the saved autonomous states bit-for-bit, and the same frozen bridge reproduces every saved autonomous pixel and stopping probability.

Across the 713 later reference positions per layout, fixed-cell writing has median foreground F1 0.573840, with 139/713 positions reaching 0.95 and 59/713 exact canonical bitmaps. Proportional inputs give median 0.985185, 475/713 at least 0.95 and 251/713 exact bitmaps. Neither layout passes both prospective requirements: median at least 0.99 and at least 90% of later positions at F1 0.95. First-position mean F1 remains 0.479076 fixed and 0.658382 proportional, with 0/32 and 4/32 exact first images. Thus conditional errors persist before any generated feedback as well as under supplied correct history.

At indices present in both histories, fixed-cell mean F1 is 0.393464 with autonomous history and 0.614716 with reference history over 738 positions. Proportional means are 0.445851 and 0.912711 over 598 positions. Seven fixed and 147 proportional target positions missing from autonomous outputs remain missing; extra autonomous output is also retained. These shared-index comparisons do not align alternative semantic answers or estimate a language-accuracy gain. Whole-response readings and all earlier language scores remain unchanged.

Relative to the supplied boundaries, fixed-cell stopping crosses 0.5 at 27/745 nonterminal positions in eleven cases and recognizes 24/32 terminals. Proportional stopping crosses at 14/745 positions in nine cases and recognizes 30/32 terminals. The proportional position-rate and terminal-rate diagnostic passes, but a supplied-history terminal is not a test of natural autonomous ending. Visual review covers all sixteen boards and 8,561 original displayed tiles. It shows clearer later proportional writing while retaining damaged forms and uncertainty. No

reference repairs a transcription.

The descriptive common-prefix analysis verifies identical states while image history remains identical. A first differing image precedes the first changed state in each of 63 affected traces. This establishes dependence on visible history, not that each divergence causes failure. All four cases with nearly exact but nonidentical first pixels have readable original autonomous answers. A universal account in which small pixel differences necessarily cause collapse is unsupported. The supplied-history benefit does not identify a unique repair, and a feedback-only explanation is insufficient.

Adapting continuation preserves frozen prompt memory in a real preflight

A successor separates source prompt interpretation from answer computation. The imported reader prefills the prompt without the new residuals. Its detached keys and values remain immutable; a learned continuous query and re-encoded glyph images drive all 36 native answer layers, each followed by a trainable rank-eight residual. Every answer layer is recomputed during fitting, rather than reusing lower-layer answer states computed under older parameters. The lexical embedding table and output projection remain absent from inference.

A real single-4090 preflight retains four authentic complete targets of 1, 74, 184 and 513 glyphs, with prompts of 328, 1,027, 695 and 128 glyphs, respectively. All 772 target positions and four terminal boundaries receive supervision. All 36 answer layers receive actual gradients. After four updates, recomputed prompt states and every prompt key/value fingerprint remain bit-exact. Changing a final target leaves all preceding predictor states unchanged; changing the first target leaves its initial predictor unchanged while later states respond. Zero-residual split/native execution differs by at most 0.000223160; updated serial/parallel execution differs by 0.000070572, both within the fixed 0.002 tolerance. Different execution shapes are not asserted to be bit-exact.

The independently opened checkpoint contains 85 Adam states at step four, with 20,212,225 trainable parameters. Peak allocated memory is 18.122546 GiB. Prompt freezing and full-body version checks are assertions from the bound GPU execution; full source tensors are not duplicated for a second comparison. Initial residual-down tensors were not archived in this preflight, so their updates are evidenced by gradients, moments and the execution checks. This is a feasibility and implementation result. A one-glyph training-case inference exercise supplies no held-out language evidence.

The registered full fit discards all preflight updates and retains 13,766 complete sequences: 9,004 QA records and 4,762 book records. Its three epochs prescribe 41,298 updates and 6,108,741 target positions across two layouts, with five fixed snapshots and 732 evaluation streams. A logging failure in the first execution discards one unsaved update; the corrected execution starts from bit-exact fresh parameters, optimizer and random state. Actual step-four restoration and step-five/six replay match the uninterrupted state exactly. These launch checks establish reproducible execution, independently of language quality. The complete fit and all 732 prescribed streams have since finished. The final proportional model produces 7/32 strict

Table 1 | Complete autonomous QA across the fixed continuation checkpoints. Each entry gives exact/readable/both-correct counts. Each layout has 32 questions and sixteen paired-question groups. All values are measured; the intermediate peak does not replace the prescribed final checkpoint.

Update	Fixed-cell prompts	Proportional prompts
0	0/0/0	4/5/0
10,324	1/1/0	4/8/0
20,649	2/3/0	6/13/0
30,973	2/4/1	9/16/1
41,298	2/4/1	7/14/0

complete answers, 14/32 readable answers and 0/16 both-correct pairs; no normal book arm receives a coherent complete continuation. Both registered pilot audits reject it. Figure 24 reports all fixed checkpoints and controls. The subsequent joint writer's autonomous language remains unmeasured; its completed physical comparison is reported below.

Complete continuation adaptation does not establish sustained language

The full continuation model completes three epochs over 13,766 complete sequences, comprising 9,004 QA and 4,762 book records. The 41,298 updates cover 6,108,741 scheduled target glyph positions. Evaluation retains all five fixed checkpoints and 732 autonomous streams, including erased prompts and matched 640-glyph book controls. All 183 output boards and 96 visible normal-book prefixes are reviewed. Readings are frozen before scoring against the new case mapping and canonical references. Review uses one model reviewer with prior cohort familiarity, without independent human adjudication.

Some short answers improve relative to the actual initial continuation checkpoint, but the final model fails the registered complete-language pilot (Table 1). Both fixed successes and six of eight required proportional successes remain. Final question-erased and passage-erased exact counts are zero and one, respectively, in each layout. The proportional normal-minus-better-erased gap is six, below the required eight. The original scorer and a separate audit of all four QA-erasure arms both reject the pilot. Canonical mismatch is not, by itself, semantic falsity; uncertain and unreadable outputs remain uncredited rather than repaired.

No normal book arm at any checkpoint receives a readable coherent complete continuation. All twelve final 128-cap books reach the cap. At a 640-glyph cap, the final candidate stops on eight of twelve normal prefixes and the matched balanced parent stops on seven; neither produces a credited coherent complete continuation. Recognizing a prepared record's end is also distinct from a natural whole-document ending. Learned stopping without readable language does not qualify the model.

The final system retains 7,578,390,003 deployed parameters, including a 7,522,463,984-parameter inherited body. Its 20,212,225 trainable parameters therefore do not define a compact foundation. Cumulative training-child execution is 75,209.326 s, including the preserved failed first attempt; peak allocated memory is 18.568 GiB. Candidate and matched-parent evaluations cost 6,338.799 and 609.626 cumulative child seconds, respectively. Preparation,

admission queues, CPU review and inherited training remain separate. These measurements establish bounded single-device execution, not greater efficiency than a conventional language model at matched quality.

Saved streams develop dense writing while prompt states still differ

A separately registered retrospective analysis authenticates all 732 saved streams and all 2,036,247 target positions in the 13,766 eligible training records. It makes no model call and changes no language score. Clean QA and book glyphs have median ink counts of 188 and 201 of 1,024 pixels; their 99th percentiles are 267 and 276. The context-independent optimum of the actual sequence-balanced writing loss has 504 black pixels after thresholding. This is a restricted baseline, not an optimum over conditional models.

For the final normal 640-cap books, 98.12% of emitted tiles exceed the clean book 99th-percentile ink count, compared with 5.94% initially. Equal-stream early/tail ink fractions are 0.4375/0.4718, and foreground F1 against the constant-image template is 0.8990/0.9386. These disjoint eight-tile windows use streams of at least 32 tiles: twelve final streams and four initial streams. The different subsets prevent treating those window averages as a matched longitudinal effect. All twelve final early/tail rows are visually inspected; dense blocks appear at the beginning and persist later.

All 416 paired-question and normal/erased first-state contrasts are unequal. All 413 contrasts in which both streams emit also have unequal first images. At the final proportional checkpoint, paired-question relative state L2 and first-image Hamming distance average 1.0428 and 0.2302. The measured failure therefore coexists with representation changes across prompts. These distances do not establish question understanding, identify the loss as the unique cause, or prove a stable dynamical attractor.

Correct history separates QA continuation from book prediction

The new initial/final control retains 248 complete supplied-history traces and 25,194 target positions. Long parallel/incremental comparisons pass the fixed 0.002 state tolerance at both checkpoints. At the final checkpoint, proportional development QA has mean first-image F1 0.6942. Across its 713 later positions, mean F1 is 0.9827 and 656/713 positions reach 0.95; 512 are exact bitmaps. The twelve sampled training QA records give later mean F1 0.9905 and 358/375 positions reaching 0.95. These supplied-history fractions pass the descriptive 90% later-writing condition. They supply answer information and do not replace the failed autonomous complete-answer results.

Book prediction remains inaccurate even with reference history. The twelve sampled training books have later mean F1 0.6025 with fixed prompts and 0.6022 with proportional prompts; only 9.03% and 9.01% of later positions reach 0.95. The twelve development books give later mean F1 0.5587, with 4.61% reaching 0.95. Respectively 75.32%, 75.34% and 81.91% of all predicted tiles exceed the source-kind clean-ink threshold. Original reference, supplied-history and autonomous early/tail images for all twelve development books are inspected

together, without changing their frozen language grades. These results reject a feedback-only explanation of this final model's book failure, while exposing a stronger supplied-history QA capability than its autonomous writing. They do not identify a unique replacement architecture.

A joint physical writer improves frozen credits but does not qualify language

Replacing independent pixel marginals with an autoregressive physical-image density increases the original frozen review credits from 599/1,040 for the point control to 970/1,040 for the joint writer (Fig. 25). The prespecified prerequisite nevertheless fails: development books receive 91/108 credits (84.26%) and training QA with fixed cells 90/101 (89.11%), both below the unchanged 90% requirement for every group. The remaining five groups meet that numerical threshold. The trial therefore stops before autonomous evaluation; no complete-language score is assigned.

All 41,298 retained updates cover three complete epochs of 13,766 sources and 6,108,741 scheduled target glyph positions. The independent CPU audit verifies the complete source schedule, inverse-packed target images, training-only normalization, all five fixed snapshots and actual Adam counters. A real step-three restore and replay reproduces the optimizer, model and random states; two additional executed updates do not add retained exposure. The new image head has 2,642,368 parameters. Prompt reading, continuation dynamics, feedback and stop prediction remain fixed and retain more than 7.522 billion inherited visual and contextual parameters. The small head does not establish a compact foundation model.

The original physical review covers 33 opaque boards and 2,080 selected images, with readings frozen before the owner mapping was opened. A subsequent binary pixel audit exposes a material review limitation: 17 blank flags disagree with the saved images, including four empty images that received credit (three candidate, one control). Thirteen nonempty images were marked blank. Every reviewed PNG matches its authenticated array. Original readings and scores remain frozen and are reported with these errors; they are not corrected physical accuracy. In the development-book group, fifteen of the seventeen uncredited outputs are empty and two are nonempty; two other empty outputs were credited (Fig. 25b). This audit checks ink presence without supplying new linguistic transcriptions.

The complete probability diagnosis evaluates all 124 cases and 12,597 positions without new generation or optimization. Of 349 false blanks, only sixteen prefer the source image to the empty image by the fixed 0.001-nat margin. Among the selected positions, one of fourteen book false blanks and none of four training-fixed-QA false blanks prefer the source, below the registered minimum counts of seven and one. The first-nonzero-prefix bound supports the empty image as a global mode for fifteen of 42 development-book false blanks, subject to finite-precision arithmetic. Saved empty outputs agree with parallel zero-prefix argmax checks. These measurements reject the search-only decision rule without proving that search can never change an image.

Coordinator-measured child execution is 40,002.756 seconds for cache preparation, in-

cluding the preserved failed attempt and recovery verification; 1,982.697 seconds for fitting; 184.130 seconds for audit; and 133.027 seconds for physical qualification. The fit worker's 0.644-GiB peak allocation measures the head-only fit, not complete deployed inference. Preparation, inherited training and earlier reader adaptation remain separate costs. The subsequent context-gradient trial is separately registered and running. Its real-source computational preflight passes within 17.795 GiB, but its completed fit and autonomous language performance remain unmeasured.

Discussion

Conditional writing and complete language

The complete continuation trial does not resolve the visual language problem. Three epochs over complete targets yield some correct short answers, but the final proportional model gives 7/32 strict complete answers, no both-correct pair and no credited coherent complete book continuation. The three-quarter checkpoint's 9/32 exact answers do not replace the prescribed final model. The final model also loses two required proportional successes and fails the normal-minus-erased contrast. Both pilot audits reject it. More unchanged epochs or selecting an intermediate peak would not explain this failure.

The supplied-history controls narrow the interpretation. At the final checkpoint, 656/713 later proportional QA predictions reach foreground F1 0.95 under correct preceding images and supplied lengths. Sampled training books reach that level on only about 9% of later positions. These are physical comparisons against canonical targets, with predicted stopping bypassed; they do not qualify autonomous answers or align alternative correct wording. They also make a feedback-only explanation insufficient: book writing remains inaccurate even when generated history is removed. Reading access, the relation between contextual states and writing, uncertainty over plausible continuations and stopping can still contribute to failure.

The saved trajectories reveal dense writing rather than a loss of every prompt-dependent state difference. Final normal long-book streams place 98.12% of emitted tiles above the clean training-book 99th-percentile ink count. All available first-state contrasts remain unequal. The restricted context-independent optimum of the balanced writing loss produces a dense thresholded template, but this is not the optimum of the conditional neural model and does not prove an attractor or unique cause. A state difference likewise does not establish question understanding. Complete outputs and controlled prompt interventions remain the decisive language evidence.

The subsequent head-only comparison replaces thresholded marginal pixels with a normalized distribution over each complete glyph, preserving the final reader, continuation dynamics, feedback encoder and stop head. Its 128 physical eight-pixel masks enumerate local patterns rather than character identities; all 2^{1024} binary rasters remain representable. Original frozen physical credits rise from 599/1,040 to 970/1,040, but two candidate groups fail the unchanged prerequisite. The later pixel audit finds seventeen blank-flag disagreements, including four

credited empty images. Original scores remain frozen with that limitation, not interpreted as corrected physical accuracy. No autonomous language result is assigned to this failed actor.

The exact probability diagnosis further narrows the intervention. Only sixteen of 349 false blanks prefer their source over the empty image. The selected book/training-QA search-only criterion fails, and a first-nonzero-prefix bound supports an empty mode for fifteen of 42 development-book false blanks, subject to finite-precision arithmetic. These measurements do not prove that every search alternative is bad, but they do not justify search alone as the next repair. The new controlled fit instead lets the query and continuation residuals learn through the same complete-image likelihood, with the balanced stop loss retained. Exact microbatch state gradients match ordinary autograd in the real-source preflight, including two complete 513-glyph book cases within 17.795 GiB. All preflight updates are reset. The current construction has 5,005,760 trainables within 7,553,448,370 total parameters; its completed fit, full-body autonomous evaluation and language remain unmeasured. A conditional-density factorization and a computational gradient check establish neither a lightweight foundation nor semantic understanding.

Resource claims concern the entire deployed system. The evaluated full point-writer model retains 7,578,390,003 parameters, including its inherited 7,522,463,984-parameter visual/contextual body. Its bounded one-4090 execution does not show that this body can be learned on that device or that it is more efficient than a text model at matched language quality. Data preparation, inherited pretraining, failed attempts, training, generation and review remain separate costs. The small compact-model trials also fail their language tests; low memory use alone does not satisfy the objective.

The completed continuation comparison reuses a development cohort, with one model reviewer familiar with that cohort and no independent human adjudication. Strict canonical mismatch is not necessarily semantic error, and uncertain readings remain uncertain. The current font and cell resolution do not establish handwriting transfer, raw-page reading, arbitrary historical-form generation, general instruction following or etymological accuracy. Any development gain must survive untouched works and passage groups, natural stopping, independent execution and whole-model accounting. Extending verified language to historical and unencoded writing remains the intended application, not a consequence of pixel support or supplied reconstruction alone.

Earlier interfaces and conditional-writing controls

The results separate a useful visual predictive state from a useful visual language generator. The established causal reader learns context-dependent next-writing information, while subsequent writers can erase that information or model the marginal appearance of glyphs. A complete visual language model must bind meaning and sequence to the exact marks it generates.

The compact complete-sequence fit adds a larger authentic target stream while keeping the contextual foundation fixed. It completes training within one 4090 but does not produce a credited readable complete answer on its fixed development cohort. Its independent cold replay

verifies execution boundaries, not comprehension. The velocity projection has a provable noise subspace; terminal normalization also matters when repairing its width. The completed wider-writer comparison removes the tested restriction but retains zero complete exact QA answers at every checkpoint. Correct-history diagnostics show writing failure before generated feedback, while a separate reader restriction limits native-target alignment. These results reject the unchanged joint recipe and a sampler-only repair. The next model decision requires a matched-cohort readable control, followed by fresh matched earlier-actor control retains two exact answers and one fully correct question pair on the same cohort. Full proportional prompts raise exact answers to six but lose that pair and an earlier readable answer, failing the registered retention rule. This establishes presentation sensitivity and limited inherited transfer, without establishing a consistently better interface or new learning. The matched native continuation gives ten exact proportional answers from the same adapted first states, while its lexical decoder remains outside the ILM. Native reading also depends strongly on layout. All first target forms have primitive coverage and supplied reconstruction is nearly exact, yet some original first forms are already damaged. Canonical pixel mismatch can also reflect readable strokes or different wording. Neither missing first forms nor later feedback alone explains every failed answer. A direct-context conditional motor gives seven proportional exact answers but retains only five of the earlier six, failing its registered pilot. Its cached-state writing errors persist on training positions. The paired late-time decoded loss also fails both physical gates. Its local endpoint estimator is trained on noise/target interpolants; this does not directly supervise the endpoint reached by its own numerical rollout. Differentiating through that full rollout improves first-position writing but still fails the later-position gate. A separately qualified deterministic writer reproduces all 64 selected training glyphs exactly, yet retains six proportional exact answers and no both-correct pair. All its initial reader states match the parent. This training-state pass does not establish transfer to autonomous language, and the complete comparison fails its pilot. Reader access, conditional writing, stopping and generated-history drift remain distinct possible restrictions; these controls do not identify a unique cause. Further model changes require tests that separate them, followed by untouched source groups and independent review after any development improvement.

An image interface enlarges the representable output space, including the possibility of forms absent from a character encoding. It does not remove the need to learn syntax, knowledge or long-range dependencies. This study has not demonstrated general instruction following, etymology answers, handwriting transfer or arbitrary historical-script generation. It also lacks independent training-seed replication and a capability-matched text-model comparison. Small allocated memory in a bounded experiment establishes feasibility on one GPU, rather than a general efficiency advantage over language models.

Supplied-target reconstruction is preserved by an aligned inverse-image path; the free image writers still fail reliable new prompt-directed copying. The separate source-pixel component passes its new-copy test but fails natural-reading qualification. The remaining test is whether

context selects and generates a coherent written future. Low teacher-forced image NLL can reflect dependencies among strokes already revealed within the target, while generated output still loses global identity. Joint training improves proper likelihood beyond matched extra density training; the raw generated continuations still fail the intended language test. Conditional continuation and short answers require counterfactual prompts and the same context effect during autonomous sampling and visible feedback. Broader books and offline teacher-generated exercises can supply later training data while leaving the final inference path independent. A larger corpus or longer training requires improvement on the intended generation task, beyond the conditional-likelihood gain established here.

Broader pixel training reproduces 495 of 512 answers but still fails the supplied unseen reference images. The matched native reader retains useful answer capability, directing the intervention toward the learned writing interface. The complete-glyph loop supports restricted input-dependent date composition, with actual-image feedback independently replayed. It remains unreliable on broader questions: eleven of sixteen normal Chinese streams cap. Wider semantic readout coverage and robust generated histories remain necessary; clear familiar glyphs and low training loss are insufficient.

Recent visual generators study mixture assignment and the geometry of distribution matching[13,14]. These results motivate more expressive writing distributions, but marginal image fidelity cannot certify an answer. A generator that swaps the correct images for two equally frequent questions preserves the answer-image distribution while answering both incorrectly. Conditional supervision and prompt interventions therefore remain necessary; natural-image gains do not establish a shortcut to language learning.

An inherited visual/contextual successor provides a separately measured execution route. With its vocabulary module removed, the complete 7.548B actor trains 25.58M adapter/writer parameters within 17.181 GiB allocated memory. Causal, backward and resume checks pass after explicit precision repairs. Eight short source-engine sanity cases support the contextual initialization; no image-language ability follows from that check. Its continuous writer fails its reconstruction test (Supplementary Note 23). The separate qualified codec preserves supplied writing, while the completed predicted-code actor fails its autonomous language test (Supplementary Notes 25–28). Poor first-block and exact-history predictions, including on training cases, show that generated-history drift cannot be the sole explanation. Conditional feature information, readout capacity, the objective and optimization remain possible limitations. The current evidence does not select a successful density, optimizer or semantic-transfer method. External pretraining remains part of the model's provenance.

Methods

Compact continuous glyph feedback

The compact experiment receives ordered 32×32 binary ink tiles, with one fixed Noto Sans CJK face during data preparation. The fixed codec maps each tile I_j to $z_j \in \mathbb{R}^{768}$. A 384-channel

image encoder uses learned boundaries to group at most sixteen glyphs. It predicts one to three continuous updates $u_{i,r} \in \mathbb{R}^{2048}$ per span. The frozen 28-layer contextual body consumes those vectors directly and supplies the state c_i after each span. During training, native source alignment supplies span ends and update counts; at inference both are derived from visible pixels.

The writer predicts length $K \in \{0,\dots,16\}$ from c_i. Length zero ends generation. For a nonempty span, a conditional rectified flow starts from $z_0 \sim \mathcal{N}(0,I)$ and applies eight Euler steps,

$$z_{k+1} = z_k + \tfrac{1}{8} v_\theta(z_k, k/8, c_i), \qquad k = 0,\dots,7. \tag{5}$$

The codec decodes the endpoint into binary ink. The same image encoder reads these actual emitted pixels to update contextual history. No font renderer, OCR service, lexical head or answer lookup supplies generated content. The 128-glyph budget can stop before a full predicted span would exceed the cap; that stream is recorded as budget-limited rather than naturally stopped. Raw page segmentation and arbitrary reading order are outside this tile-stream experiment.

Six complete-sequence objectives

The reference-image training history differs from generated-image history at inference. Each complete sequence supplies all continuation/answer spans and its terminal decision. For predicted feedback vector $\hat{u}$ and compact source embedding e, the relative regression objective is

$$\mathscr{L}_{\text{fb}} = \mathbb{E}\frac{\|\hat{u}-e\|^2}{\max(\|e\|^2, 10^{-8})}. \tag{6}$$

Training-only source embeddings and native span alignment come from the authenticated source archive; they do not enter deployed inference. The encoder also minimizes update-count cross entropy $\mathscr{L}_{\text{count}}$ and binary span-boundary cross entropy $\mathscr{L}_{\text{boundary}}$. The writer minimizes span-length cross entropy $\mathscr{L}_{\text{length}}$, including the zero-length terminal. For clean codec span y, fresh noise ε and $t \sim U(0,1)$, its velocity objective is

$$z_t = (1-t)\varepsilon + ty, \tag{7}$$

$$\mathscr{L}_{\text{flow}} = \mathbb{E}\|v_\theta(z_t,t,c) - (y-\varepsilon)\|^2/d, \qquad d = 768. \tag{8}$$

Only valid glyph positions contribute. An offline Qwen3-VL teacher[17] supplies causal context states $h^T \in \mathbb{R}^{4096}$. A training-only linear projection $A : \mathbb{R}^{2048} \to \mathbb{R}^{4096}$ gives

$$\mathscr{L}_{\text{teacher}} = \mathbb{E}[1 - \cos(Ac, h^T)]. \tag{9}$$

The implemented objective is the equal-weight sum

$$\begin{aligned}\mathscr{L} = {} & \mathscr{L}_{\text{fb}} + \mathscr{L}_{\text{count}} + \mathscr{L}_{\text{boundary}} + \mathscr{L}_{\text{length}} \\ & + \mathscr{L}_{\text{flow}} + \mathscr{L}_{\text{teacher}}.\end{aligned} \tag{10}$$

The corpus has 7,255,223 training glyph positions including repeated QA passages, and 2,576,239 continuation/answer target glyph positions. Its 13,785 sequences cover fourteen training works and 9,023 QA pairs. Two whole works and separate CMRC passage groups form the development partition. Exact source, text and input-pixel overlap checks pass. Pretraining exposure of either inherited foundation remains unknown.

The offline teacher body supplies 2,380,620 state targets in 108 authenticated shards. The fit uses seed 202611152, AdamW at learning rate 2×10^{-4}, weight decay 0.01, gradient norm capped at 1.0 and a 1,024-position padded batch budget. Trainable parameters remain FP32. Frozen contextual matrices are dequantized from the authenticated GGUF archive into BF16, and neural computation uses BF16 autocast. All 8,172 updates complete one pass. Five snapshots preserve initial, quarter, half, three-quarter and final shard exposure, recording their actual optimizer steps. Training resumes only with the original source, corpus, teacher-cache and configuration identities.

Compact evaluation, review and provenance

The prospective development cohort contains 32 QA prompts from sixteen passage groups and twelve prefixes from two whole-held-out works. The five snapshots use seed 202611153, eight flow steps, the same normal prompts and a 128-glyph budget. The initial and final snapshots additionally run question-erased, passage-erased and book-prefix-erased controls, giving 372 pages. One model reviewer transcribes opaque original images before their mapping and scoring references are opened. Prior dataset familiarity is disclosed. Uncertain whole transcriptions, blank responses and caps stay in the original complete answer denominators. This is not independent human adjudication.

Exact QA success requires a reliable nonempty transcription matching the canonical answer and a learned stop. Both-correct pairs require both answers and distinct output images. Book coherence is judged against the visible prefix; an uncertain whole transcription cannot establish readable coherent continuation. Pointwise 95% intervals use 10,000 whole-passage-group bootstrap replicates, seed 202611154, retaining both questions per group. The same groups are resampled for final-minus-initial and normal-minus-erased differences. With constant zero observations the interval degenerates; it cannot establish zero population success. Two book works do not support a book-level interval. This small development cohort also requires fresh confirmation after any model-selection improvement.

The unchanged native-text reference uses the same 44 normal prompts with Qwen3 1.7B on CPU; its book coherence remains unrated. A traced fresh-process replay separately checks the final image actor's fixed book output, exact display pixels, termination and emitted-pixel feedback against its suite run. The child receives only the visible input, authenticated foundation/codec and deployable image heads. Access auditing distinguishes an unused vocabulary inside source-archive metadata from materialized lexical model modules or answer access. Preparation, fit and inference costs are recorded separately.

Wider writer and frozen failure localization

The controlled wider writer uses 1,024 hidden channels for a 768-dimensional velocity. All other registered training choices remain fixed, including seed, initial encoder, length head and teacher projection. CPU-only construction preserves the GPU random state. Initial and final output, centered-output and local Jacobian ranks are checked separately from language quality. The same five-checkpoint protocol, 372 opaque images and complete-answer definitions apply. The cold child's original traces are retained, including its parent-wrapper interruption and CPU-only verification recovery. No inference rerun supplies that receipt.

For failure localization, source groups are ordered by a fixed hash, taking eight QA and eight book records per partition. The two development works contribute four records each. Both final checkpoints remain frozen. The six writer arms use correct image history at eight or 64 flow steps, native source-embedding history, an incorrect reference context, emitted history with supplied update counts, or learned prompt grouping with emitted history. All receive reference output lengths and identical per-span initial noise. The direct codec control reconstructs the targets without the writer. Native history is an explicit distribution-changing bypass. Gold history, span lengths and counts are diagnostic supplies, not available autonomous answer information. First-span measurements precede generated feedback.

Pixel metrics are independently recomputed from persisted binary targets and outputs, including blank-target and nonblank-target exact counts. Original PNGs, noise hashes and feedback hashes are authenticated. Latent errors are not independently recomputed because separate target latents were not saved. The first registered case of each split/kind supplies the raw displays; that deterministic display rule was introduced during analysis, not preregistered. These selected diagnostics have no semantic grading or generalization interval.

For the reader, write D for the fixed output scaling and μ for its mean. With the final LayerNorm parameters γ, β, projection W, b and $C = I - \mathbf{1}\mathbf{1}^\top/384$, every feedback vector lies in

$$a + \mathrm{col}(A), \qquad A = DW\,\mathrm{diag}(\gamma)C, \tag{11}$$

$$a = \mu + D(W\beta + b). \tag{12}$$

Both final heads have $\mathrm{rank}(A) = 383$. If Q is an orthonormal basis of its column space, relaxing all preceding network constraints gives the target error floor

$$\ell_{\min}(e) = \frac{\|(I - QQ^\top)(e-a)\|^2}{\max(\|e\|^2, 10^{-8})}. \tag{13}$$

The reported mean weights selected source positions uniformly. It is not the original objective's batch weighting or a lower bound on language error.

The first diagnostic stops at its original prefix-consistency tolerance. A separate one-prefix probe compares full and truncated sequence execution, same-shape future erasure, cache/mask controls and FP32 arithmetic using the same BF16-rounded weights. Its twelve contextual calls and

twenty-four velocity calls are recorded separately. The completed diagnostic uses matching prefix prefill and cached updates for reference and generated history, retaining the original tolerance. The stopped attempt, probe, completed run and CPU preparation costs remain in the evidence. No optimizer update or new autonomous language result is attributed to this diagnostic.

Matched earlier actor and full-prompt presentation

The earlier actor uses a separate inherited 36-layer visual/contextual body, four adapted tail layers and a complete-glyph CNN encoder/decoder. Its previously fitted checkpoint remains fixed throughout these comparisons. All 32 compact-cohort QA questions, both erased-input controls and twelve book prefixes with their erased controls give 120 fixed-cell inputs. All 32 QA questions and their 64 erased controls give the 96 proportional inputs. No earlier successful pair, short reference or favorable font-coverage subset is selected. Recorded adaptation data contain 3,176 admitted questions, with no current question, normalized passage or exact training-PNG overlap. Inherited pretraining and all earlier ancestors remain uncertified.

CPU preparation authenticates each original complete source prompt. The proportional view reuses the exact earlier Pango renderer and Noto CJK face, 24-pixel font size and 32-pixel strip height. Each source newline becomes one ordinary space; all non-whitespace codepoints remain unchanged. Capacity is measured from the full prompt without shrinking or truncation. Blank-column wrapping preserves and exactly recovers the complete rendered strip. Passage, question and structural ink masks are isolated without changing shaping, must be disjoint and must reconstruct every normal ink pixel. Erased views whiten only the original field pixels in place, retaining the normal wrap ranges, canvas and all other ink. All 96 images pass exact strip/page roundtrips and the unchanged host preparation. Preparation takes 4.16 seconds.

The native visual observer, numeric task frame, deterministic contextual numerics, learned stop threshold and 128-glyph cap remain unchanged. The deterministic writer produces binary glyph images and rereads their actual pixels. Each complete output retains its original signed tensors, displayed PNG, states, feedback hashes and cap/stop decision. The deployed answer path has no lexical output module, tokenizer, font renderer, reference answer or external teacher. Zero new optimizer updates occur in either comparison.

One model reviewer inspects all opaque outputs and ambiguous enlargements before the corresponding mapping or references are opened. Proportional outputs use presentation-specific opaque identifiers, preventing reuse of the earlier display mapping. Frozen readings give three complete readable fixed-cell outputs and eleven proportional outputs; the remaining 117 and 85 whole responses are uncertain, with no blank outputs. Reviewers have prior dataset and earlier-output familiarity. No independent human adjudication is performed. The original complete canonical exact-match, learned-stop and distinct-pair requirements remain fixed. Noncanonical wording is not relabeled as factually false, and no favorable semantic score is added after inspecting answers.

Pointwise intervals use 10,000 whole-passage-group paired bootstrap draws, seed 202611154. Constant zero groups give degenerate sample intervals, without proving zero

population success. The registered development rule requires gains in exact answers, correct pairs and readability while retaining the three original readable cases and not increasing QA caps. Its paired gain and exact-answer retention conditions fail. An additional conservative core-prefix retention check was implemented during generation before new output inspection; it is not described as preregistered and is unnecessary to reject the original rule. Figure 19's three raw example pairs were chosen after scoring to show the lost area, retained location and first newly exact quantity answer. They do not substitute for all registered denominators.

The fixed-cell and proportional runs include 80.20 and 97.89 seconds loading, with 1,484.92 and 1,060.67 seconds of generation, respectively. Both record zero resource-pressure wait. Earlier adaptation, inherited pretraining and corpus preparation remain excluded costs. The fixed-book cold replay takes 84.89 seconds at 17.757 GiB, and the proportional fixed-question replay takes 79.08 seconds at 17.693 GiB. Each independent traced invocation checks actual signed pixels, states, stopping and feedback. The unchanged command-line viewer lays out a long row rather than the suite grid; exact tensors and display pixels, rather than PNG layout bytes, verify the proportional replay. No observed forbidden corpus, reference or font access or IP connection occurs. These are fixed-case boundary checks, separate from language accuracy.

Matched native continuation and first-writing audit

The native diagnostic keeps all 32 questions, both full image presentations, normal inputs and both erased controls. Each image case runs with the learned contextual adapters and with their output matrices set to zero, giving 384 image calls. Complete text prompts give 32 calls per adapter condition, for 448 calls total. One shared body executes the conditions serially. All 192 adapted image-prefill vectors match the actual glyph actor's saved first states exactly. The source-adapter control changes that contextual condition; it does not authorize replacing the deployed image-output actor.

The native tokenizer, vocabulary projection and previous-symbol embedding lookup remain helper-only variables. The BF16 source projection contracts with each FP32 contextual vector in ascending 4,096-row chunks using FP32 arithmetic, with lower source index breaking exact ties. Greedy continuation ends at a source terminal symbol or 128 source units. This cap differs from the glyph actor's 128 emitted images, so neither output lengths nor costs form a matched-unit efficiency comparison. Raw source IDs, selected logits, decoded texts, hidden states and ending decisions are retained. CPU scoring verifies every case and original glyph-score binding. A score-receipt schema lookup is corrected without regeneration or any change in references or matching. Whitespace-only normalization, natural ending and distinct paired answers retain the original strict scoring rules. No semantic regrading is added.

Native and glyph differences use 10,000 paired whole-passage bootstrap draws with seed 202611154 and pointwise 95% intervals. The same sixteen passage groups define all comparisons. Unknown inherited pretraining exposure, development reuse and prior output familiarity remain limitations. Native execution takes 2,838.924 seconds, including 158.819 seconds

setup and 180.172 seconds pressure waiting, at 18.947 GiB peak allocated GPU memory. It makes no optimizer update or new ILM image; acquisition and earlier adaptation remain excluded costs.

The subsequent CPU preparation authenticates all 192 original QA streams, first states, stop decisions and signed-image/PNG/feedback bindings. Canonical target compatibility is checked against the original Noto CJK SC collection face 2, size 26, baseline 26 and 32-square binary rasterization. All 4,096 selected primitive characters reproduce their 4,087 unique image arrays, and all 745 source target tiles reproduce exactly before neural scoring. Character and exact-bitmap membership are counted separately over all first and later targets, without removing absent forms. The evaluator-only inventory never enters inference.

For target ink image I, the answer-supplied physical control computes

$$\widetilde{I} = \mathbf{1}\left[\sigma(D(E(1-2I))) \geq \tfrac{1}{2}\right],$$

where E is the frozen image encoder and D includes the frozen motor projection and image decoder. Batches of 32 use FP32 CPU arithmetic and two threads. Original GPU emissions are never replaced by this reconstruction. Per-image foreground F1 is $2\mathrm{TP}/(2\mathrm{TP}+\mathrm{FP}+\mathrm{FN})$; means average images rather than pooling pixels. Two empty arrays receive one; an absent emission receives zero and is distinct from a blank tile. Pixel exactness requires identical arrays. First-pixel disagreement can reflect different wording or readable learned strokes and does not imply semantic error. All original whole-answer readings and strict scores remain fixed.

An independent CPU recount verifies every counter, coverage flag and outcome subgroup from saved arrays, plus all 128 tiles in the two labelled comparison boards. Those boards are inspected with prior cohort familiarity, without independent human review. Preparation takes 6.539 seconds without a neural call; supplied reconstruction takes 1.502 seconds at 1.078 GiB peak process RSS. CUDA remains uninitialized, and there is no contextual call or new optimizer update. Figure 20's first-tile examples are retrospective selections from these completed controls and are explicitly excerpts rather than full answers.

Direct-context motor, physical gates and decoded objective

For target ink image I, the frozen encoder and motor projection provide $z_{\mathrm{raw}} \in \mathbb{R}^{64\times4\times4}$. Coordinate means μ and scales s use training targets only, giving $z = (z_{\mathrm{raw}} - \mu)/s$. A 24,477,696-parameter conditional head receives actual reader state $h \in \mathbb{R}^{4096}$, Gaussian noise e and uniform $t \in [0,1]$. The probability-path regression follows flow matching[25]:

$$x_t = (1-t)e + tz, \tag{14}$$

$$\mathscr{L}_{\mathrm{FM}} = \mathbb{E}\left[\|v_\theta(x_t,t,h) - (z-e)\|_2^2/1024\right]. \tag{15}$$

The 1,536-wide head sums projected noisy motor, layer-normalized context and a 128-to-

512-to-1,536 time branch, then applies three residual layer-normalized SiLU blocks and a full 1,024-coordinate output projection. Training-coordinate standard deviations have floor 0.05. This bypasses the earlier deterministic semantic point. A head fit reuses 3,176 first and 20,086 later authentic training states from 3,176 training questions. Each batch contains 64 first and 64 later samples. AdamW, seed 202611171, learning rate 2×10^{-4}, Adam coefficients $(0.9,0.99)$, weight decay 0.01, gradient-norm cap one, 200-update warm-up, cosine decay to 10% and EMA decay $\min(0.999,(1+k)/(10+k))$ run for 12,000 updates. Each pool supplies 768,000 exposures. Only final EMA weights are evaluated. The inherited 7.522B visual/contextual parameters, learned adapters, encoder/decoder and stop predictor remain fixed; no development or native-helper answer supplies training.

One fixed per-case SHA-derived seed supplies one sample per emitted glyph. Sampling uses 32 integration intervals with Heun updates and a final Euler step. The unchanged stop decision and 128-glyph cap supply no target length. Only actual emitted binary pixels enter subsequent history. All 120 fixed and 96 proportional streams include normal inputs and their existing erased controls. The complete language pilot requires proportional exact QA at least 8/32, both-correct pairs at least 2/16, readable QA at least 11/32, and retention of all six earlier proportional and both fixed exact cases. Frozen whole readings precede reference scoring. Bootstrap intervals use 10,000 whole- passage draws and seed 202611154, with the same sixteen groups and strict scoring rules as earlier comparisons.

The physical diagnosis orders training indices by SHA-256 of the fixed `motor-capacity-v1`, pool and index tuple, retaining 32 first and 32 later positions. Position-bound seeds are fixed before sampling. It reuses all 64 saved normal development first emissions without resampling. Targets, parent, sampled flow and supplied reconstruction are compared by integer foreground counts and exact arrays. Medians and counts at F1 at least 0.95 are reported separately for each 32-position pool. Each pool requires median at least 0.99 and at least thirty qualifying tiles. These fixed training-state gates test writing sufficiency, not semantic accuracy. Stopping is recorded but bypassed for this one-glyph diagnosis. All 128 scores and 512 displayed tiles are independently recounted, and all eight boards are inspected.

For the paired decoded objective, let D return frozen ink logits and set

$$\widehat{z}_1 = x_t + (1-t)v_\theta(x_t,t,h), \tag{16}$$

$$\mathscr{L} = \mathscr{L}_{\mathrm{FM}} + \mathbb{E}[\ell_{\mathrm{bal}}(D(\mu + s\widehat{z}_1),I) \mid t \geq 0.75]. \tag{17}$$

Writing b_p for per-pixel binary cross entropy with logits, the balanced per-image term is

$$\ell_{\mathrm{bal}} = \tfrac{1}{2}\left[\frac{\sum_p I_p b_p}{\max(1,\sum_p I_p)} + \frac{\sum_p (1-I_p) b_p}{\max(1,\sum_p (1-I_p))}\right].$$

Its coefficient is one. Eligible images are averaged per batch; an empty set contributes differentiable zero. Gradients pass through D without updating its parameters. The endpoint is

an estimator on a noise/target interpolant, not the endpoint of a numerical rollout. This auxiliary objective need not preserve the original flow optimum or improve language. Initialization, 12,000-update schedule, EMA, targets and all per-position sampling remain paired. Actual sampled-index digests and final CPU/CUDA random states match the original control exactly; all optimizer parameter steps and finite weights are checked. Frozen decoder parameter versions are unchanged. The same 128 supplied states are assessed once. Either failed training gate withholds the full 216-stream language trial, leaving that quality unmeasured. The second eight boards and every tile are independently checked.

The original head fit takes 109.485 seconds at 0.989 GiB peak allocated GPU memory. Fixed/proportional generation takes 3,442.507/2,720.582 seconds at 17.777 GiB, including 73.438/74.581 seconds actor loading. Pressure waiting is 0/260.247 seconds. Their endpoints are 12 stops/108 caps and 25 stops/71 caps. A missing deterministic CUBLAS setting causes a zero-update first launch (0.299 seconds after stage creation; full validation wall time unmeasured). A separate tensor-version setup failure takes 74.902 seconds before any answer; both failed attempts are retained. Cached-state diagnosis takes 7.815 seconds at 0.311 GiB. The paired fit takes 130.299 seconds at 1.169 GiB; its 128-position assessment takes 22.577 seconds at 0.311 GiB. All these small-head/diagnostic stages record zero pressure wait. The full reader is not loaded for head fitting or supplied-state diagnosis. Inherited pretraining, original adaptation and offline cache construction are excluded from these stage costs, precluding a foundation-training efficiency claim.

A cold replay of the fixed first book initially reproduces pixels and nested values but fails its access gate because shared admission opens old ownership ledgers. The retained attempt takes 125.884 seconds. Standalone admission is then changed to live process/memory/pressure and mutex checks, leaving input, weights, seed and sampling fixed. The independent invocation reproduces all pixels, reader states, stopping, feedback, noise and motor values without observed forbidden artifact access or IP connections. It takes 100.755 seconds, including 68.654 loading and 31.435 generation, at 17.748 GiB. Bound loader provenance excludes lexical parameters from the actor; the source archive's unused vocabulary payload remains disclosed. A file trace alone cannot prove tensor-level exclusion. The reproduced book is still capped and unreadable, so cold execution does not establish language.

Realized-sampler supervision and the qualified-writer control

Full-sampler endpoint supervision uses pathwise differentiation through the numerical generation procedure, as in direct differentiable-reward fine-tuning[24], while retaining the original conditional velocity loss[25]. Let $x_0 = e$, $\delta = 1/32$ and $t_j = j\delta$. For $j = 0, \ldots, 30$, the deployed

Heun update is

$$a_j = v_\theta(x_j, t_j, h), \tag{18}$$
$$\widetilde{x}_{j+1} = x_j + \delta a_j, \tag{19}$$
$$b_j = v_\theta(\widetilde{x}_{j+1}, t_{j+1}, h), \tag{20}$$
$$x_{j+1} = x_j + \tfrac{\delta}{2}(a_j + b_j). \tag{21}$$

The last interval is Euler: $x_{32} = x_{31} + \delta v_\theta(x_{31}, t_{31}, h)$. Writing $S_\theta(e,h) = x_{32}$, the candidate minimizes

$$\mathscr{L}_{\mathrm{FM}} + \ell_{\mathrm{bal}}\big(D(\mu + sS_\theta(e,h)), I\big).$$

The balanced BCE coefficient is one. The sampler receives e, h only, without reference pixels, target motors or target lengths. Its forward uses 63 velocity calls. Nonreentrant checkpointing recomputes activations during backward; gradients traverse all intervals and the frozen decoder without a detached prefix, truncated gradient or binary straight-through estimator. Seven numerical/runtime checks include actual-network finite differences, bit-exact recomputed gradients and transmission through the frozen decoder. Both GPU arms verify the differentiable forward against the deployed sampler.

Both arms begin from the same completed original-flow EMA. Seed 202611181, batch 32 with sixteen first and sixteen later positions, AdamW learning rate 2×10^{-5}, 100-update warm-up, cosine decay to 10% and EMA decay 0.99 are fixed for 2,000 updates. Each arm receives 32,000 first and 32,000 later exposures. Sampled-index digests and final CPU/CUDA random states match exactly. An independent CPU audit checks all thirty optimizer parameter steps, finite weights/moments and unchanged normalization. Reader, contextual adapters, encoder/decoder and stopping remain frozen. The same 128-position diagnosis and both per-pool median/count gates apply. Neither arm passes both gates; full 216-stream language generation is withheld and unmeasured.

The frozen qualified-writer comparison authenticates its earlier 18,000-update fit and full physical qualification before any new output. Only twelve readout/stop tensors differ from the parent; 26 codec/normalization tensors and all contextual adapter tensors remain bit-exact. This trial performs zero new updates and deterministic writing with no seed selection. The same 128 supplied states and all 216 original image inputs are retained. Generated binary pixels alone provide subsequent history, with stop threshold 0.5 and cap 128. No lexical decoder, tokenizer, font, reference length or teacher enters its answer path. Every actual initial state is compared with its bound parent tensor. Its prospective language pilot requires ten proportional exact answers, two correct pairs, sixteen readable answers, all eight union proportional exact cases and both original fixed cases. Review precedes reference scoring, with uncertain partials unrepaired. Both scoring and per-answer aggregates are independently recounted. Pointwise intervals retain the earlier 10,000 passage-group draws and seed 202611154. There is no broad book interval from twelve prefixes in two works. No new book cold replay is performed;

the earlier different-cohort 64-glyph replay is not substituted.

Matched extra-FM/full-rollout fitting takes 103.775/875.962 seconds, including 80.072/80.073 seconds pressure waiting, at 1.165/1.166 GiB peak allocated GPU memory. Their shared assessment takes 39.623 seconds at 0.402 GiB; terminal wall time including boards is 40.027 seconds. Balanced supplied-state assessment takes 1.301 seconds at 0.217 GiB, or 1.677 seconds including board export. Its fixed/proportional complete generation takes 1,534.527/1,140.482 seconds at 17.786 GiB, including 82.055/77.201 seconds loading, with zero recorded pressure waiting. Actual generation sums to 1,426.712/1,041.959 seconds. The inherited 7.522B body, earlier fits, preparation, CPU validation and review remain excluded costs. These measurements do not establish a foundation-training or matched-quality efficiency advantage.

Reference-image history intervention

The unchanged balanced checkpoint receives each of the 64 normal physical QA prompts once, with no optimization or external native decoding. After prompt prefill, the next-writing state is computed at every complete reference boundary using the canonical preceding glyph images. The supplied reference length determines the number of diagnostic steps; the stopping predictor is observed without terminating the trace. These teacher-forced traces must not be combined with autonomous answer counts. Source images, initial states and the bridge remain unchanged.

Each layout contains 745 target glyph positions and 32 terminal boundaries. The first position of each case is reported separately, leaving 713 later positions. Foreground F1 is $2\mathrm{TP}/(2\mathrm{TP}+\mathrm{FP}+\mathrm{FN})$, with value one for two empty ink masks. Exact equality compares the complete binary bitmap. Punctuation and spaces remain in the pools. Summaries aggregate positions unless explicitly identified as equal-case means. Shared autonomous and reference indices are retained only where both predictions exist, with missing and extra positions reported separately. A shared numeric index is not semantic alignment when wording or output lengths differ.

The later-writing gate requires both median F1 at least 0.99 and at least 90% of positions reaching 0.95. The stopping diagnosis separately requires the prospectively fixed nonterminal and terminal rates. All 1,490 reference positions, initial-state comparisons and saved autonomous pixel/stop replays are independently checked. Review then inspects all 8,561 displayed tiles. The common-prefix state comparison is post hoc and descriptive. All cases are reused development data, with reviewer familiarity and no independent human adjudication. The diagnosis takes 460.834 seconds, including 70.067 seconds pressure waiting, at 17.786 GiB peak allocated GPU memory. Loading is 137.167 seconds; loading and waiting intervals may overlap. Its 1,554 contextual calls do not create new autonomous answers or change original language scores.

Immutable prompt memory and visual continuation

Let X be the physical prompt image and $Y_0, \dots, Y_{N-1}$ the complete target glyph images. Each signed image has shape $1 \times 32 \times 32$, with ink at -1 and background at $+1$. Source prompt prefill produces detached memory $C_\ell(X) = (K_\ell^P, V_\ell^P)$ for each native layer $\ell = 0, \dots, 35$. The new residuals are bypassed during this computation. Source vision, contextual parameters and the original zero-output source adapters are frozen. The answer input sequence is

$$a_0 = q, \qquad a_{t+1} = S(E(Y_t)), \quad 0 \le t < N, \tag{22}$$

where E is the frozen image encoder and S its fixed affine denormalization into the imported 4,096-dimensional contextual coordinates. The learned query q initially copies the final fixed numeric task-control vector. It contains no case-specific answer or character identity.

With $h_t^0 = a_t$, the answer transition is

$$\begin{aligned} u_t^\ell &= F_\ell(h_t^\ell; C_\ell(X), h_{<t}^\ell), \\ h_t^{\ell+1} &= u_t^\ell + \frac{\alpha}{r} B_\ell A_\ell u_t^\ell. \end{aligned} \tag{23}$$

Here F_ℓ is the frozen native layer, $A_\ell \in \mathbb{R}^{8\times 4096}$, $B_\ell \in \mathbb{R}^{4096\times 8}$, $r = 8$ and $\alpha = 16$. Each A_ℓ uses Kaiming-uniform initialization and each B_ℓ begins at zero. The residual is applied after the native layer, including to the query. Prompt and answer position offsets preserve the imported rotary convention. The causal mask allows the complete source prompt and current/preceding answer inputs. Thus state t predicts Y_t before that target image enters the history. Final native normalization produces z_t.

Only frozen prompt keys and values are cached during fitting. The cache update is a pure concatenation of prompt memory with the current answer keys/values; activation recomputation cannot append to or lengthen the prompt. All 36 answer layers are recomputed with their current residuals using nonreentrant checkpointing. Incremental inference maintains a separate answer cache for the current output stream and leaves the prompt memory untouched. No mutable answer cache is captured during gradient-enabled execution.

Physical writing loss and autonomous feedback

The trainable readout R and frozen decoder D predict pixel logits $L_t = D(R(z_t))$. A separate trainable head predicts stopping logit s_t. Writing $b_{ti} = (1 - Y_{ti})/2$ for reference ink at pixel i, the implemented balanced reconstruction objective is

$$\begin{aligned} \mathscr{L}_{\text{image}} = \frac{1}{2}\Bigg[&\frac{\sum_{t,i} b_{ti}\, \operatorname{softplus}(-L_{ti})}{\max(1, \sum_{t,i} b_{ti})} \\ &+ \frac{\sum_{t,i} (1 - b_{ti})\, \operatorname{softplus}(L_{ti})}{\max(1, \sum_{t,i} (1 - b_{ti}))} \Bigg]. \end{aligned} \tag{24}$$

The sums cover every pixel of all N complete targets, rather than balancing each glyph independently. The final teacher-forced state z_N predicts the terminal boundary without an

additional image target. The balanced stopping loss is

$$\mathscr{L}_{\text{stop}} = \frac{1}{2}\left[\frac{1}{N}\sum_{t=0}^{N-1}\text{softplus}(s_t) + \text{softplus}(-s_N)\right]. \tag{25}$$

Training minimizes $\mathscr{L}_{\text{image}} + 0.1\mathscr{L}_{\text{stop}}$. Frozen encoder, decoder, source weights and affine statistics are not updated. Image-loss gradients still pass through the frozen decoder into readout, query and residual parameters. No decoded reference identity or lexical output projection supplies these gradients.

At autonomous inference, evaluate stopping before another emission. If $\sigma(s_t) \geq 0.5$, end the answer; otherwise emit

$$\widehat{Y}_{ti} = 1 - 2\mathbf{1}\{\sigma(L_{ti}) \geq 0.5\}. \tag{26}$$

Only $S(E(\widehat{Y}_t))$ supplies the next input. An initially stopped stream is empty; a stream reaching its image cap is retained as capped. Reference length, font rendering, a tokenizer, OCR, lexical logits, a live teacher and an answer bank are absent from this path. The physical pixel factorization and teacher-forced objective do not by themselves guarantee semantic learning or stability under generated histories.

Preflight evidence and the registered complete fit

The real preflight selects four complete eligible training cases and excludes the earlier feedback-ceiling diagnostic records and source groups. Its authentic targets have 1, 74, 184 and 513 glyphs, totaling 772 positions. All 36 answer layers receive actual gradients. An independent audit opens the saved checkpoint and verifies four Adam steps for all 85 trainable tensors, FP32 moments, original targets and prompt pixels. Prompt key/value, causality and frozen-body version checks come from the registered GPU execution. Initial residual-down tensors are not archived; their gradients, moments and bound update checks are retained. Numerical split/native and serial/parallel comparisons use fixed absolute tolerance 0.002 rather than asserting shape-independent equality.

Trainable parameters comprise 2,359,296 residual coordinates, a 4,096-coordinate query and 17,848,833 readout/stopping parameters. The deployed system also retains the 7,522,463,984-parameter inherited visual/contextual body and frozen image interfaces. The 150.811-second preflight includes 20.020 seconds pressure waiting and peaks at 18.122546 GiB allocated memory; loading takes 113.573 seconds and may overlap waiting. These measurements cover the four complete cases, not arbitrary page sizes, complete foundation training or matched-quality efficiency.

The subsequent full trial starts fresh, without the preflight updates. Complete eligible QA and book targets provide 2,036,247 positions per epoch. For a book record of n glyphs, expose the first $\min(128, \max(1, \lfloor n/4 \rfloor))$ glyphs and predict the remainder. The 13,766 records are presented once per epoch for three epochs; each gets one fixed-cell and two proportional prompts. Forty-five font/layout rejections retain the complete original raster. All source pixels,

target boundaries, ordering, exclusions and evaluation controls are registered before fitting. With one complete sequence per update, FP32 AdamW uses no weight decay, peak learning rate 10^{-4}, 500-update linear warm-up, cosine decay to 10^{-5} and global norm clipping at one. Initial and 10,324/20,649/30,973/41,298-update snapshots are fixed. The 732 prescribed autonomous streams, output-first review, complete-language pilot, retained successes and untouched confirmation determine any subsequent language claim. The final proportional pilot requires at least 12/32 exact answers, 4/16 both-correct pairs, 24/32 readable answers and 4/12 coherent complete long-book continuations, retaining all eight union proportional and both fixed exact cases. Each QA-erasure arm may score at most two exact answers, with normal minus the better erased arm at least eight. The completed comparison fails this pilot at the prescribed final checkpoint; the full four-QA-erasure audit also rejects it. Figure 24 reports the measured outcomes. Neither the preflight nor a finite training loss substitutes for complete-language assessment.

The single-device fit requires 48 GiB available host memory and 20 GiB free GPU memory on admission, with 24 GiB host reserve and a 22 GiB allocated-device ceiling at runtime. These host requirements are separate from the 24-GiB GPU. The cumulative fit wall budget is 48 hours, including the failed first child's 139.558657 seconds. Pressure waiting is bounded within that budget. Offline data preparation, inherited pretraining, earlier interface fits, evaluation and review remain additional costs; feasibility does not establish an efficiency advantage.

Saved-trajectory measurements and the constant-image baseline

The retrospective analysis is registered after complete language scoring and before computing its new physical/state statistics. It authenticates each saved output tensor and binary raster and retains all 732 streams, including empty, short, erased and capped outputs. Equal-stream summaries average a metric within each stream before averaging streams. Pooled density counts instead divide the number of qualifying emitted tiles by all emitted tiles. Early and tail analyses use eight tiles each, with a separate disjoint-window population restricted to streams of at least 32 tiles. No new semantic grades are assigned.

For complete target sequence i, let n_i be its tile count, c_{ij} its number of black pixels at coordinate j, and $B_i = \sum_j c_{ij}$, $W_i = 1024 n_i - B_i$. For a predictor constrained to emit the same probability image at every context and position, define

$$a_j = \frac{1}{M} \sum_i \frac{c_{ij}}{\max(B_i, 1)}, \qquad b_j = \frac{1}{M} \sum_i \frac{n_i - c_{ij}}{\max(W_i, 1)}. \tag{27}$$

The sequence-balanced BCE restricted to this family is

$$\begin{aligned} \mathscr{L}_{\text{const}}(q) &= -\frac{1}{2} \sum_j \left[a_j \log q_j + b_j \log(1 - q_j) \right], \\ q_j^* &= \frac{a_j}{a_j + b_j}. \end{aligned} \tag{28}$$

The expression uses the implemented denominator clamps, including blank targets. Common epoch and layout multiplicities cancel because they repeat the same targets. All 13,766 complete targets contribute once, not one representative glyph per source. Independent autograd tests include unequal ink masses and all-white/all-black targets. The measured optimum loss is 0.437380, and its binary comparison uses the deployed $q_j \geq 0.5$ tie rule. This derivation does not characterize the optimum of the conditional neural model.

Physical measurements include ink count, adjacent Hamming distance, exact adjacent repetition and foreground F1 against the binary constant template. Saved-state measurements include relative adjacent L2 and cosine distance. First-state contrasts cover 160 paired-question and 256 erasure comparisons. Relative L2 divides the difference norm by the mean of the two state norms, with an epsilon denominator clamp. An unavailable first-image comparison remains unavailable when either stream is empty. State/image inequality is not a semantic test, and a small tail change is not a stability proof.

The CPU analysis uses two threads and no initialized CUDA state. Successful execution takes 18.879 s including its wrapper, with 0.593 GiB peak RSS and at least 51.210 GiB available host memory. A preserved earlier attempt returns SIGTERM/143 without a report. Its cause is unknown and its wall-time upper bound is 242.049 s; exact cumulative execution cost is unknown. These costs are distinct from model fitting, generation and review.

Initial/final clean-history comparison

The new diagnostic fixes 124 complete prompt/history jobs per checkpoint: all 32 development QA cases in both layouts, twelve fixed-layout development books, and twelve training QA plus twelve training books in both layouts. Training sources are selected by source-ID hash, with the longest complete training book included. All target rasters are recovered from the authenticated corpus; answer text is not rendered. The original initial and final checkpoints are fixed before execution. Each supplied-history state predicts the current image before that target enters its inputs. Predicted stopping is recorded but bypassed, with reference lengths supplied. All 25,194 target positions remain in the comparison.

At each checkpoint the complete 513-glyph training book is also evaluated incrementally with the identical reference history. Maximum state errors 0.000534058 and 0.000148773 are below the unchanged 0.002 limit. Every original development observer geometry is reproduced; maximum first-state difference is 0.000120163. Differences across execution shapes are numerical comparisons, not bit-exact identities. All parameters retain their recorded versions.

The physical description of accurate later writing requires at least 90% of later target positions to reach foreground F1 0.95. Training/development, QA/book and layout remain separate. Source-kind ink thresholds are the clean training 99th percentiles. The independent CPU export recomputes every metric from all saved binary predictions and original targets. No new autonomous stream, semantic grade, source head, teacher or optimizer update is produced. The successful single-GPU stage takes 648.971 s, including 30.029 s of pressure waiting, with 18.124 GiB peak allocation and at least 49.819 GiB available host memory. An earlier

missing-environment launch fails before model loading; its 0.341-second measured stage and unknown whole-process overhead are preserved separately.

Conditional joint writing over physical pixels

The completed head-only comparison changes the image writer while retaining the final continuation model's visual reader, answer dynamics, initial query, image feedback encoder, affine statistics and stop predictor. Its physical review misses the prespecified prerequisite, so its autonomous language remains unmeasured. A separately registered comparison trains the continuation query and residual adapters through the same image likelihood; its completed fit and language result remain unmeasured. Autoregressive image densities and vector-conditioned image decoders are established formulations[6,7]; their use here does not guarantee readable characters, correct answers or improved efficiency.

Let X denote the visible prompt and $Y_t \in \{0,1\}^{32\times 32}$ the tth emitted image, with one indicating ink. The interface signs its pixels as $S_t = 1 - 2Y_t$. Let $M_0(X)$ be immutable prompt memory and F_κ the frozen image-feedback function. The next contextual state is

$$h_t = D_\theta(q_\theta, F_\kappa(S_{<t}); M_0(X)). \tag{29}$$

Here F_κ acts on each preceding image in order, and the notation selects the last state of the causal continuation computation. The trainable part of θ comprises the initial query and answer-layer residual adapters; native body parameters remain fixed. The head-only control fixes $\theta = \theta_0$. The actual feedback is $F_\kappa(S) = \sigma_e \odot E_\kappa(S) + \mu_e$, with the existing image encoder and fixed semantic affine buffers. A target image does not enter its own predictor. At generation time these inputs are the model's emitted pixels, rather than a retrieved glyph or a rendered transcription.

Partition each image into sixteen rows and eight columns of 2-by-4 pieces. For $0 \le a < 16$, $0 \le b < 8$ and $j = 8a + b$, encode the piece by

$$r_{t,j} = \sum_{u=0}^{1} \sum_{v=0}^{3} 2^{4u+v} Y_t[2a+u, 4b+v]. \tag{30}$$

Each $r_{t,j}$ ranges from zero to 255. The mapping is exactly invertible and imposes no glyph inventory: every binary 32-by-32 image has a representation. These discrete physical values encode pixels rather than Unicode characters or linguistic tokens. Representability alone establishes neither legibility at this resolution nor generalization to historical or unencoded writing.

The new head defines a normalized conditional distribution within each image,

$$p_\phi(Y_t \mid h_t) = \prod_{j=0}^{127} p_\phi(r_{t,j} \mid h_t, r_{t,<j}). \tag{31}$$

Every factor is a 256-way softmax. During fitting, piece j receives only preceding target pieces;

its own value enters the input at $j+1$. Causal attention prevents future pieces from entering that factor. During generation, the head chooses each mask greedily from its own emitted prefix. The same inverse packing produces the final image without a font, lexical output head, OCR, answer lookup or live teacher. Greedy local choices need not yield the most probable full image or a valid character. The head makes 128 sequential decisions per glyph; reduced training parameter count does not establish a wall-time advantage.

Image history and variable-length outputs

Let $s_t = \mathrm{sigmoid}(g_\psi(h_t))$ be the retained stop probability. The likelihood of a finite image stream with an end event is

$$p_{\theta,\phi}(Y_{1:T}, \mathrm{end} \mid X) = s_{T+1} \prod_{t=1}^{T} (1 - s_t) p_\phi(Y_t \mid h_t). \tag{32}$$

This expression permits an empty response when the first decision stops. It does not prove almost-sure stopping or a natural document ending. The deployed rule stops before appending a glyph when $s_t \geq 0.5$; otherwise it appends the generated image and rereads it through F_κ. Fixed caps of 128 and 640 glyphs bound the registered comparisons, with caps retained as failures to stop rather than silently converted into natural endings. Reference output lengths never enter autonomous generation.

Only the image head is fitted in the completed head-only comparison. Its within-image likelihood cannot train the fixed continuation state or stop predictor. In the new comparison, the parameters ψ remain fixed but the stop loss supplies gradients to h_t. In either case, clean training histories do not establish stability under generated histories. The interpretation of the prompt, image-form fidelity and complete-language behavior therefore require separate evidence.

Complete-sequence fitting and state normalization

For the head-only control, source i with T_i target images and layout ℓ provides cached h_{it}^{ℓ} after the query and preceding reference images, before target t enters history. Retain all $T_i + 1$ states losslessly in FP32, but exclude the terminal state from image loss. Both layouts of all 13,766 sources give 27,532 cache records and 4,072,494 target-predicting states. The loss used for one complete source presentation is

$$\mathscr{L}_{i\ell}(\phi) = -\frac{1}{128 T_i} \sum_{t=1}^{T_i} \sum_{j=0}^{127} \log p_\phi(r_{it,j} \mid h_{it}^{\ell}, r_{it,<j}). \tag{33}$$

The fixed $1/128$ factor reports mean physical-piece NLL, proportional to the complete-image negative log likelihood. Averaging within each full source prevents its length from determining its optimizer weight. Accumulating microbatches of at most 64 images preserves this exact reduction; a complete source presentation makes one optimizer update.

The three-epoch schedule presents every source once per epoch, with one fixed and two

proportional layouts overall. Its empirical objective is

$$\mathscr{L}(\phi) = \frac{1}{N}\sum_{i=1}^{N}\sum_{\ell\in\{f,p\}} w_\ell \mathscr{L}_{i\ell}(\phi). \tag{34}$$

Here $N = 13{,}766$, $w_f = 1/3$ and $w_p = 2/3$. The completed comparison retains 41,298 updates and 6,108,741 target-image exposures. Development histories contribute to neither fitting nor normalization. Compute the head's conditioning mean and second moment from training predictor states, giving each source equal weight and the same 1:2 layout ratio. Exclude terminal states and floor each resulting standard deviation at 10^{-4}. These conditioning statistics are distinct from the retained image-feedback affine buffers μ_e, σ_e.

The 4,096-dimensional standardized condition projects to width 192. Four causal layers, each with six attention heads, process 128 physical positions. Each preceding mask projects from $2\,\mathrm{bits}(r) - 1$, where bits recovers the eight bits in Eq. 30's order. Learned start and positional vectors complete the inputs. The component has 2,642,368 parameters. FP32 AdamW uses a peak learning rate 2×10^{-4}, 500-update warm-up, cosine decay to 2×10^{-5}, weight decay 0.01 and gradient norm clipping at one. All five fixed head snapshots are retained; no development result selects a snapshot or schedule. They retain the same final reader and continuation parameters. The subsequent context-gradient comparison reuses this exact normalization, fresh head draw, original point-final continuation initialization and complete source/layout schedule. Its conditioning statistics remain fixed even while the continuation states change; they are not estimated from development data or updated during the fit.

Exact gradients through the continuation state

The head-only control fits ϕ on immutable states learned under the earlier pixel objective. The new comparison recomputes the differentiable continuation graph for every complete source. Write $H_\theta^\ell = (h_{i1}^\ell, \ldots, h_{i,T_i+1}^\ell)$ and let $a_{it}^\ell = g_\psi(h_{it}^\ell)$. Its implemented training objective is

$$\begin{aligned} \mathscr{J}_{i\ell}(\theta,\phi) &= \mathscr{L}_{i\ell}(\phi; H_\theta^\ell) + 0.1\,\mathscr{S}_{i\ell}(\theta), \\ \mathscr{S}_{i\ell}(\theta) &= \frac{1}{2T_i}\sum_{t=1}^{T_i} \mathrm{softplus}(a_{it}^\ell) \\ &\quad + \frac{1}{2}\mathrm{softplus}(-a_{i,T_i+1}^\ell). \end{aligned} \tag{35}$$

Here $\mathscr{L}_{i\ell}$ is the complete-source mean physical-piece NLL in Eq. 33, and $\mathrm{softplus}(a) = \log(1 + e^a)$. The stop term balances the mean nonterminal loss against one terminal loss. These reductions and the weight 0.1 mean that Eq. 35 is not the unweighted negative log likelihood of Eq. 32. Only the image density is supervised at target-image positions; the terminal state has no image target. Reference images supply history during fitting, never during autonomous generation.

The empirical context-gradient objective uses the same source/layout weights as Eq. 34,

replacing $\mathscr{L}_{i\ell}$ by $\mathscr{J}_{i\ell}$. No completed-source or layout exposure is added.

To limit writer activations, create a numerically equal detached FP32 leaf $\bar{H}$ from H_θ^ℓ, retaining the original continuation graph. Accumulate image-loss microbatches through $\bar{H}$ and ϕ, then backpropagate the balanced stop loss through $\bar{H}$. With $\mathscr{B}$ a disjoint partition of the T_i image positions into batches of at most 64, the gradients are

$$\begin{aligned} V &= \nabla_{\bar{H}} \mathscr{J}_{i\ell}(\bar{H}, \phi), \\ \nabla_\phi \mathscr{J}_{i\ell} &= \sum_{B \in \mathscr{B}} \nabla_\phi \mathscr{L}_B(\bar{H}, \phi), \\ \nabla_\theta \mathscr{J}_{i\ell} &= \left(\frac{\partial \operatorname{vec} H_\theta^\ell}{\partial \theta} \right)^\top \operatorname{vec} V. \end{aligned} \tag{36}$$

In this expression $\mathscr{J}_{i\ell}(\bar{H}, \phi)$ means the loss in Eq. 35 with its contextual states replaced by the independent leaf $\bar{H}$. Reverse-mode differentiation applies the final vector-Jacobian product without constructing a Jacobian matrix. Every $\mathscr{L}_B$ retains the same denominator $128T_i$; a short last batch does not receive extra weight. The stop term has no direct dependence on ϕ under reference-image history. Apply the accumulated state gradient once to the original continuation graph, with checkpointed native answer layers. This is the ordinary chain rule, without a learned gradient or a gradient through discrete sampling. The actual training path calls the teacher-forced density directly; the inference-only greedy writer is not a training surrogate. All gradients are accumulated before a single norm clip and AdamW update for that complete source.

The real-source preflight compares this construction with unsplit autograd on a complete five-glyph QA in both layouts, across all 131 trainable tensors. Its maximum relative gradient error is 2.408×10^{-6}. Both complete 513-glyph book layouts yield finite nonzero gradients and an actual optimizer step. The peak allocated VRAM is 17.795 GiB. All preflight trainables are restored exactly, and neither preflight update enters the fit. These are computational checks, not language demonstrations. The new fit trains 5,005,760 parameters: 2,363,392 continuation parameters and the 2,642,368-parameter image head. The actual construction contains 7,553,448,370 unique parameters, including 7,522,463,984 inherited visual/contextual parameters. Prompt memory, native body, feedback, stop parameters and all four affine buffers remain fixed.

Completed prerequisite and subsequent evaluation

Conditional writing first requires frozen opaque review of the prescribed positions in all 124 complete cases. At least 90% of selected candidate forms must be confidently readable and nonblank in every training/development, source-kind and layout group. A legible alternative book glyph can pass this component gate without matching the canonical next glyph. Original pixels, uncertainty and all control groups remain intact. The head-only comparison fails this numerical gate. Its original frozen credits are reported with the subsequent 17 blank-flag

disagreements, including four credited empty images. Readings and scores are preserved; they are not corrected physical accuracy. The new evaluation additionally verifies that every credited candidate image contains ink. That mechanical check cannot establish visual legibility or meaning. Physical qualification does not replace the 708-stream autonomous comparison, paired questions, erasures, retained successes, coherent long books or subsequent untouched confirmation.

The small fitted head retains the inherited 7.522-billion-parameter visual and contextual body and the previously trained continuation/interface components. The complete new deployment, cached preparation, inherited pretraining, fitting, generation and review must all enter resource accounting. A vocabulary-free runtime image boundary does not establish language acquisition from images alone. No compact-foundation, matched-quality efficiency or useful complete language claim follows from the conditional density or computational preflight.

Initial raster control data

The public Chinese stream contains 7,017 provenance-tracked records. Records are assigned deterministically to training, development or a sealed partition before rendering. Original and simplified views inherit their source split. Both generation experiments use one fixed Noto Sans CJK font, 26-pixel writing inside 32-pixel cells, without image augmentation. A 65-cell training segment supplies the first 60 cells as context and cells 61–64 as future writing. The remaining cell is unused. No string, source identifier or symbolic target enters a model method. Sealed records remain unopened.

Initial raster control training

In the diffusion experiment, the reader is frozen. Copied glyph-prior weights initialize the denoiser, which is then optimized as a velocity predictor. AdamW uses a learning rate of 5×10^{-5} for transferred layers and 2×10^{-4} for new modules, 400 warmup updates, cosine decay to 10%, batch size eight, BF16 arithmetic and gradient clipping at one. EMA decay is 0.999. Checkpoints retain model, optimizer, RNG, dataset offset and provenance hashes. Interrupted and continuous smoke runs produce byte-identical model and EMA weights.

The field objective applies an orthonormal transform directly to floating signed predictions. Thresholding is confined to output and identity evaluation. This distinction is verified by an isolated pixel-gradient test. The primary velocity MSE is supplemented by foreground, Dice, field and representation losses at weight 0.10 each, plus an ordered-history margin at weight 0.25. Supplementary Information gives the complete objective and gates.

The exact-density experiment uses 64-cell causal histories and 16 stratified positions in each of eight streams per update. A quarter of selected images receive supplied-image motor conditions; the remainder use predicted states. Binary observations are thresholded once before unweighted Bernoulli NLL. AdamW uses learning rate 2×10^{-4}, 400-update warmup, cosine decay, BF16, norm clip one and EMA 0.999. All 8,000 updates complete in 207.8 s with 0.400 GiB peak allocated CUDA memory. This timing excludes the frozen reader's earlier training. Exact

resumed weights and EMA match uninterrupted smoke training. Strict-causality gradient tests and enumeration of a four-bit subproblem verify the probability masks.

The matched continuation reinitializes optimizer and EMA from identical exact-density weights. Both arms make 8,000 additional updates, using the same data and training generator. The joint arm uses reader learning rate 2×10^{-5} and density rate 2×10^{-4}, with the same schedule and motor mixture. It trains 18,711,840 participating parameters; the frozen arm trains 2,434,592. Joint and control times are 342.4 and 218.0 s, with peaks 0.648 and 0.400 GiB. Frozen-reader weights remain identical to initialization, while joint reader weights change. Both final training RNG states match. Development opens only after both arms finish.

The subsequent decoder retains eight partial images after each binary pixel expansion, ranked by accumulated Bernoulli log probability. Row conditions are gathered along beam-parent indices within each row. Stable ties prefer ink, matching the original width-one rule. Full FP32 matmuls are used for both new controls after a documented numerical preflight; model weights are unchanged. The maximum production incremental-score/parallel-NLL error is 2.10×10^{-5} nat, below the unchanged 0.001 tolerance. The complete inference audit takes 269.3 s at 0.154 GiB allocated CUDA memory.

The assignment experiment keeps the native eight-expert architecture and trains all parameters from the same V62 EMA. Both arms score the same four streams for every sampled pair. AdamW, sample order, update budget and final EMA selection match. The auxiliary penalty has temperature 0.01 and weight one for aligned, zero for control. Complete details, independent runtime replay and the unchanged conjunctive gates appear in Supplementary Note 22.

Separate continuous writing coordinates

The successor uses a visual reader E_r, contextual body F_θ, output projection A_θ and frozen writing codec (E_w, D_w). For observed prompt image X and completed answer blocks $Y_{<t}$,

$$h_t = F_\theta(E_r(X), E_r(Y_{<t})), \tag{37}$$

$$\widehat{z}_t = N(A_\theta h_t), \qquad z_t^* = \mathrm{sg}(E_w(Y_t)), \tag{38}$$

$$\widehat{Y}_t = D_w(\widehat{z}_t). \tag{39}$$

Here h_t contains two structural queries, N is nonaffine LayerNorm with $\varepsilon = 10^{-5}$, and sg stops label gradients. Each writing state comprises two continuous 768-vectors for two physical 32×32 tiles, without character segmentation. The encoder already normalizes its codes; the state loss explicitly normalizes both arguments again:

$$\begin{aligned} \mathscr{L} = {} & \mathrm{MSE}(N(\widehat{z}_t), N(z_t^*)) \\ & + \mathrm{MSE}(D_w(\widehat{z}_t), Y_t) + \mathscr{L}_{\mathrm{stop}}. \end{aligned} \tag{40}$$

Uniformly sampled block losses retain the complete teacher-forced history. A separate class-balanced stop loss supervises termination after the complete answer. Frozen decoder weights

still transmit image-loss gradients into predicted codes and contextual adapters. Generation uses prompt pixels, learned stopping and only its own completed output images. This regression objective is not a normalized language likelihood or a guarantee against averaging distinct conditional writing modes. Execution and reconstruction are evaluated separately from language (Supplementary Notes 26–28). The complete instruction run fixes 10,000 updates and two examples per update under fresh full-corpus permutations, with final raw weights and no EMA. Its autonomous audit scores all 192 fixed normal prompts and their matched blank controls, preserving full widths and learned stops. Neither the actor nor its evaluator repairs generated pixels. The subsequent exact-history diagnostic uses the same final weights and no optimizer updates.

Sampling, controls and statistics

The diffusion experiment uses 32 registered DDIM schedule levels and classifier-free scale two. Four future cells start from Gaussian noise. The same noise is reused for ordered, suffix-preserving shuffled and zero-condition histories. Only one sample contributes to the primary block result. Generated cells are appended as visible rasters for a second four-cell block.

An evaluator-only canonical bank measures nearest-raster identity after generation. Targets outside the bank receive no identity credit, and bank coverage is reported per position. This measurement can assign a nearest identity to an illegible pseudo-glyph, so direct images and surface statistics accompany every result. Pixel F1 alone is also insufficient because many Chinese glyphs share strokes. No OCR is used by the model or the reported identity evaluator.

The exact-density audit uses 512 NLL targets, 256 single-image generations and 64 four-step feedback streams. Its sampler uses 32 row-transformer evaluations and 1,024 masked-head evaluations per image. Ordered and shuffled runs share the same Bernoulli uniforms. Its post-generation bank includes all distinct training forms plus every audit target, including punctuation: 9,932 unique binary rasters and 100% target coverage. This enlarged bank differs from the earlier Han-only bank. Zero condition is an out-of-distribution intervention, not a separately trained unconditional baseline. One standalone export embeds the complete reader and density; image-prompt inference reproduces the training-checkpoint continuation exactly under the same seed and greedy mode.

Confidence intervals use 2,000 bootstrap resamples of source records, retaining all windows from a record together. Paired differences use identical windows and initial noise. Intervals describe development-record variability for this trained checkpoint, not variability across training seeds. The current development partition has informed a sequence of experiments and is not a final independent generalization test.

The assignment comparison retains the same fixed 256 primary questions and 393 paragraph pairs. Its paired 95% intervals use 2,000 whole-article resamples, seed 20266329. Reference-energy ranking supplies accepted answers and remains a separate diagnostic. One seed, reused development data and external foundation training limit both this result and efficiency claims.

Data and code availability

Implementation, protocols, evidence summaries and scripts are available in https://github.com/lachlanchen/ImagizedLanguageModel. Manifest and checkpoint hashes identify the exact local research artifacts. Restricted reference books are not redistributed. The complete research record and Supplementary Information distinguish earlier foundation provenance from the present raster-only deployment experiment.

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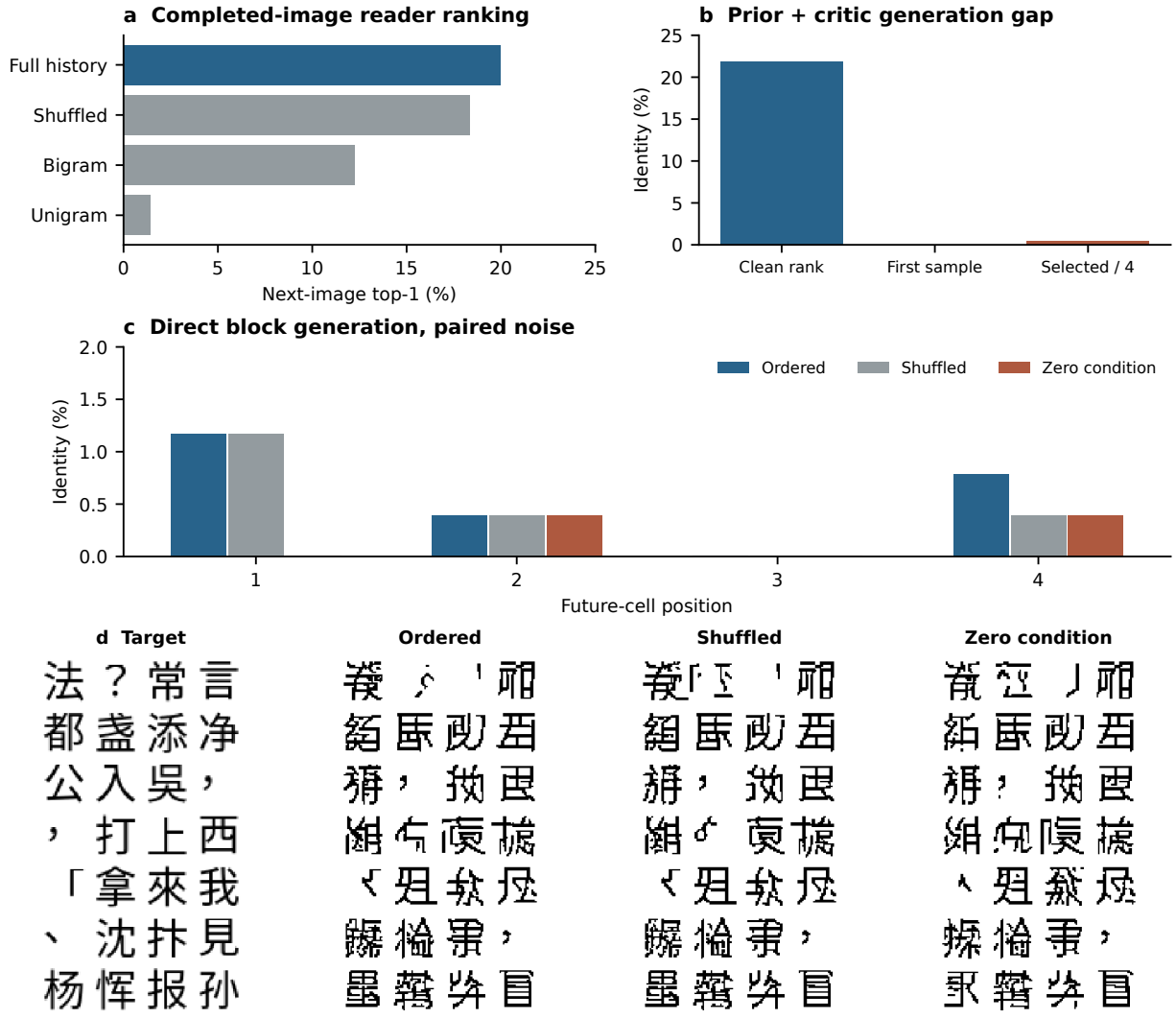


Fig. 2 | Prediction and generated-image evidence. a, The established causal reader ranks completed next images above unigram, bigram and shuffled-history controls on 2,048 windows. These are ranking measurements, not generated text. **b**, Prior-plus-critic guidance retains completed-image ranking but almost never generates the target identity. **c**, Direct conditional block generation reports all four positions under ordered, shuffled and zero conditions. Bars use the same starting noise and single primary samples. **d**, Fixed, unselected examples show the target and all generated controls. Images are raw model outputs with nearest-neighbour enlargement. No OCR or image-bank replacement is applied.

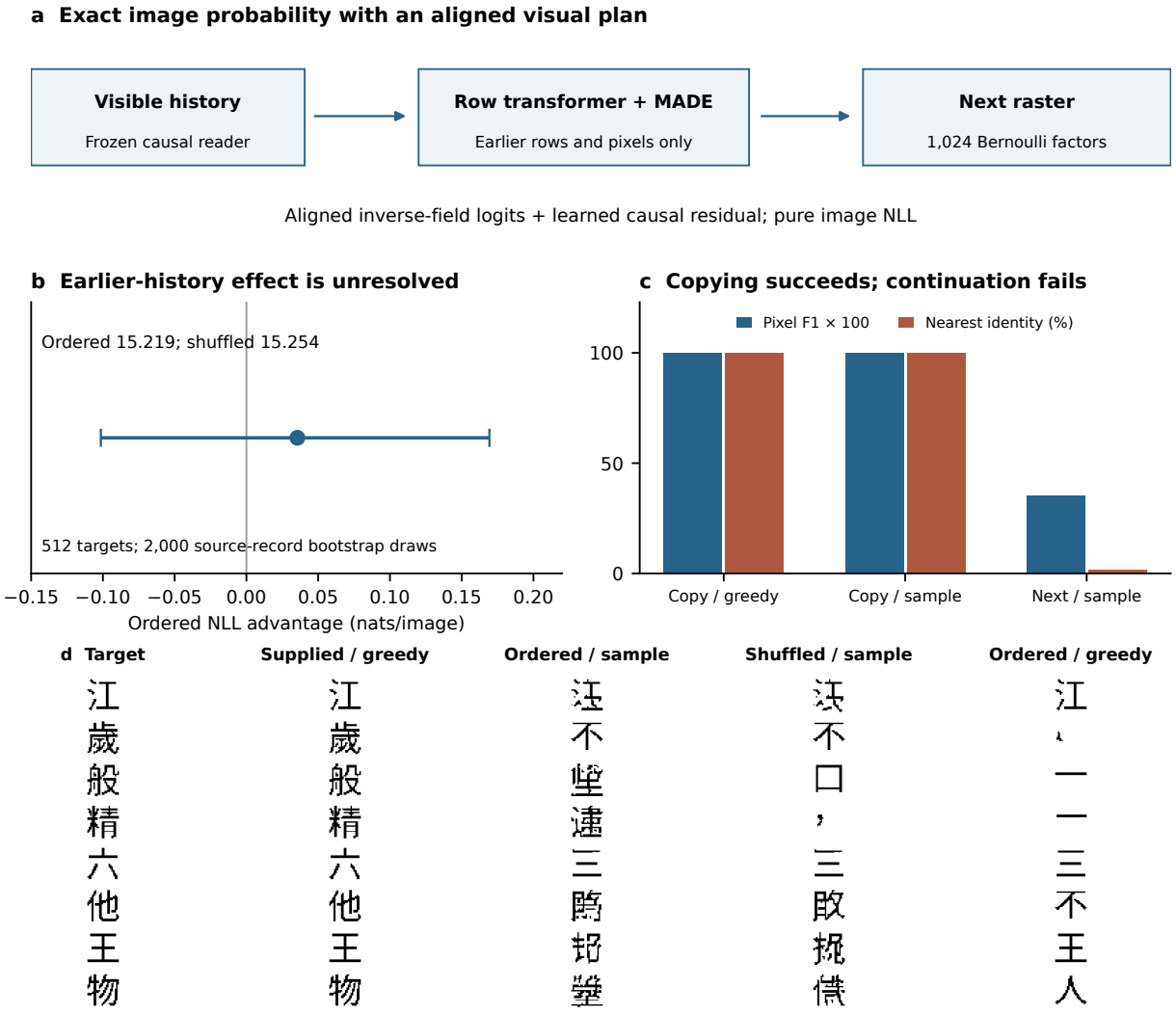


Fig. 3 | Exact image likelihood separates motor success from language failure. a, A frozen image reader supplies an aligned plan and continuous history state. Strict row and pixel masks give a normalized raster law. **b**, Ordered-versus-shuffled NLL advantage on 512 development targets; the interval uses 2,000 source-record bootstrap draws and crosses zero. **c**, Supplied-image copying and autonomous next-image sampling on 256 targets. The copy controls reveal the answer. **d**, The first eight audit rows, shown without selection, compare target, exact supplied-image copying and autonomous controls. Every displayed output is the actual model raster. The 9,932-image evaluator bank is created after generation, covers all targets and never selects or replaces an output. Its identity rates are not directly comparable to earlier 1,024-image-bank results.

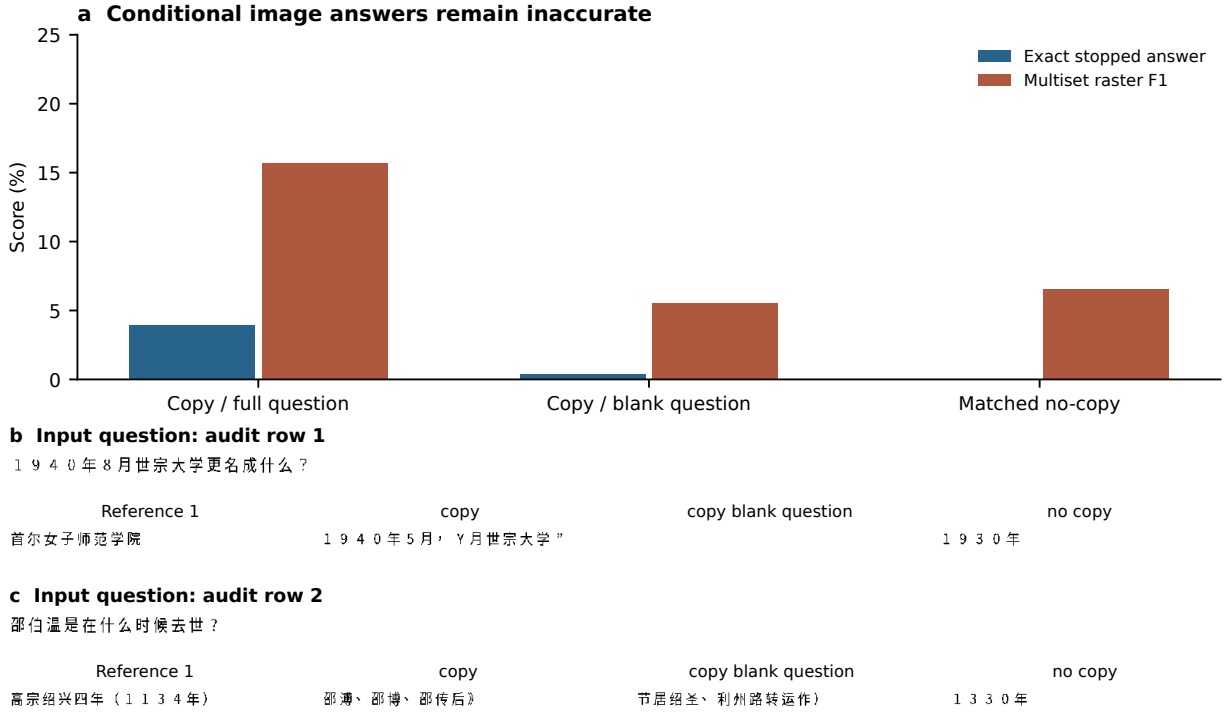


Fig. 4 | Question-dependent image answers remain inaccurate. a, Exact stopped answer and multiset raster F1 on 256 fixed questions from unseen articles, using full questions, blanked question pixels and the matched no-copy arm. F1 does not enforce order or correct termination. **b,c**, First two audit questions and raw answer images, with the first retained reference. The complete paragraph is also supplied to the model. No examples are selected or repaired; white output denotes a zero-cell answer. Question images are host renderings, and outputs are actual model rasters. All primary copy-arm answer cells are selected from the visible prompt.

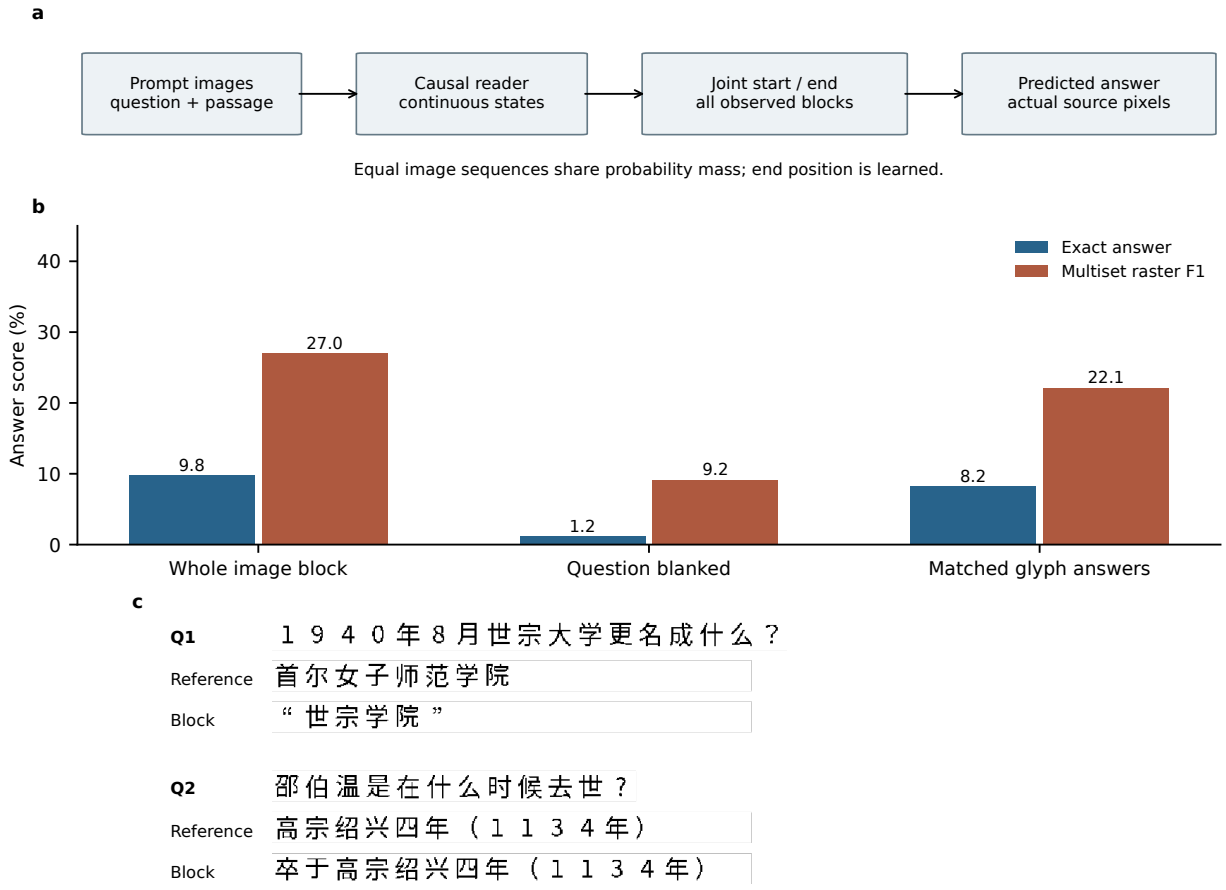


Fig. 5 | Predicting whole observed image answers. a, The reader assigns probability to variable-length source image blocks, marginalizing identical sequences. **b**, Fixed development answers after matched additional training. F1 counts exact-raster multiset overlap; exact answers also require order and termination. **c**, First two audit questions, retained reference images and raw block answers, without example selection or repair. The full passage also enters the model. Block copying preserves forms but does not establish that the selected text answers the question.

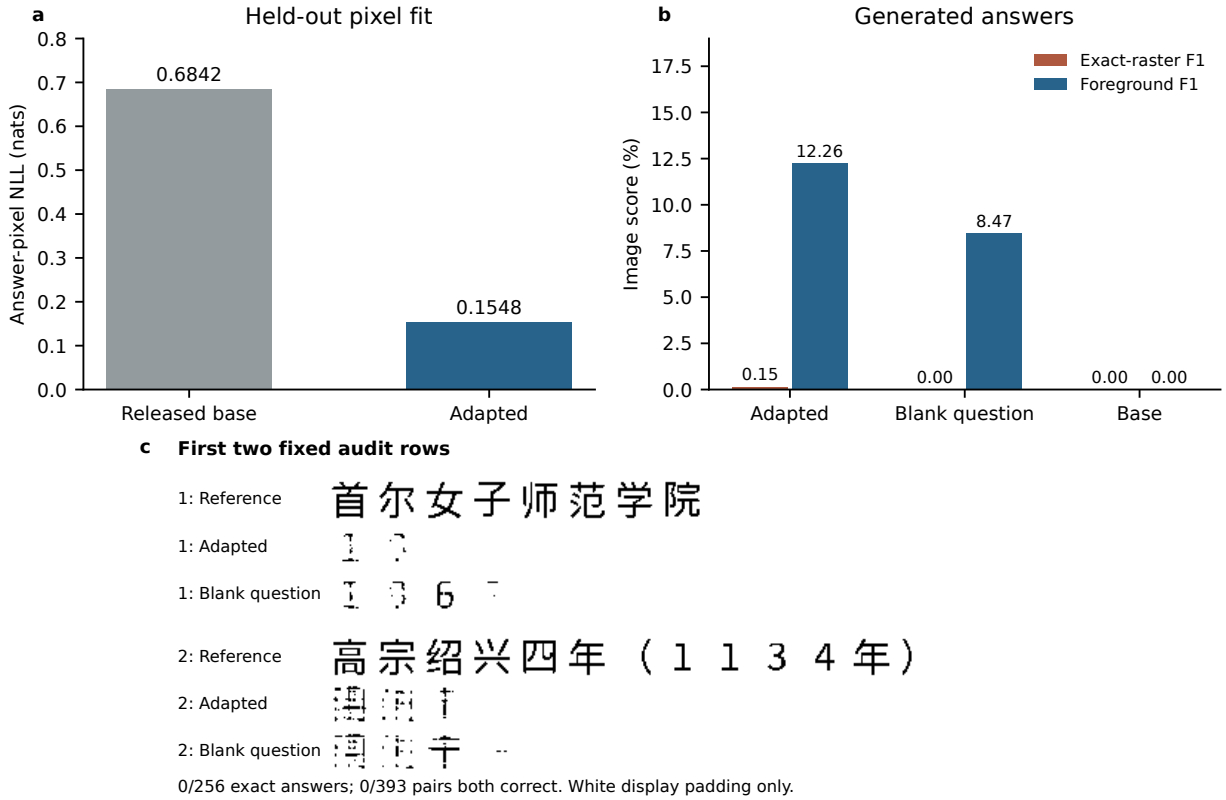


Fig. 6 | Better pixel likelihood does not yield useful free writing. a, All-development answer/termination-pixel NLL, released base and final adapted EMA. **b**, Exact-raster multiset F1 and aligned foreground F1 on 256 fixed primary questions, with question-blanked and released-base controls. Foreground overlap does not establish a correct character or answer. **c**, First two fixed audit rows, with retained references and unmodified generated pixels. White padding aligns the display and never changes a nonblank output. All exact answers and both-correct pairs fail.

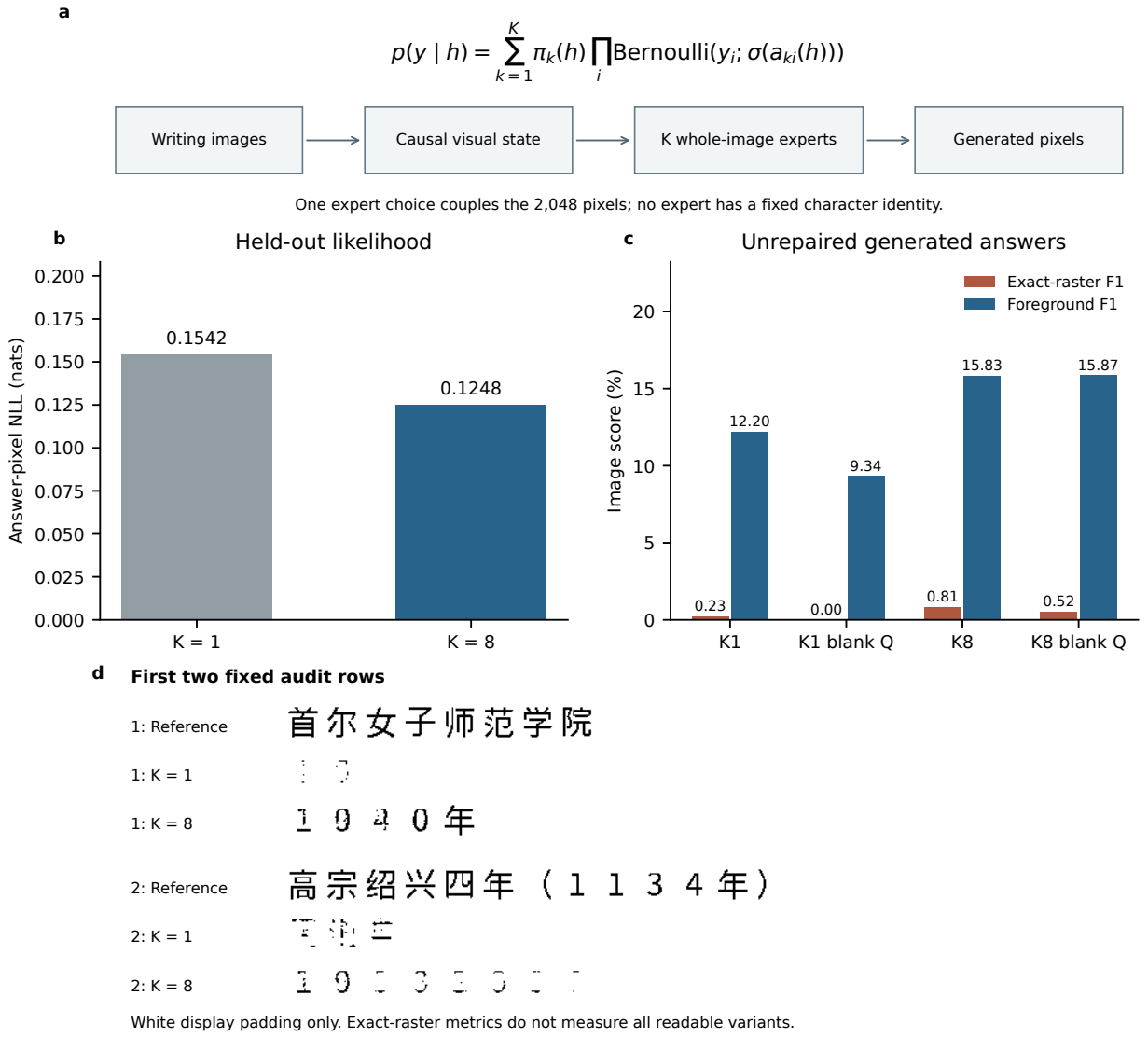


Fig. 7 | A whole-image mixture improves pixel fit but fails the question-dependent writing test. a, Conditional image law and inference boundary. One latent expert couples all pixels; experts are context-dependent neural functions. **b**, All-development answer/termination-pixel NLL after matched-budget adaptation. **c**, Strict-raster and aligned foreground F1 on normal and blank-question prompts. **d**, First two fixed primary references and raw outputs, with white display padding only. No nonblank pixel is repaired or removed. Parameter counts and output computation differ between arms; both use externally pretrained weights.

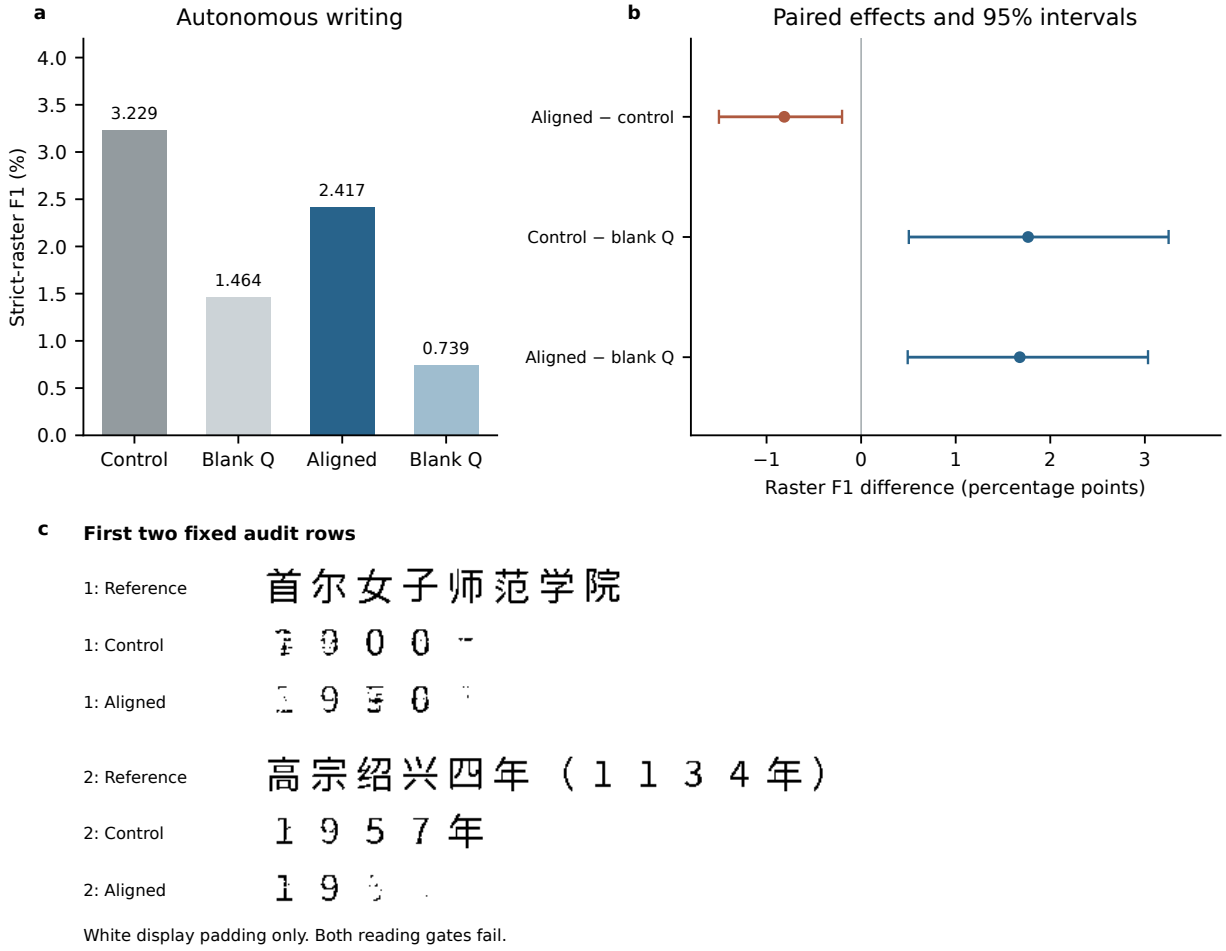


Fig. 8 | Question use and reference compatibility do not establish correct free writing. a, Strict-raster F1 on the fixed 256 primary questions and their blank-question controls. Bars show means. **b**, Paired differences with 95% intervals from 2,000 article resamples. The assignment arm scores lower than control despite positive question effects in both. **c**, First two fixed primary references and unmodified generated answers. White padding aligns the display; no nonblank output pixel is removed or repaired. Strict raster agreement does not credit every readable alternative form.

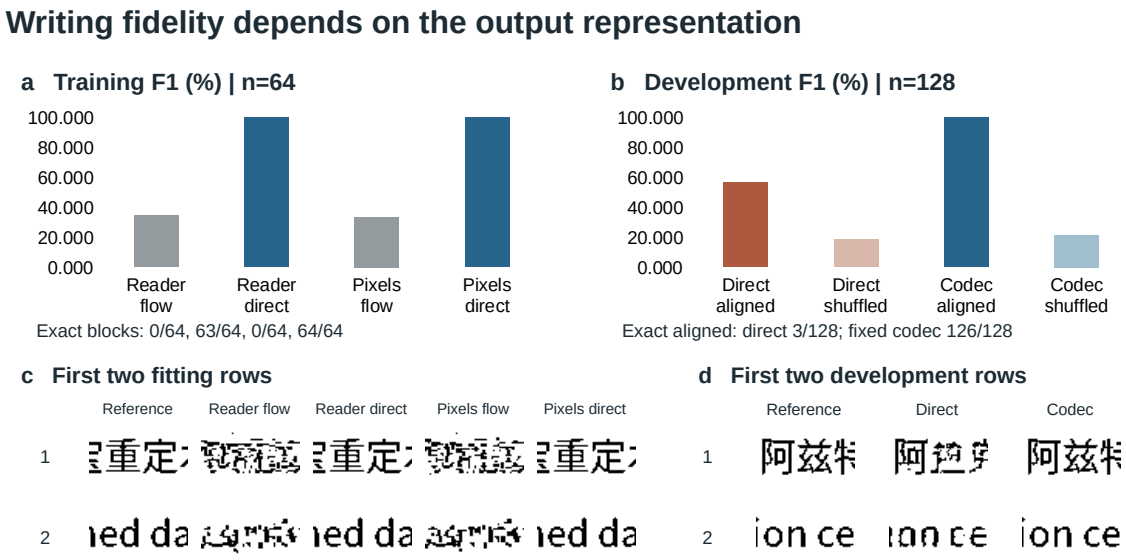


Fig. 9 | Faithful supplied-code writing does not establish answer selection. a, Foreground F1 for four matched fitting arms on 64 training blocks. The pixel condition is an answer-supplying control. **b**, Direct and fixed-codec reconstruction on the same 128 reused-development blocks, with their shuffled-condition controls. Training histories differ; this is not a matched efficiency comparison. **c,d**, The first two fixed fitting and development rows, with original source pixels and unmodified reconstructions. No row is chosen by quality, cropped or repaired.

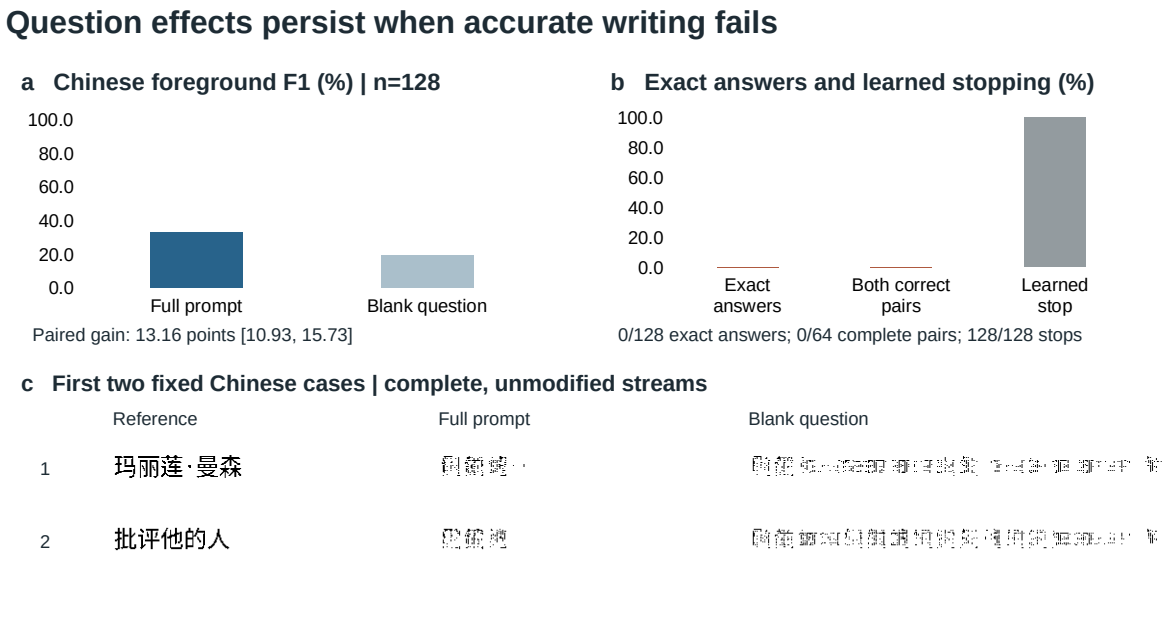


Fig. 10 | **Question dependence persists when accurate answer writing fails. a**, Chinese foreground F1 on 128 fixed development questions with full and matched question-blank prompts. The paired difference uses 2,000 bootstrap draws of 64 source articles. **b**, Exact full answers, both-correct question pairs and learned stopping. The original answer-quality gates fail, although question dependence and stopping pass. **c**, Complete reference and unmodified generated streams for the first two fixed Chinese cases. Images retain their physical width and aspect ratio; no nonblank pixel is removed or repaired. These are reused development data.

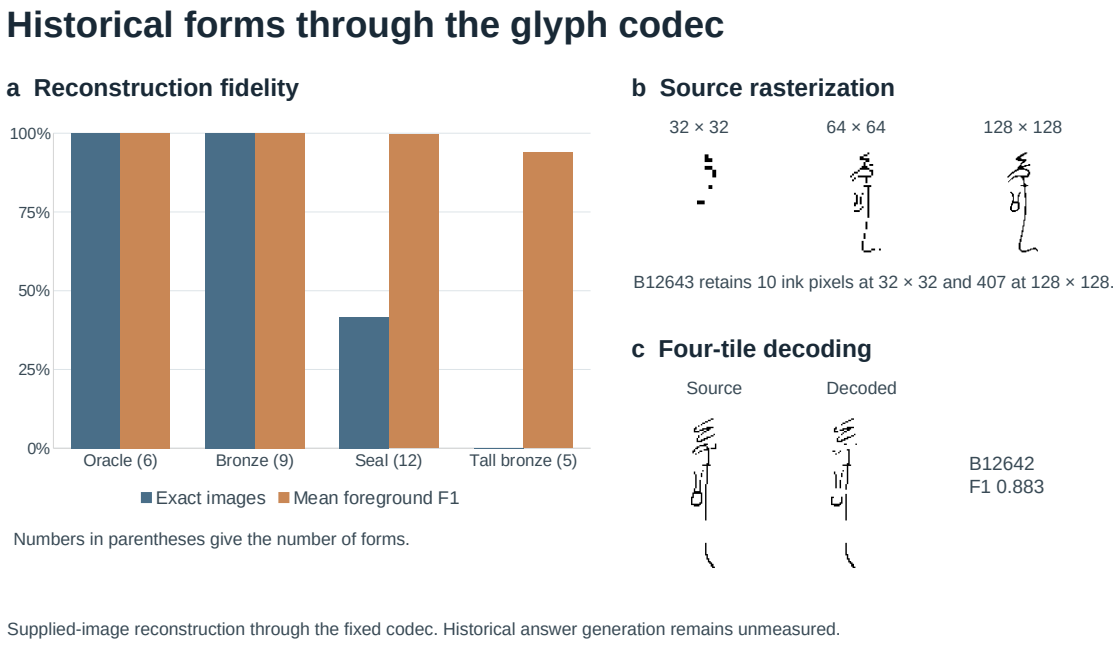


Fig. 11 | Historical forms through the fixed glyph codec. a, Exact reconstruction fraction and mean foreground F_1 for the 27 single-tile images and five tall bronze crops. Group sizes are in parentheses. **b**, The same source SVG, B12643, rasterized at three resolutions. It is the sparsest input among the 27 single-tile images. **c**, B12642 before and after decoding four 32×32 tiles. It had the lowest foreground F_1 of the five tall crops. All panels use supplied historical images. The source identities and measurements are recorded in the archived evaluation receipt.

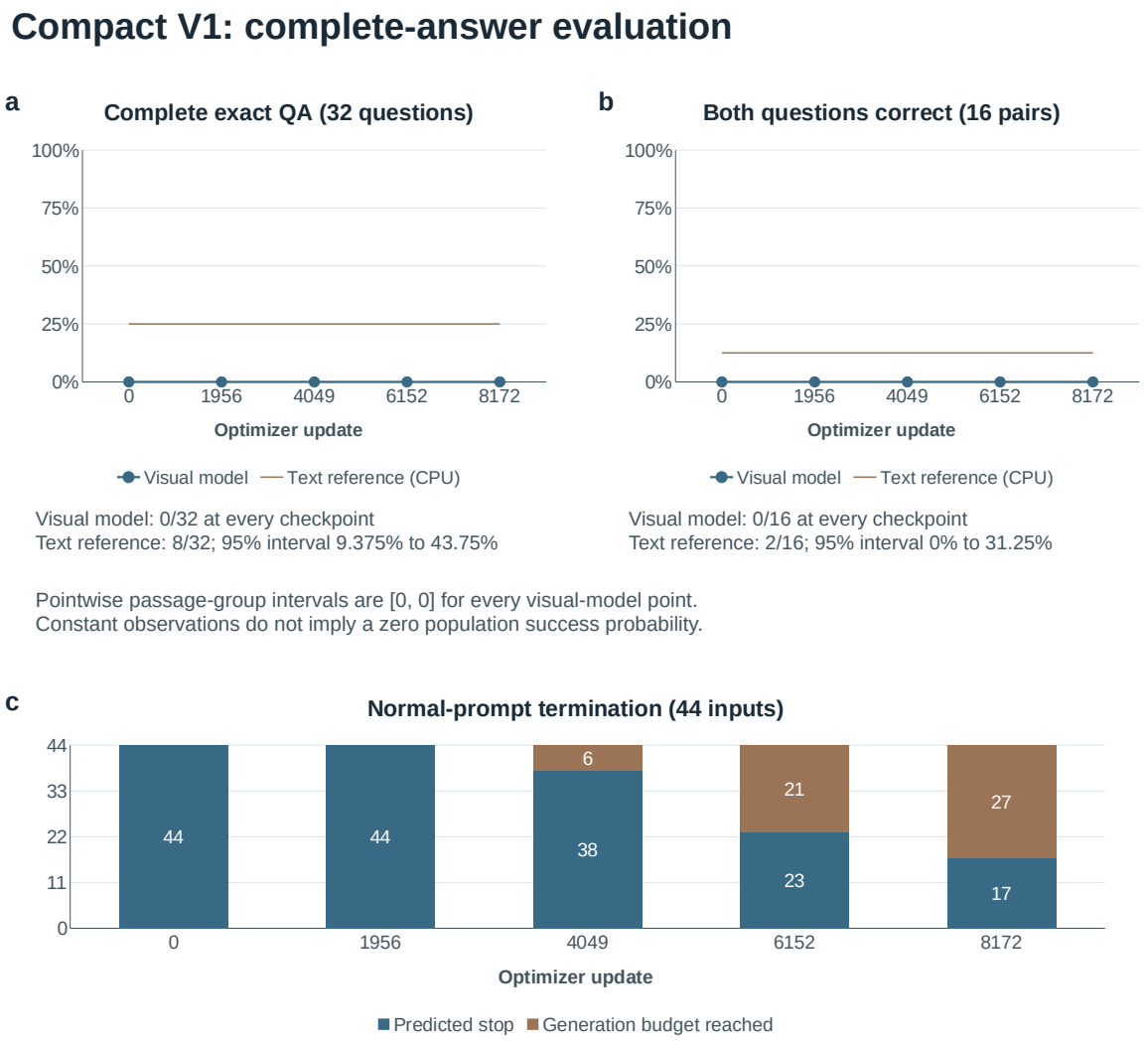


Fig. 12 | Complete-answer performance across the compact full-sequence fit. a, Complete exact QA answers among 32 questions at five fixed checkpoints. **b**, Both different questions correct within sixteen passage groups. The native-text reference ran on CPU and is shown for task quality. Its displayed intervals resample whole passage groups. Every visual complete-answer interval is [0, 0], which does not imply zero population success. **c**, Predicted stops and budget-limited streams for 44 normal inputs. Blank responses are included among stops. All 372 generated pages were retained. Readability judgments come from one model reviewer without independent human adjudication.

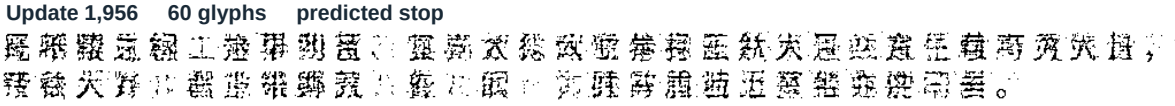
Fixed QA question A: raw outputs

Visible input: last three rows; the complete source PNG is retained in the evidence

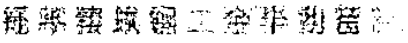
。下图显示英格兰48个名誉郡的分布情况。东萨塞克斯东北与根德郡相邻，南临英伦海峡，西与西萨塞克斯郡相邻，北与舒梨郡相邻。　問題：东萨塞克斯管辖哪些区域？　回答：

Update 0　0 glyphs　predicted stop (blank output)

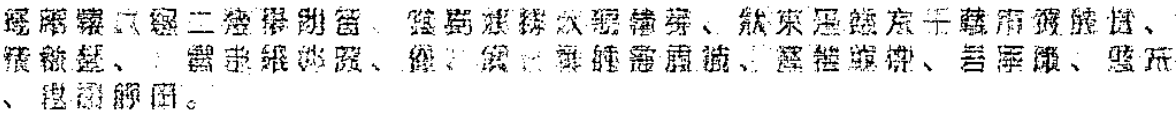
Update 1,956　60 glyphs　predicted stop



Update 4,049　11 glyphs　predicted stop



Update 6,152　70 glyphs　predicted stop



Update 8,172　126 glyphs　predicted stop



The first QA pair and first book prefix were fixed before fitting. All five outputs are shown.
Original answer PNGs are embedded without cropping, reconstruction, OCR replacement or repair.

Fig. 13 | Raw writing for the first fixed QA question. The input preview shows its last three rows; the complete input is retained in the evidence. All five original answer PNGs are shown, including the blank initial response. Labels give actual update, emitted glyph count and predicted stop or budget status. The case was selected before fitting. No answer pixels were cropped, reconstructed, replaced by OCR or repaired. Recognizable fragments do not establish a readable complete answer.

Fixed QA question B: raw outputs

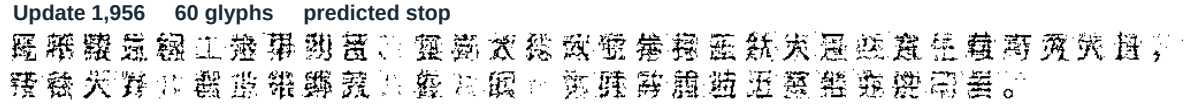
Visible input: last three rows; the complete source PNG is retained in the evidence

。下图显示英格兰48个名誉郡的分布情况。东萨塞克斯东北与根德郡相邻，南临英伦海峡，西与西萨塞克斯郡相邻，北与舒梨郡相邻。　問題：东萨塞克斯在哪里？　回答：

Update 0　21 glyphs　predicted stop



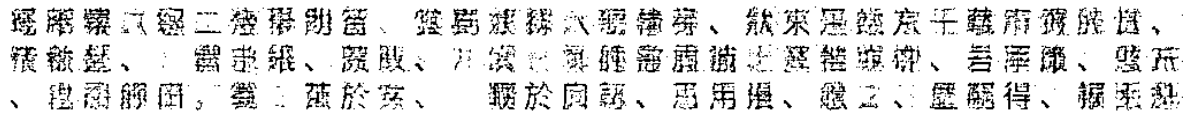
Update 1,956　60 glyphs　predicted stop



Update 4,049　11 glyphs　predicted stop



Update 6,152　97 glyphs　predicted stop



。

Update 8,172　128 glyphs　generation budget reached



The first QA pair and first book prefix were fixed before fitting. All five outputs are shown.
Original answer PNGs are embedded without cropping, reconstruction, OCR replacement or repair.

Fig. 14 | Raw writing for a different question about the same passage. The second member of the first fixed QA pair changes the question while retaining the passage. All five original answer PNGs and termination labels are shown. The input preview contains its last three rows, with the complete input preserved separately. Similar output prefixes are visible across the two questions. This display supplies the registered examples, without selecting successes or repairing ambiguous marks.

Fixed book continuation: raw outputs

Visible input: last three rows; the complete source PNG is retained in the evidence

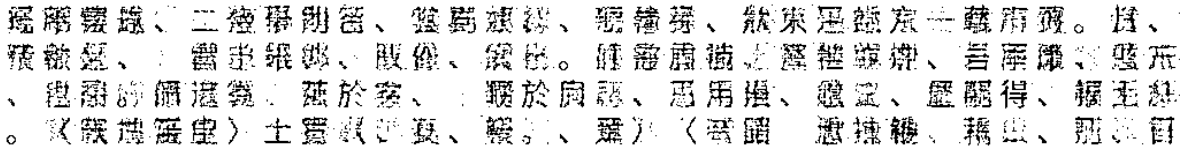
，奈何？」行者道：「莫成是丫頭的頭，和尚的身子？弄的這等不男不女，卻怎生是好？你可佈起罡來。」他就吹他一口仙氣，果然即時把身子變過，與那孩兒一般。

Update 0 **0 glyphs** **predicted stop (blank output)**

Update 1,956 **0 glyphs** **predicted stop (blank output)**

Update 4,049 **0 glyphs** **predicted stop (blank output)**

Update 6,152 **128 glyphs** **generation budget reached**



Update 8,172 **128 glyphs** **generation budget reached**



The first QA pair and first book prefix were fixed before fitting. All five outputs are shown.
Original answer PNGs are embedded without cropping, reconstruction, OCR replacement or repair.

Fig. 15 | Raw writing for the first fixed book continuation. The preview contains the last three rows of the visible prefix. All five original outputs are retained, including the three early blank responses. Later nonblank writing does not establish readable coherent continuation. The complete input, original pixel streams, source hashes and all endpoint controls remain in the evidence.

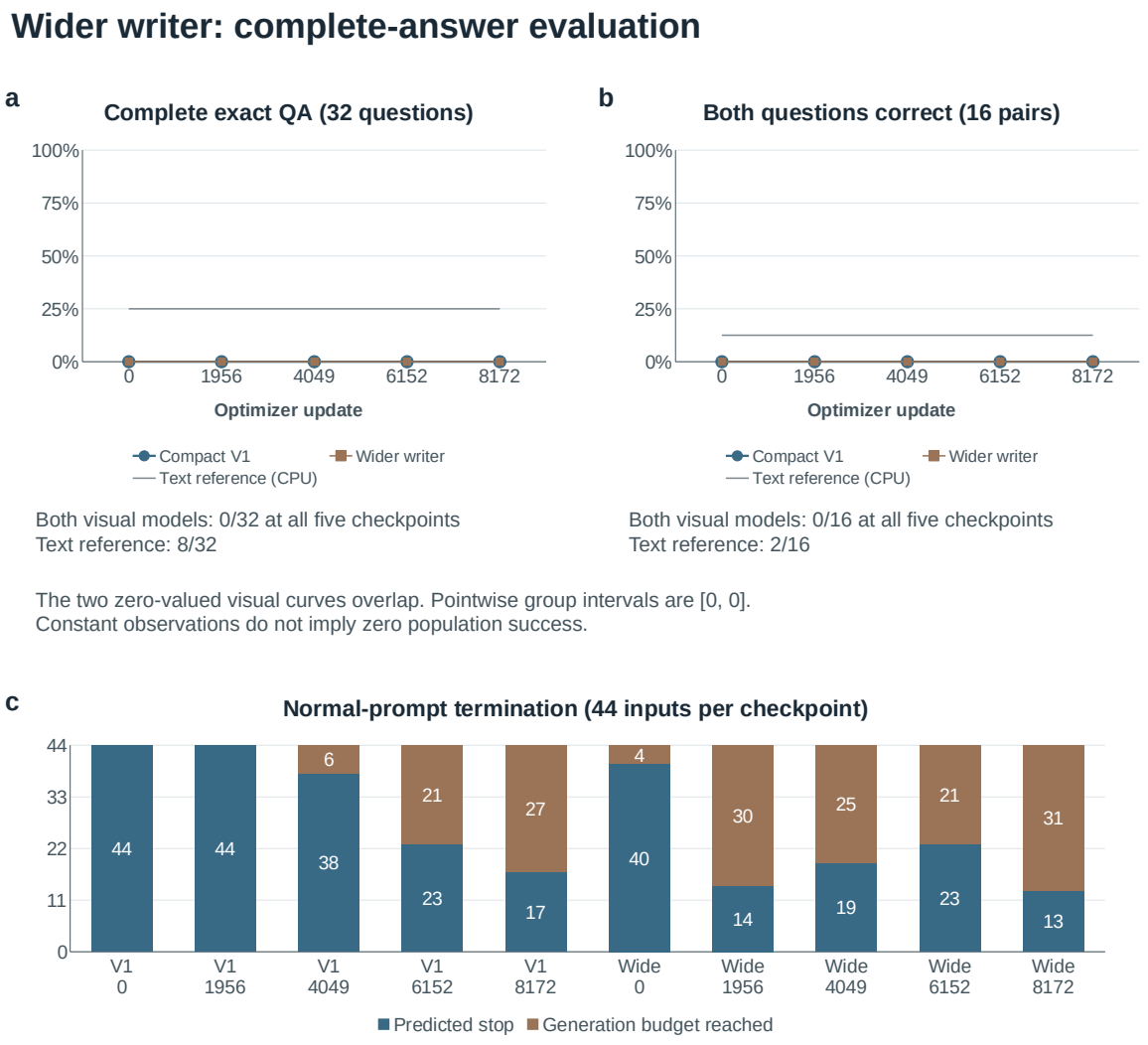


Fig. 16 | The wider writer preserves the complete-answer failure. a,b, Complete exact QA and both-correct passage pairs at all five fixed checkpoints. The two visual curves overlap at zero. The same native-text CPU reference is shown for task quality. Constant group-bootstrap observations give [0, 0] intervals, without establishing zero population success. **c**, Predicted stops and budget-limited streams in all 44 normal inputs per model/checkpoint. Blanks remain included. Both fits retain all 372 outputs; each review has one model reviewer with prior dataset familiarity and no independent human adjudication.

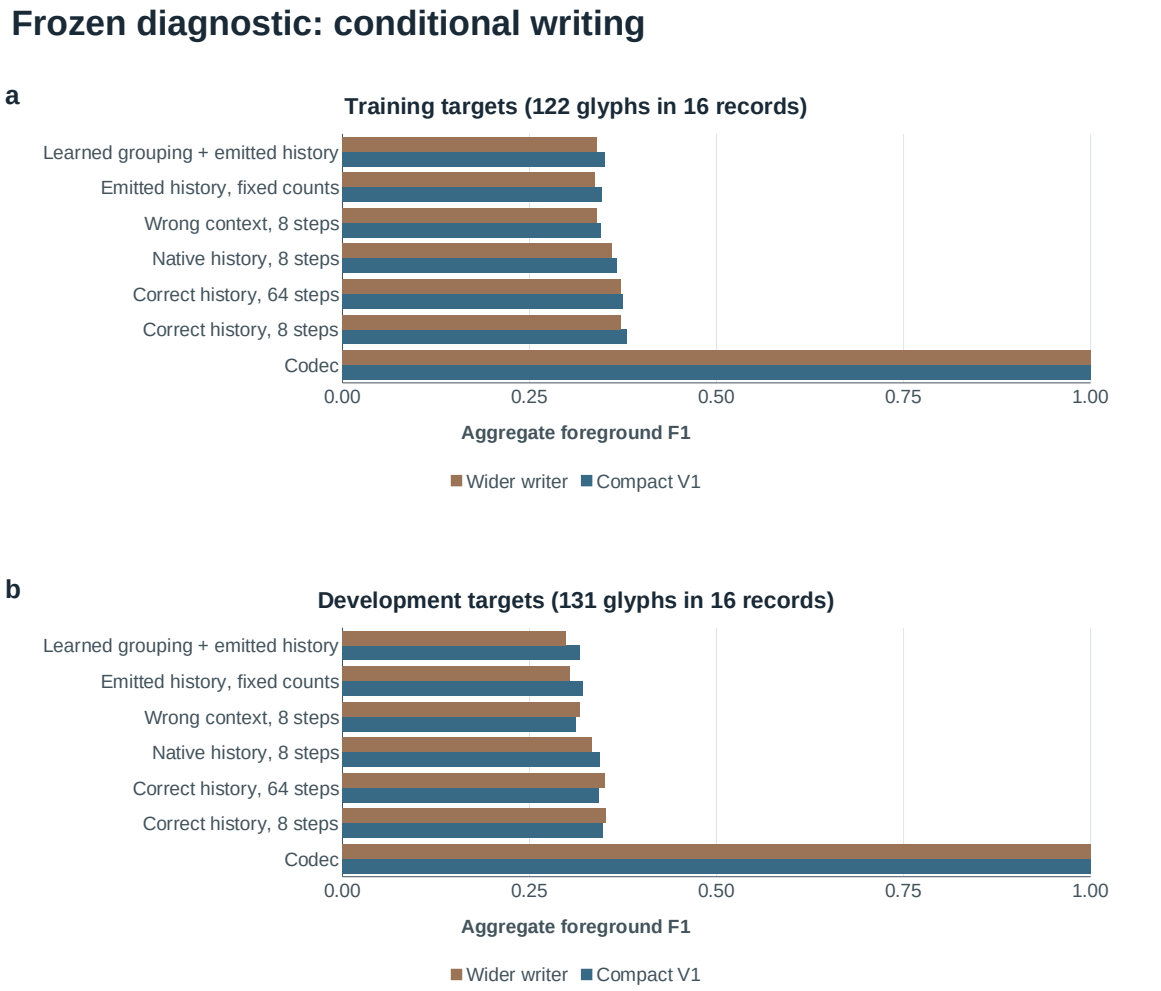


Fig. 17 | Frozen-weight writing with explicit reference supplies. a,b, Aggregate foreground F1 against the 122 training and 131 development target glyphs. Every writer arm receives reference output lengths. History and update counts vary as labeled. Native history is an explicit source-embedding diagnostic bypass, not deployed image inference. Both models fail the first target span before emitted feedback. Codec fidelity and the supplied-reference measurements are separate from autonomous language accuracy. These selected diagnostics have no generalization interval.

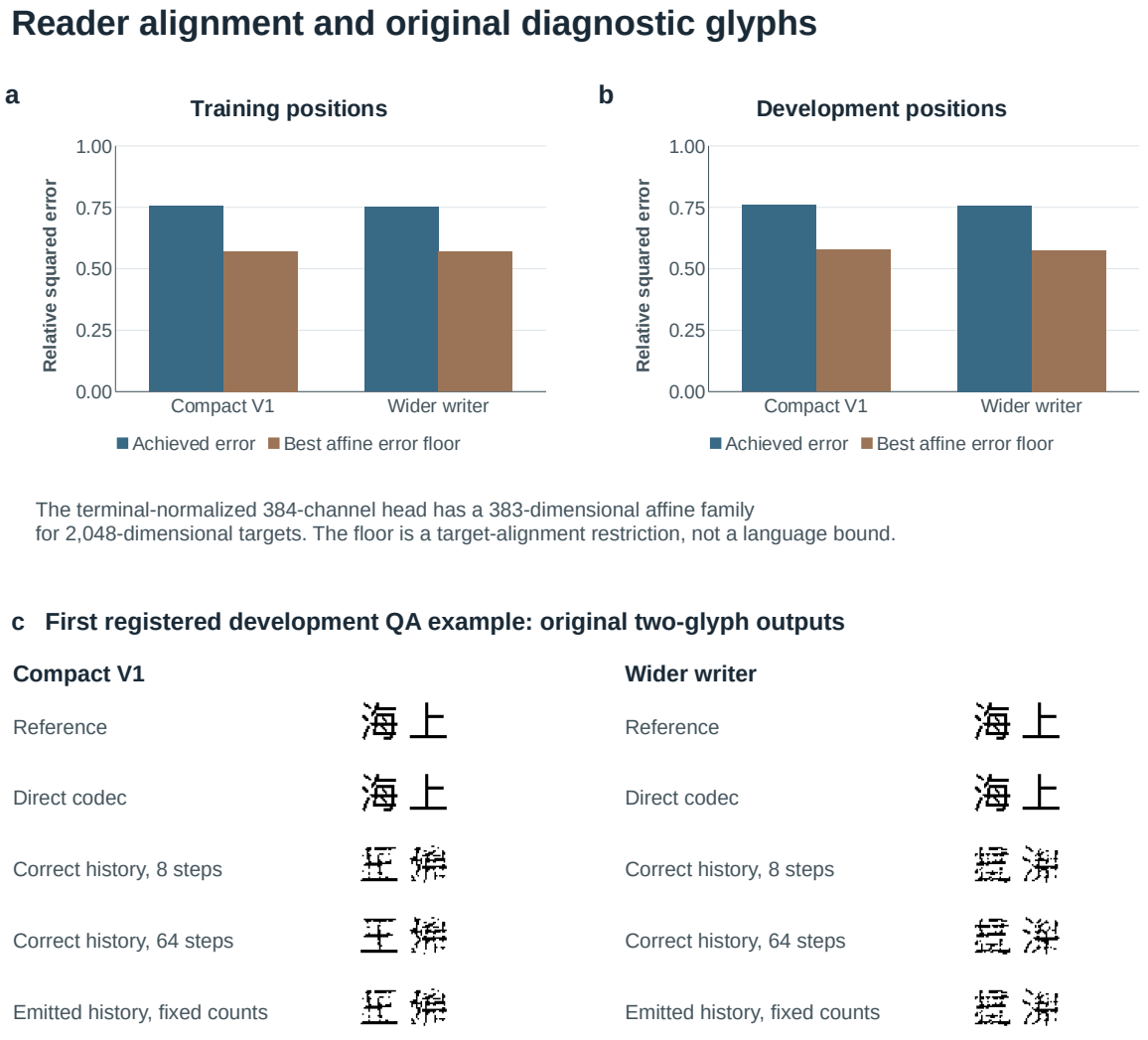


Fig. 18 | Reader alignment and unrepaired diagnostic outputs. a,b, Native-position-weighted achieved relative squared error and the best affine-family error floor for each trained head. The floor relaxes preceding image-network constraints and is not a language bound. **c**, The first registered development QA example, showing original reference, codec and writer pixels at exact nearest-neighbour enlargement. The deterministic display rule was added during analysis, not preregistered. Reference supplies and incorrect glyphs remain visible. No OCR replacement or output repair was applied.

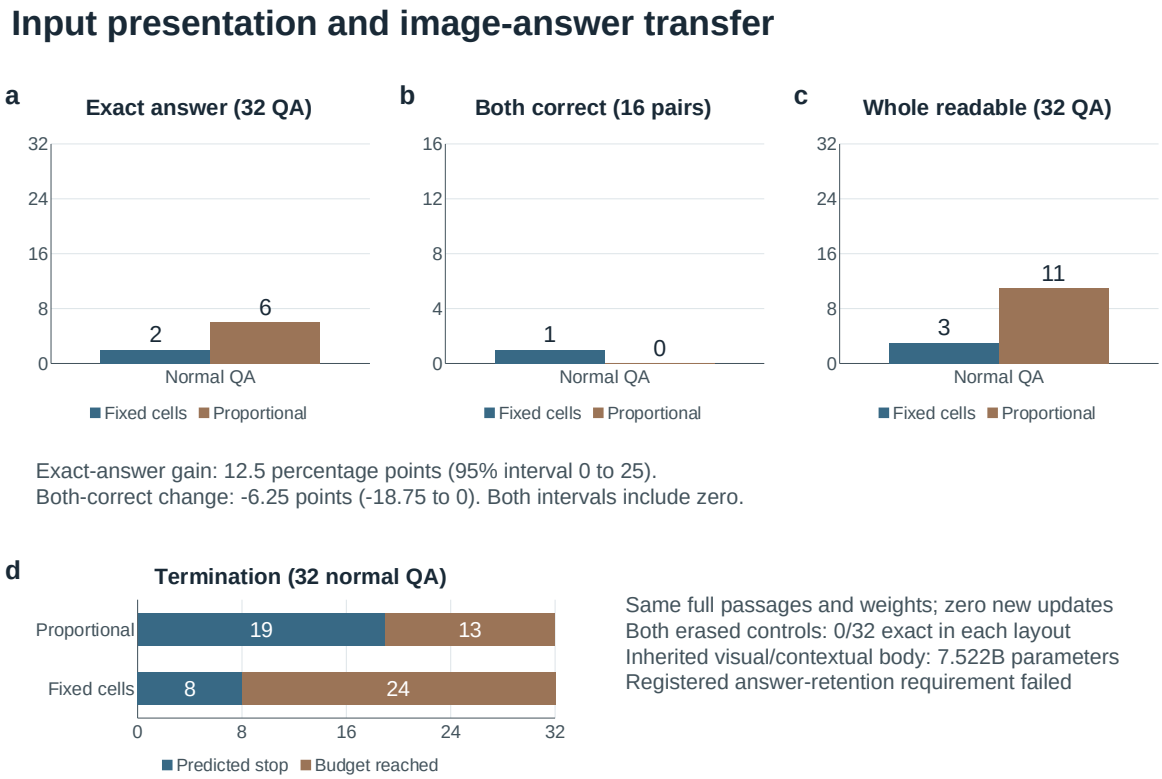


Fig. 19 | Proportional prompts increase exact answers but lose a correct pair. a–c, Complete exact QA, both-correct question pairs and whole-response readability on the same 32 questions. Input presentation changes with weights, full passage content and actual-pixel output feedback fixed. Counts retain every registered case. The displayed pointwise difference intervals resample sixteen whole passage groups; both include zero. **d**, Predicted stops and caps for the 32 normal QA inputs. Neither layout has an empty normal answer. **e**, Complete original outputs for the lost area answer, retained location and a newly exact quantity answer. These three matched examples were chosen retrospectively after complete scoring. Each pair is fitted at a common physical scale with no image crop, stroke reconstruction or OCR replacement. Opaque readings come from one model reviewer with prior cohort familiarity and no independent human adjudication. The registered retention rule fails, and no new training is performed.

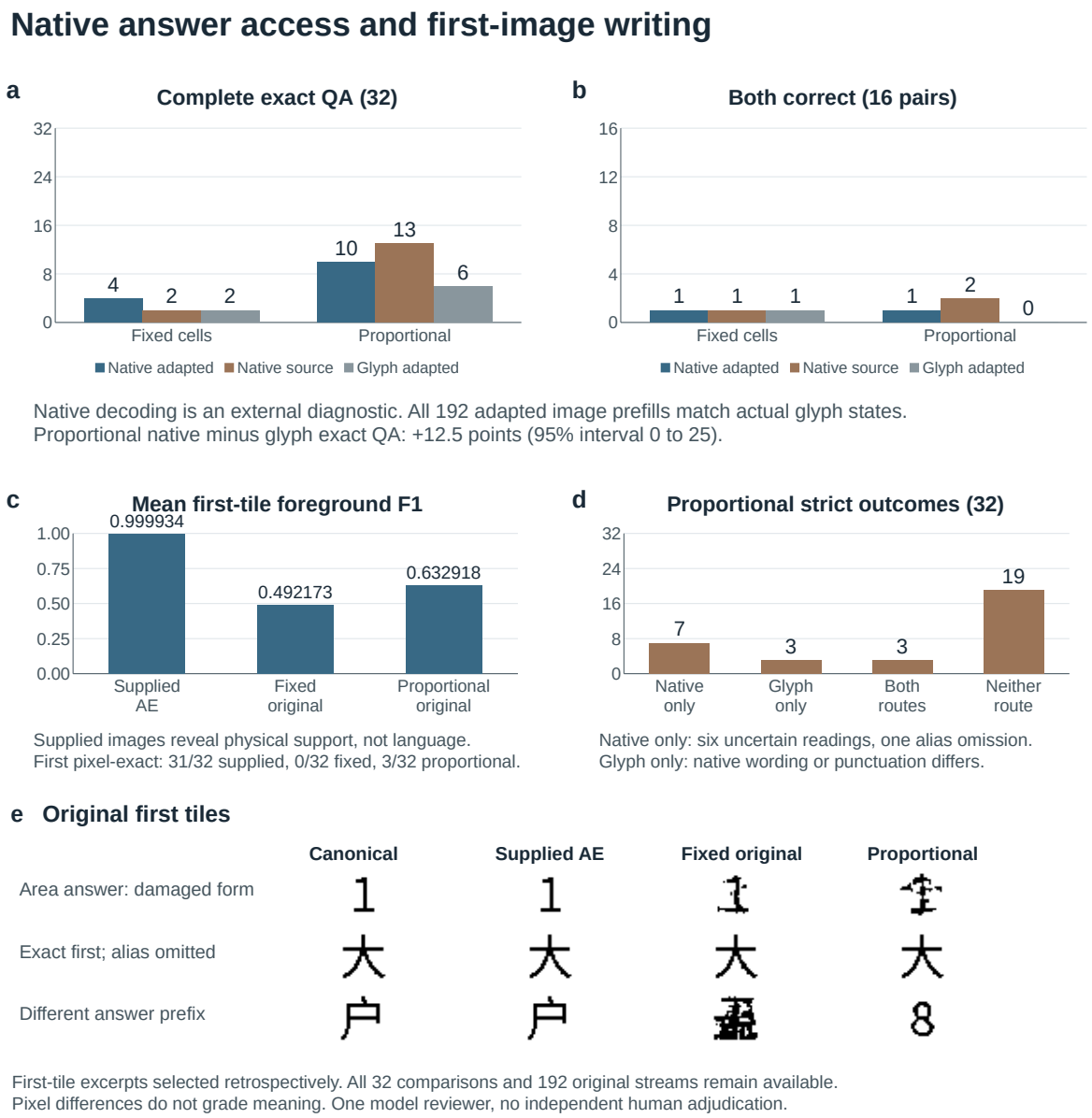


Fig. 20 | Retained answer access and physical support do not ensure glyph language. a,b, Complete strict QA and both-correct pairs for the external native decoder and original glyph actor. Native source disables contextual adapter outputs; it is not a separately evaluated image-output model. All original questions remain. Native source-unit and glyph budgets use different units. **c**, Mean foreground F1 of the first image against its canonical tile. The supplied-image encoder/decoder receives the answer. First-pixel agreement does not grade character identity, meaning or complete-answer quality. **d**, Strict native/glyph outcome groups for the 32 proportional cases. Native-only includes six uncertain glyph readings and one alias omission; glyph-only includes native wording/punctuation differences. **e**, Twelve original first-tile excerpts for three retrospectively selected cases: early damage, an exact first image with a later alias omission, and a different answer prefix. Tiles retain original pixels at a common scale, without stroke repair. All 32 labelled first-image comparisons and 192 complete original streams remain in the evidence. There is one model reviewer with prior cohort familiarity and no independent human adjudication. No new ILM generation or training is performed by these controls.

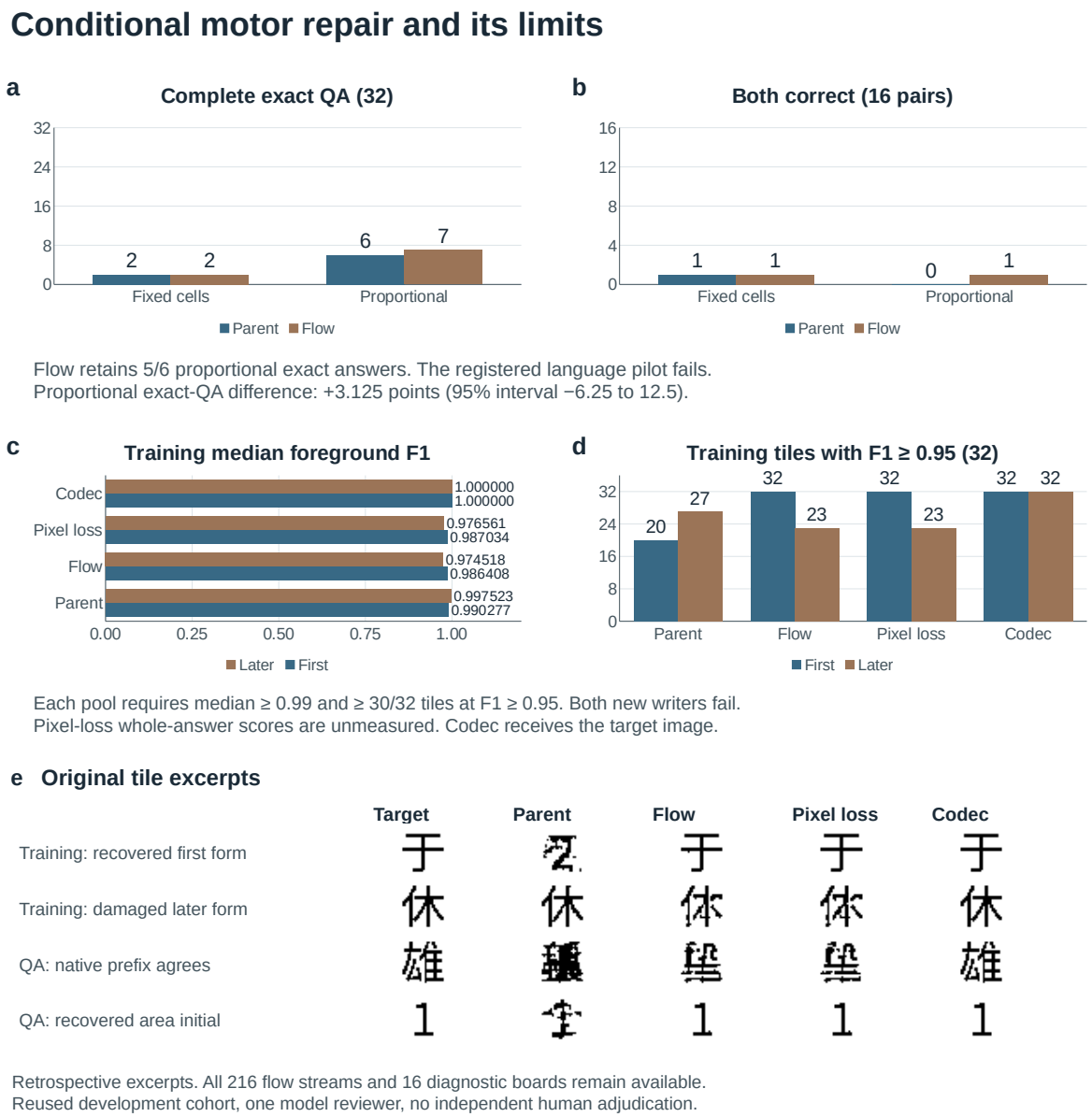


Fig. 21 | Conditional writing remains a restriction after retained answer access. a,b, Complete exact QA and both-correct passage pairs compare the unchanged parent with the original direct-context flow writer. All original questions remain. The gain loses one earlier proportional exact answer and fails the registered pilot. The pointwise interval resamples sixteen passage groups; it does not establish a general improvement. **c,d**, Median foreground F1 and counts reaching 0.95 for 32 first and 32 later training states. Pixel loss is the paired late-time decoded objective. Codec receives the target image. Both new writers fail the per-pool median/count gates. F1 values display six decimals; full precision remains in the bound data. These are model comparisons, not a measured language-learning trajectory. The pixel-loss model's complete-answer quality is unmeasured. **e**, Twenty original 32-square tiles at equal geometry, selected retrospectively to expose training recovery, later damage, a damaged native- agreeing prefix and an area-answer initial. QA rows are first-tile excerpts, not whole answers. All 216 flow streams and sixteen complete diagnostic boards remain available. No pixels are repaired. There is one model reviewer and no independent human adjudication.

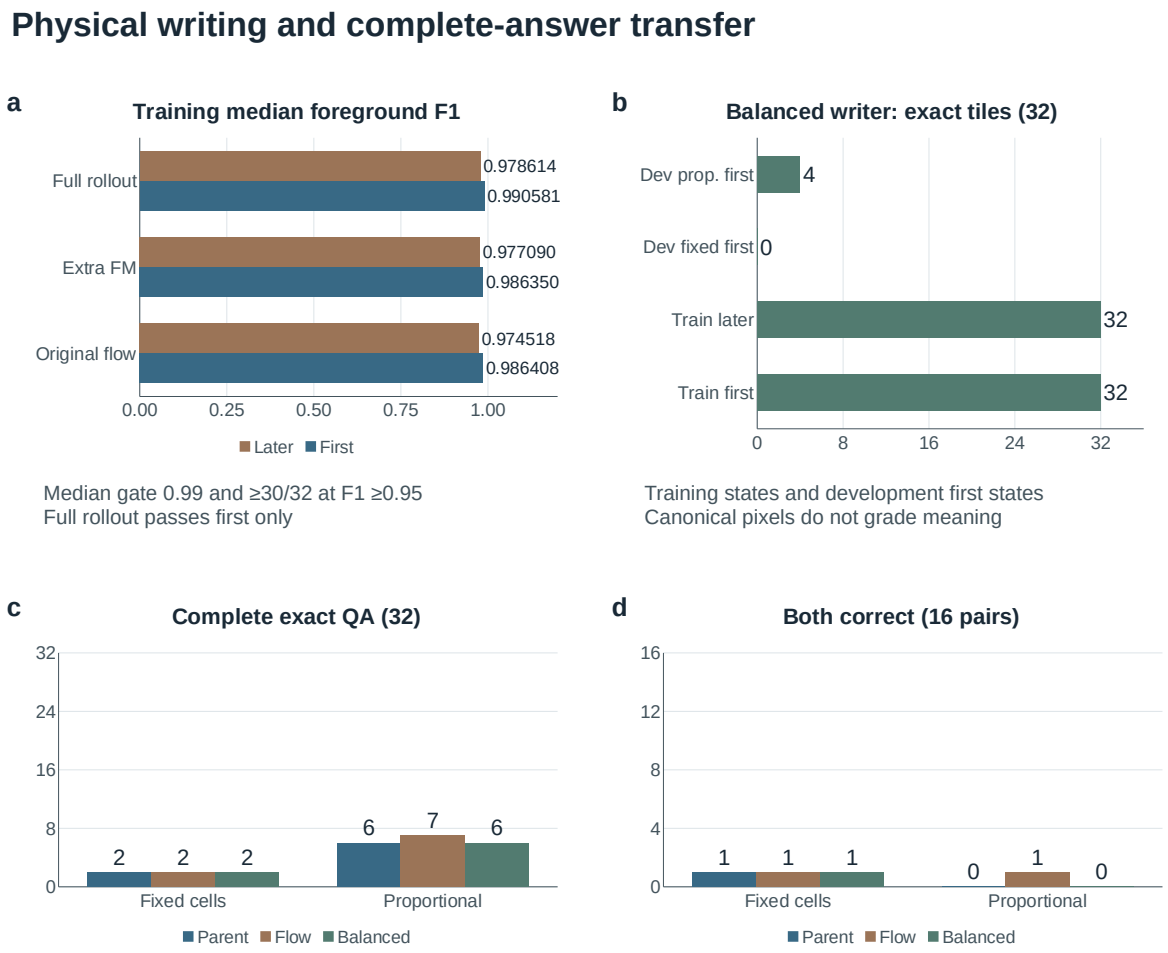


Fig. 22 | Faithful sampled training glyphs do not establish autonomous language transfer. a, Median foreground F1 for 32 first and 32 later training states. Extra FM and full rollout are matched 2,000-update continuations from the original flow EMA. Full rollout differentiates decoded loss through every sampling interval. The first-only median/count pass leaves both new arms' complete-language quality unmeasured; no language bar is assigned to them. **b**, Exact balanced-writer tiles across four unchanged 32-position pools. Training fidelity and canonical development agreement are distinct from semantic accuracy. **c,d**, All 32 normal QA questions and sixteen pairs per layout compare the parent, original flow and frozen balanced bundle. The balanced model retains all parent exact answers but only six of eight required proportional union cases, failing its pilot. Its readout and stopping predictor differ together; reader, codec and normalization stay unchanged. All 216 initial states match the parent. Values and workbook cells derive from complete records, with F1 displayed to six decimals. These fixed model comparisons are not a measured language-learning trajectory. One model review has prior cohort familiarity, without independent human adjudication or untouched confirmation.

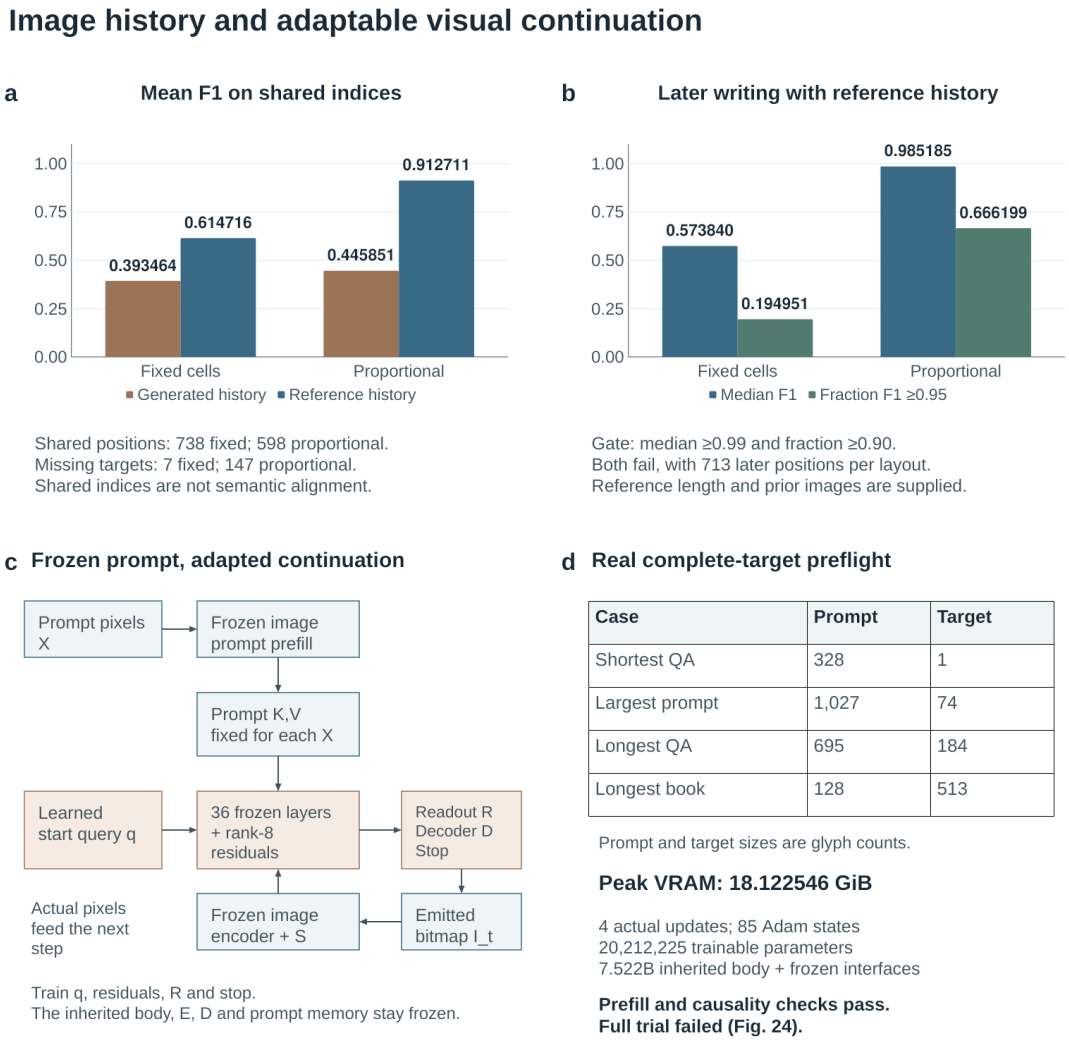
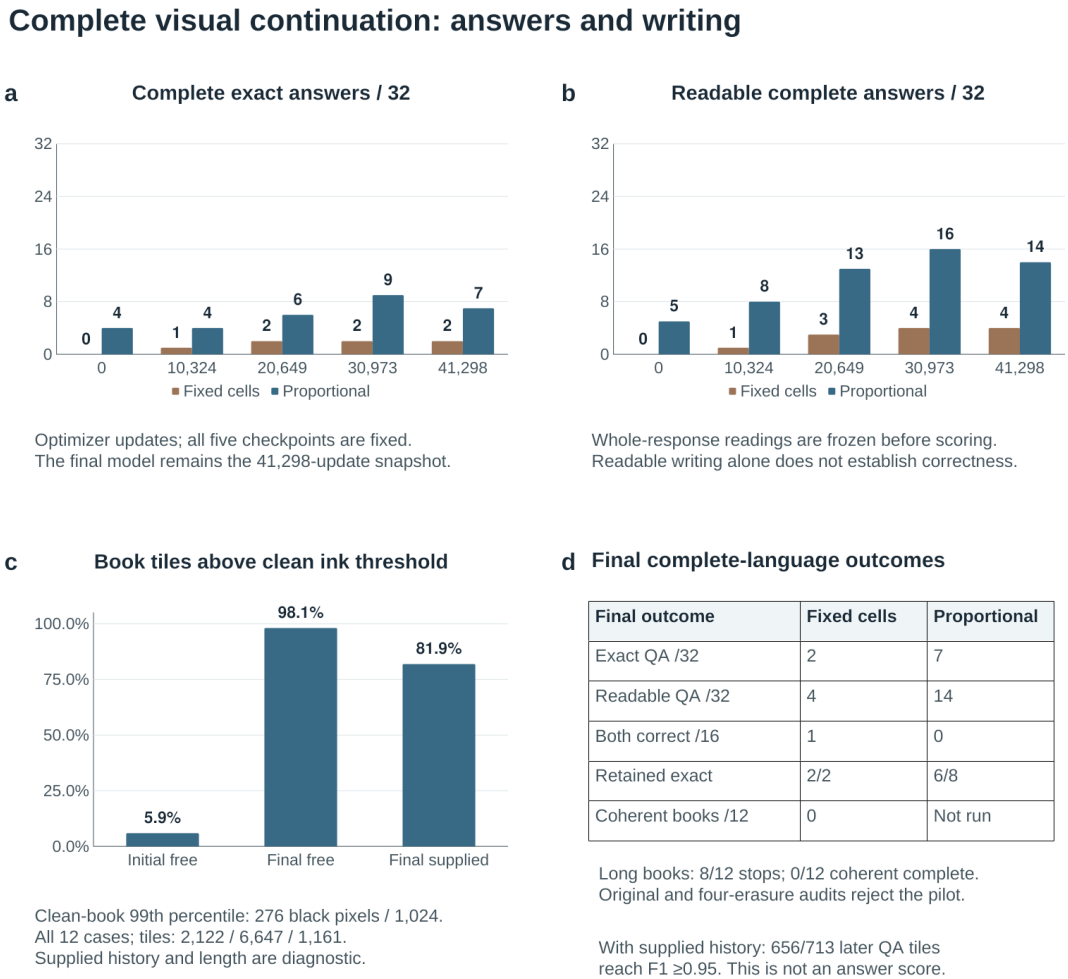


Fig. 23 | Correct prior images improve canonical writing without qualifying autonomous language. a, Mean foreground F1 at target indices present under both generated and supplied reference-image history: 738 fixed-layout and 598 proportional positions. Missing and extra autonomous positions remain in the complete records. Numeric index agreement does not align alternative semantic answers. Reference history supplies answer information and length, with stopping bypassed. **b**, Median later F1 and the fraction of later positions reaching 0.95, using all 713 later reference positions per layout. Both layouts fail the joint median/count gate. These canonical image metrics are not language accuracies. **c**, Implemented continuation computation. Frozen prompt prefill creates immutable prompt keys and values. The query and all 36 native answer layers continue from visible glyph images; new residuals, readout and stopping train, while the inherited body, image encoder/decoder and affine transform remain frozen. The stopping head is separate from decoded pixels. Only actually emitted images supply subsequent inputs; every answer layer is recomputed during fitting. **d**, Four complete preflight cases retain all 772 target positions and four terminal boundaries. Four real optimizer updates peak at 18.122546 GiB allocated GPU memory. Independent audits check targets, optimizer states and numerical boundaries; prompt/frozen-body checks come from the registered GPU execution. Initial residual-down tensors were not archived in that preflight. The 7.522B inherited body and frozen interfaces remain deployed costs beyond the 20,212,225 trainable parameters. F1 chart cells and labels round to six decimals; complete precision remains in bound records. The supplied-history diagnosis and preflight are separate from the completed full fit, whose final language pilot fails (Fig. 24). The later joint writer has no measured language result here. Reused development data and one model reviewer with prior familiarity lack independent human adjudication or untouched confirmation.



Final outcome	Fixed cells	Proportional
Exact QA /32	2	7
Readable QA /32	4	14
Both correct /16	1	0
Retained exact	2/2	6/8
Coherent books /12	0	Not run

Fig. 24 | Complete visual-continuation adaptation improves some short answers but fails sustained writing. a,b, Frozen whole-response exact and readable counts at all five fixed checkpoints, with 32 questions and sixteen paired passage groups per layout. The three-quarter peak does not replace the prescribed final model. **c,** Fractions of book tiles above the clean training-book 99th-percentile ink count, 276 of 1,024 pixels. Initial and final autonomous pools contain 2,122 and 6,647 tiles; the final supplied-history pool contains 1,161 target positions. Each pool retains all twelve prefixes, but lengths differ. Supplied preceding answer images and length, with stopping bypassed, make the third bar a physical diagnosis. Chart fractions round to six decimals for workbook storage and percent labels to one decimal; exact counts and full precision remain in the source data. Excess ink alone does not establish its cause or a stable attractor. **d,** Final complete-answer, paired and retained-success counts. Book prompts use only the fixed layout; no normal book arm receives a coherent complete continuation. The 656/713 later QA positions reaching F1 0.95 use supplied history and do not replace autonomous answer scores. Both the original pilot and the four-QA-erasure audit reject the final model. The full deployment retains 7.578 billion parameters; one-GPU execution does not establish a matched-quality efficiency advantage. All results use reused development data and one model reviewer, with no independent human adjudication. The subsequent joint writer has no language result in this figure.

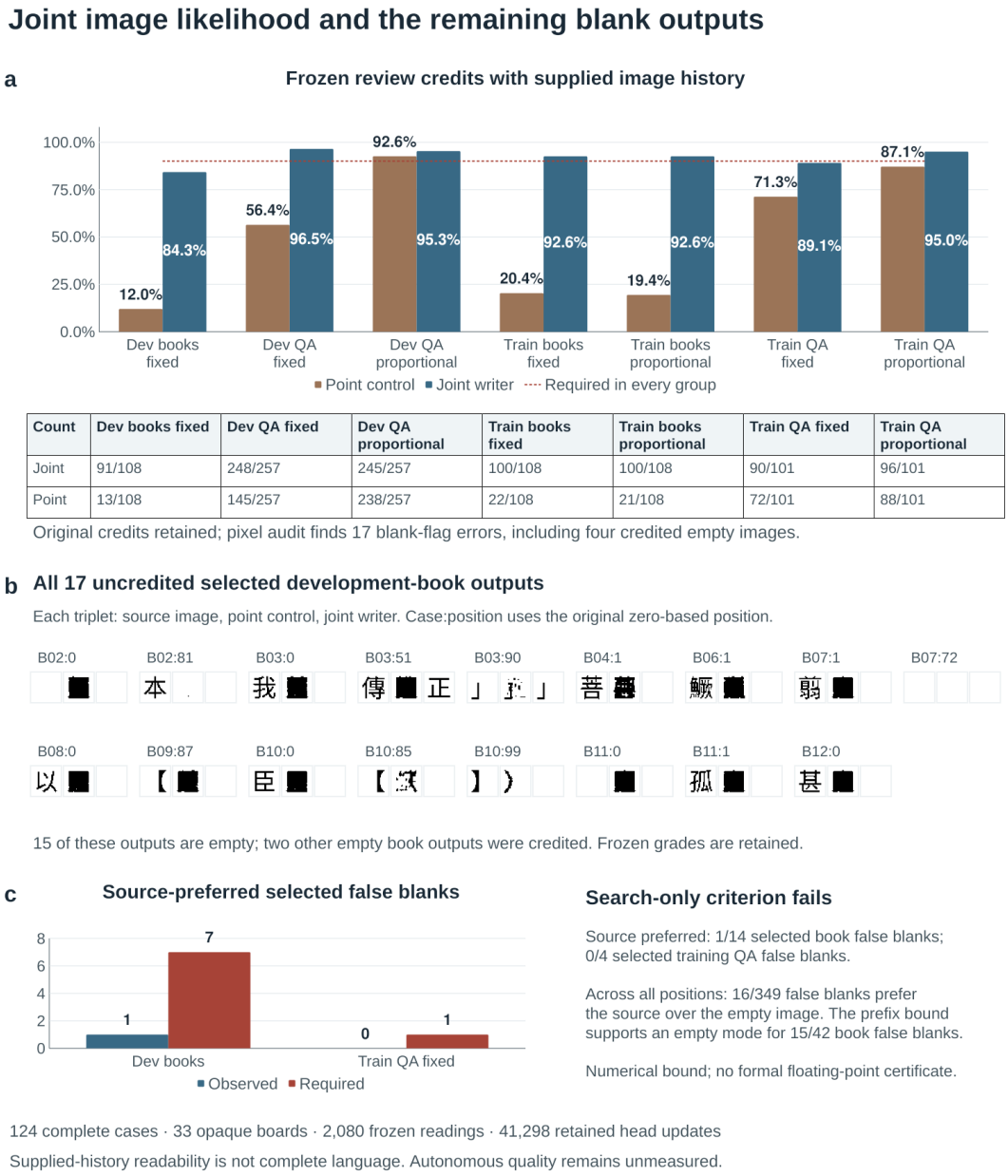


Fig. 25 | Joint image likelihood changes conditional writing but fails physical qualification. a, Original frozen readability credits for the joint writer and unchanged point-writer control in all seven supplied-history groups. The dashed line is the prespecified 90% prerequisite for every candidate group. Exact counts accompany the chart; chart fractions are rounded to six decimal places. Development books and training QA with fixed cells fail. These are retained review counts, not corrected physical accuracy: a subsequent pixel audit finds 17 blank-flag disagreements among 2,080 readings, including four credited empty images. **b,** Every uncredited selected development-book output, ordered by case and original zero-based position. Each triplet shows the source, point control and joint output. Original binary 32×32 images are expanded threefold by nearest neighbour, without repair. Fifteen of these seventeen joint outputs are empty and two are nonempty; two other empty book outputs were credited. Among all 108 selected book positions, four source images are blank, three overlapping the seventeen empty outputs, giving fourteen false blanks. **c,** Complete-image likelihood prefers the source over the empty image for one of fourteen selected book false blanks and none of four selected training-QA false blanks, below the unchanged requirements of seven and one. Across all 12,597 positions, sixteen of 349 false blanks prefer their source; the prefix bound supports an empty mode in fifteen of 42 development-book false blanks. The finite-precision bound is not a formal certificate. Review used one model reviewer with prior cohort familiarity and no independent human adjudication. Frozen readings, original scores and the failed verdict are preserved. Autonomous language quality remains unmeasured.